

GENERAL COLLEGE CHEMISTRY LAB MANUAL (SEMESTER 1)



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West*

Los Medanos College



LOS MEDANOS
COLLEGE

Laboratory Manual

Chemistry 25

Traditional

2022



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Disclaimer: These experiments are intended to be used with experienced supervision in a laboratory classroom setting. Los Medanos College and the Contra Costa Community College District assumes no responsibility or liability from the use or misuse of these experimental procedures either in or out of the laboratory classroom.

In other words: if you think it is a good idea to go in your back yard (or anywhere else) and mix some random chemicals, we inform you firmly that it is a bad idea. Do not sue us if or when you get hurt. We warned you.

Several sources were used for the experiments in this laboratory manual:

Permission was granted from the Laney College Department of Chemistry to adapt some of their experiments. Experiments 1, 12, 15, 16, 17, 21, 34 and Volume Measurements from the 30A manual; Experiments 3 and 4 from the 30B manual; and Experiment 17 from the 1A manual were adapted for use.

Permission was granted from the Santa Monica College physical science department to adapt their experiments from their Chemistry 11 laboratory manuals. Experiments 1, 2, 5, 7, 12, 15, 17, and 18 were adapted.

Permission was granted from Long Island University to use images in Experiment 2510-T.

The Techniques section is modified from the LMC Chemistry 06/07 Laboratory Manual.

Additional descriptions of techniques are available from

https://chem.libretexts.org/Bookshelves/Ancillary_Materials/Demos_Techniques_and_Experiments/General_Lab_Techniques.

The 2020-2021 LMC staff of student stockroom workers were of great assistance in testing out and improving these experiments.

Los Medanos faculty and staff who assisted this effort include Melinda Capes, Dennis Gravert, Edward Haven, Nancy Peters, Girlie Sison and Paul West.

Funding for this effort was provided by the following: "Expanding the LibreTexts Project into the Next-Generation Hub for Construction, Dissemination, and Usage of Open Educational Resource Textbooks" Grant 12701438 (19-0867)

Organization Version Control

Laboratory Manual Organization

This laboratory manual is organized in three parts:

Part 1 is laboratory experiments. These are available to instructors for laboratory class assignments. There are more experiments than are necessary for most classes, so students may not do all experiments in this manual.

Part 2 is laboratory techniques. These techniques are sometimes used in multiple experiments. They are available for both reference and supporting instructions on laboratory experiments.

Part 3 is for supplemental documents. This section is for more detailed explanations, additional problems sets, and abridged reporting sheets for specific experiments. This section is intended for faculty to add customized supplemental documents.

Experiments are numbered in the sequential order they were prepared. There is no particular relevance to the sequential numbering scheme. Experiments primarily intended for the Los Medanos College Chemistry 25 course begin with a '25'. Experiments primarily intended for the Chemistry 26 course begin with a '26'. Faculty may assign experiments as appropriate across both courses. Many of these experiments have two versions: 'traditional' and 'inquiry'. The numbering scheme includes a 'T' or an 'I' to indicate which version is present. Refer to the appropriate course syllabus for which experiments are being assigned by your instructor.

Pagination is done within each experiment and technique.

Version Control

Because this laboratory manual is intended to be improved over time, all documents use version control. Users of this manual should be aware of how version control has been implemented. In the upper right hand area of all pages, there is a version number. Version numbers are changed when a revision is done, and are also incorporated into the filenames of the individual experiments. The magnitude of the difference done during an update will determine how much the version number is changed.

A minor change such as a misspelled word would cause the version to change in the third place. For example, if an experiment went from version 1.3.2 to 1.3.3, it indicates a minor change between the two versions.

A moderate change, such as adding a section or changing a diagram or photograph would cause the version to change in the second place, and resetting the third place. For example, a technique changed from version 1.2.5 to version 1.3 indicates that a significant change occurred, but that the underlying procedure is similar.

A major change, such as a full rewrite of the procedure, would be a large and significant change. For example, a change from version 2.3.4 to version 3 would indicate that the two versions are

fundamentally different.

The most current version of experiments and techniques will be offered as the individual file (available for uploading to Canvas). An annual update is planned for a single production file for easy printing.

EXPERIMENTS	
Expt #	Title
2500	Safety in Chemistry Laboratory
2501	Using Excel for Graphical Analysis of Data
2502	Scientific Method
2503	Accuracy and Precision of Lab Glassware
2504	Mass and Density Measurements
2505	Hydrates
2506	Quantitative Reactions
2507	Stoichiometry of Reactions
2508	Beer's Law
2509	Spectral Analysis of Cu^{2+}
2510	Thermochemistry
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Techniques	
Letter	Title
A	Use of Laboratory Notebook
B	Significant Figures
C	Use of Balances
D	Setup of vacuum filtration
E	Volumetric Transfer Pipet Use
F	Crucible use
G	Buret use
H	Use of the Bunsen burner
I	Use of Spec 20/Spec 200

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This text was compiled on 01/12/2026

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Licensing

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Los Medanos College
General College Chemistry Lab Manual
(Semester 1)

CHAPTER OVERVIEW

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2500 Safety in Chemistry Laboratory

SAFETY AND THE CHEMICAL LABORATORY

1.0 INTRODUCTION

Teaching and research laboratories are locations for learning and exploring chemical reactions and properties. Like all activities in daily life, a risk of injury exists. Safety precautions and work practices reduce this risk. There are some hazards that are specific for chemicals.

This document cannot cover all laboratory safety guidelines for all future chemistry operations students will encounter. A whole field of industrial hygiene exists to create safer work practices. The objectives for this experiment are to

- Establish a culture of safety awareness
- Introduce the RAMP concept
- State the general rules for working safely in a chemical laboratory
- Describe differences between a hazard and a risk and acute and chronic toxicity
- Introduce a safety data sheet (SDS) and identify information it contains
- Present three common labeling and marking systems
- Build safe work practices into good habits
- Introduce regulatory agencies
- Prepare for emergency responses for when things happen

Chemical experiments in the teaching laboratory should contain a section on safety precautions and waste disposal. It is important to review these before entering the laboratory.

Safety precautions have been developed over many decades and are based on much experience and many events. The American Chemical Society (ACS) guidelines for undergraduate laboratory safety recommend the R. A. M. P. method.

- **R**ecognize the hazards
- **A**ssess the risks of hazards
- **M**inimize the risks of the hazards
- **P**repare for emergencies from uncontrolled hazards

References and further reading

Guidelines of Chemical Laboratory Safety in Academic Institutions, ACS Committee on Chemical Safety (CCS) Task Force for Safety Education Guidelines, 2016.

National Fire Protection Association (NFPA), 2017 https://www.nfpa.org/Assets/files/Ab...4/704_FAQs.pdf (accessed 06/23/2020)

Emergency Response Guidebook (2016), U. S. Department of Transportation. <https://www.phmsa.dot.gov/sites/phmsa...cs/ERG2016.pdf> (accessed 06/23/2020)

Prudent Practices in the Laboratory, Handling and Management of Chemical Hazards, National Research Council Committee on Prudent Practices in the Laboratory, 2011

Chemical Hygiene Plan, Los Medanos College, (the current version in effect).

2.0 RECOGNIZE THE HAZARDS

2.1 CHEMICAL LABELING

To recognize the hazards of chemical, one of the most important things is to read the label, documents that are shipped with the chemical, and additional references. Several types of labeling systems are in use. Each has been developed over time and for specific reasons. Three labeling systems are described below, but there are a few more systems in specific industries:

A: National Fire Protection Association (NFPA)

The NFPA is an international non-profit organization that develops standards to reduce injury due to fire, electrical and related hazards. The NFPA developed a labeling system that is in common use from a fire service perspective. Federal, state or local regulations may

require the use of the NFPA labeling system. It is common to see it on buildings such as water treatment facilities and medical facilities where elevated levels of oxygen are present.

The NFPA labeling consists of a diamond made of red, blue, yellow and white squares:



Figure 1: NFPA Diamond

Each color represents a specific and different hazard. Red is for flammability, blue is for health, yellow is for chemical reactivity and white is for special hazards. A number is shown from 0 (low) to 4 (high) in each of the squares that indicates a general assessment of the specific hazard. The NFPA diamond for acetone (formula C_3H_6O), a common organic solvent, is shown below:

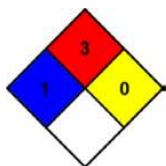


Figure 2: NFPA Diamond for Acetone

Acetone is moderately to highly flammable, and so is rated a '3' in the flammability diamond. Direct exposure can cause minor skin irritation, so it is rated a '1' in the health diamond. Acetone is not particularly chemically reactive, and so is rated as a '0' in the reactivity diamond. It has no special hazards, so the white diamond is left blank.

The NFPA labeling is designed for a quick and general overview of the safety concerns, especially for first responders. It is often displayed on buildings and storage facilities. Sometimes it can be seen on smaller containers.

B: U. S. Department of Transportation (DOT) Pipeline and Hazardous Materials Safety Administration (PHMSA)

When chemicals are transported along public highways, it is important for external packaging to display the hazards posed by the contents in the event of an accident and potential release. The PHMSA, a part of the DOT, regulates many aspects of this transportation, including but not limited to packaging, quantities and placarding. Some examples of the placards include the following:



Placards are often placed on trucks and railcars. These are often also displayed on shipping boxes for smaller quantities (in general for air transport) and for compressed gas cylinders.

Specific information for the transportation of chemicals can be found in the Emergency Response Guidebook (ERG). The ERG is updated every 4 years by the PHMSA. Most public safety agencies are required to have a hardcopy of the ERG or immediate access to the ERG in the event of an emergency.

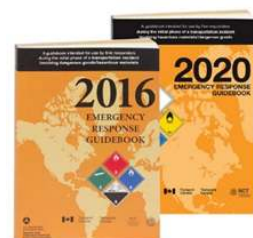


Figure 3 ERG Covers

A link to the current ERG is included in the reference section. A mobile app is available.

If the placard contains a number, it is cross-referenced in the ERG with guides to additional information about the specific hazards for the classes of chemicals present. An example would be a placard showing corrosive with the number "1830". The "1830" is an identifier for concentrated sulfuric acid (in the yellow bordered pages of the ERG). It corresponds to Guide No. 137 in the orange pages which has information about "Substances - Water-Reactive Corrosive".

The ERG is a useful resource to cross-reference trade and industry names. It provides more specific information about classes of compounds than the NFPA diamond.

C: U. S. Department of Labor (DOL) Occupational Safety and Health Administration (OSHA) Globally Harmonized System (GHS)

The GHS is an international system of hazard identification. The GHS was required by OSHA to be fully implemented by 2016. GHS labels are not mutually exclusive with the NFPA and DOT systems. It is possible to see a GHS symbol, a DOT symbol and an NFPA diamond side by side. Because GHS is used internationally, the same set of symbols should be used on the same material anywhere on the globe.

There are 3 components to the GHS:

- Hazard Classification Statements
 - o Physical
 - o Health
 - o Environmental Hazard
- Labels and Standard Symbols
- Safety Data Sheets (SDS)

The chemical manufacturer or importer is responsible for determining the hazard classification for the material. There are *physical*, *health*, and *environmental* hazard statement general categories. Some materials may fall into multiple categories.

A listing of the 9 specific hazard classes and GHS pictograms are below:



The GHS standardized the format and contents of labels on chemicals from the manufacturer. Within the laboratory environment, the GHS hazard labels will be the system most commonly encountered.

The GHS pictograms provide a way to rapidly assess the general chemical hazard of material. Additional information is contained within documents called the Safety Data Sheets (SDSs). The GHS specifies a standard set of 16 sections and content for SDSs. The chemical manufacturer is responsible for producing the SDS that accurately describes the known hazards of the material within the GHS format. Employees working with hazardous material must have access to the SDS as part of several labor laws (both Federal and State).

Several online sources of SDS are available. Labor laws require that the documentation is available and accurately describes the hazard of the material. Although it is best to have the SDS from the manufacturer, if a SDS is missing or lost, a replacement SDS can be accessed from an alternative source. Most manufacturers provide SDS free of charge from their catalog website. Some possible sources include:

<https://www.sigmaaldrich.com/united-states.html>

<https://www.fishersci.com/us/en/home.html>

<https://www.flinnsci.com/>

A full SDS for sulfuric acid is included in Appendix A. The SDS has 16 parts:

Table 1 Safety Data Sheet Sections

1. Name	9. Physical and Chemical Properties
2. Hazard(s)	10. Stability and Reactivity
3. Compositions	11. Toxicological information
4. First Aid	12. Ecological Information
5. Firefighting Measures	13. Disposal Considerations
6. Accidental Release Measures	14. Transport information
7. Handling and Storage	15. Regulatory Information
8. Exposure Controls/Personal Protection	16. Other information

One of the most important pieces of information provided in the SDS is the proper name and Chemical Abstracts Service (CAS) number for each pure component of the material. The CAS is a unique identifier for a compound and avoids confusion. Many industries use legacy names for chemicals, and these names are often used interchangeably. For example, isopropyl alcohol (also called 'rubbing alcohol') is the common name for 2-propanol. Even though these names are different, they all have the same CAS number of 67-63-0.

The Los Medanos College Chemistry Stockroom maintains a SDS for all pure compounds within the stockroom. A SDS will be provided upon request.

Reagents produced and distributed for use within the chemistry courses will have a GHS warning label, or the words 'No GHS Label Required', as appropriate.

Critical thinking skills are essential when reviewing a SDS. Unusual, rare or new compounds may have unknown hazards that are not included on the SDS. Manufacturers are conservative when describing hazards. Sometimes different manufacturers will assign different hazard classes to the same material. Consider the difference between sulfuric acid at different concentrations. Appendix B shows the SDS pages for different dilutions of sulfuric acid. While it would be expected that concentrated sulfuric acid will have the same GHS hazard class from all manufacturers, with diluted sulfuric acid there may be a judgment call made by the manufacturer.

2.2 ADDITIONAL HAZARDS

Like many work and school environments, laboratory hazards are not limited to chemical hazards. Wet floors are potential slip areas. Glassware can break, leaving sharp edges which can cut. Hot items often look like cold items and could result in accidental burns. Electrical hazards from equipment are commonly present.

Less common hazards include, but are not limited to, radioactive materials, cryogenics (cold materials), high pressure systems, vacuum systems, energetic compounds (explosives), and air or water reactive materials. It is beyond the scope of this document to comprehensively describe safety procedures for less common hazards. Specific knowledge, safety procedures, equipment and careful consultation with experienced personnel is recommended before working with any of these hazards.

3.0 ASSESS THE RISKS OF HAZARDS

The amount of information included in safety documentation can be overwhelming. By establishing 9 hazard classes, the GHS system quickly allows a general assessment of the hazard of a material. Whenever an unfamiliar chemical is to be used in a new procedure, a thorough review of the SDS is recommended to accurately evaluate the risk of hazard.

Laboratory experiments within introductory chemistry courses have been tested and evaluated to minimize the risk to students, faculty and staff. Maintaining proper labeling and documentation is part of the training for all students and staff for assessing risks of materials.

Experiments sometimes have special warnings that are set apart from the main text. Personnel should be certain to read and understand these warnings. An example is shown below:

!!Special Warnings Will be Stated Here!!

Experiments may include descriptions of specific hazards relevant to the experiment being planned. Be certain to read and understand these instructions.

!!Disposal Instructions!!

A description of how to dispose of the waste generated in the experiment should be included. Some waste may be permitted for drain disposal. Other waste must be captured for off-site disposal. Waste types are often segregated to prevent side reactions. Some waste is acutely hazardous and is easier and cheaper to manage in smaller quantities. Refer to this section in the specific experiment.

Consider the generation, management and disposal of waste generated by laboratory processes. Sometimes, the disposal costs for chemical reagents can be several times the purchase price of the original quantity.

4.0 MINIMIZE THE RISKS OF THE HAZARDS

Work practices can be designed to minimize the risks of many types of hazards within the chemistry laboratory. These measures include but are not limited to:

- Creation of experimental procedures and standard operating procedures
- Evaluation of experimental design
- Substitution of less hazardous materials when feasible
- Engineering controls such as fume hoods, separate laboratory classrooms
- Personal Protective Equipment (PPE) appropriate to the operation

By building good work habits within the chemistry laboratory course, risks are minimized. Some good habits include:

- Read and learn about the experiment before the course session

- Wear protective eyewear

- Wear protective clothing

- o Closed-toe shoes

- o Long pants

- o Lab coat

- Tie loose clothing and long hair

- Use specialized waste containers as appropriate

- o Broken glass waste containers

- o Biohazard disposal bags

- o Sharps disposal containers (for syringe needles)

- Segregate waste for disposal

- Keep areas clear from clutter

- Never leave open flames unattended

- Use fume hoods if appropriate

The creation, review, and incorporation of this Safety in the Chemistry Laboratory procedure and familiarization by personnel is part of the minimization of risk.

5.0 PREPARE FOR EMERGENCIES FROM UNCONTROLLED HAZARDS

While the previously described documentation, labels, procedures and good work practices are very effective at reducing emergencies, failures and mishaps will occur. Glass equipment may experience hairline fractures resulting in breaks. Spills may occur at any time. Electrical equipment may provide sparks and ignition sources for flammable materials.

All personnel should be prepared for emergencies and should be familiar with safety equipment including:

- Fire extinguishers
- Eyewash stations
- Safety showers
- Spill kits
- First aid kits
- Fire alarms
- Evacuation routes
- Emergency telephone numbers

Equipment should be inspected on a regular basis. Inspection tags and other documentation should be retained for a time to indicate that equipment has been inspected and is operational. At some facilities, emergency drills are used to test and exercise the emergency response system.

6.0 ACCIDENT INVESTIGATION

When an accident does occur, a report should be made to the instructor or supervisor. A review may be initiated. The review may be simple or complex and depends upon the severity of the accident. For example, a dropped beaker that breaks and does not injure anyone may need a simple cleanup, disposal and replacement. However, a chemical splash into a person's eye will require at least 15 minutes of flushing with water followed by a medical evaluation and a more comprehensive investigation.

The accident investigation process should be viewed to determine a root cause of the underlying accident. In some cases, the root cause may not be identified, or it cannot be addressed (such as with types of equipment failures). It is important for all people involved in an accident investigation to understand that the purpose is not to establish blame or responsibility, but the root cause (if any) so conditions that led to the accident can be changed to avoid future recurrences.

7.0 Activities

1. Choose one experiment in this laboratory manual that uses chemical reagents. Find a Safety Data Sheet for each reagent in the experiment. List the reagent, CAS number and GHS Warning Symbol for all reagents below:

Experiment Number			
Chemical	CAS	GHS Warning Symbol	Notes

2. Select one of the above reagents. Find a second SDS from an alternative supplier. Are there any significant differences between the two SDS documents? Would you expect any differences? Why or why not?

3. On the SDS provided in Appendix A, select an abbreviation or acronym. Find the full definition and share with your class what it is, what it means, and how should it be interpreted (e.g, LD50, IDLH, TWA, ACGIH, NIOSH, CFR, OSHA, etc.).

4. Draw a map of the laboratory classroom. Include the following:

- Where your lab station is located
- Exit/entrance doors
- Sinks
- Fume Hoods
- Eyewash station
- Safety Showers
- Fire Extinguishers
- Fire Alarm locations

- Emergency notification equipment

Appendix A: Safety Data Sheet for Concentrated Sulfuric Acid

Appendix B: First Pages of Additional SDS for Sulfuric Acid (various dilutions)

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2501 Using Excel for Graphical Analysis of Data

Using Excel for Graphical Analysis of Data

Last Name	First Name	Partner Name(s)	Date
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1.0 INTRODUCTION

In several upcoming labs, a primary goal will be to determine the mathematical relationship between two variable physical parameters. Graphs are useful tools that can elucidate such relationships. First, plotting a graph provides a visual image of data and any trends therein. Second, using appropriate analysis, graphs provide us with the ability to predict the results of any changes to the system.

An important technique in graphical analysis is the transformation of experimental data to produce a straight line. If there is a direct, linear relationship between two variables, the data may be fitted to the equation of a line with the familiar form $y = mx + b$ through a technique known as linear regression. Here m represents the slope of the line, and b represents the y-intercept, as shown in the figure below. This equation expresses the mathematical relationship between the two variables plotted and allows for the prediction of unknown values within the parameters.

Computer spreadsheets are powerful tools for manipulating and graphing quantitative data. In this exercise, the spreadsheet program **Microsoft Excel**® will be used for this purpose. Students will learn to use Excel in order to explore a number of linear graphical relationships. Please note that although Excel can fit curves to nonlinear data sets, this form of analysis is usually not as accurate as linear regression.

NOTE: The navigation prompts (commands) below are based on using MS Excel for PC (downloaded version).

2.0 PROCEDURE

Part A. Simple Linear Plot

Experimental Data: Different phosphate concentrations are analyzed using colorimetric analysis and the following data are generated.

mg Phosphate	Absorbance
4.984	0.097
9.968	0.174
19.935	0.303
29.903	0.500
49.838	0.774

1. Launch the program Microsoft Excel® in your computer. Go to the 'Start' button (at the bottom left or top right on the screen), then click Programs, followed by Microsoft Excel or look for this icon:

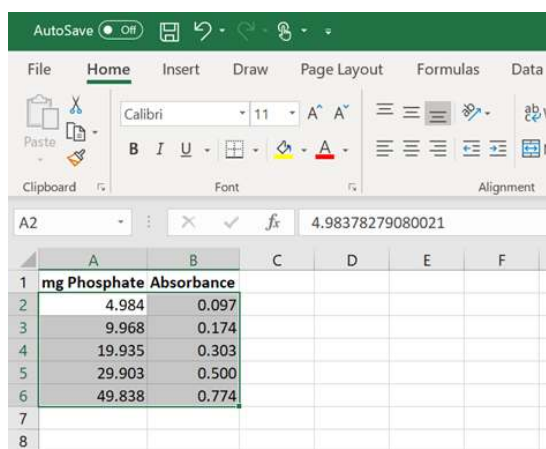


2. Select 'Blank Workbook' and enter the above data into the first two columns in the spreadsheet.

- Reserve the first row for column labels.

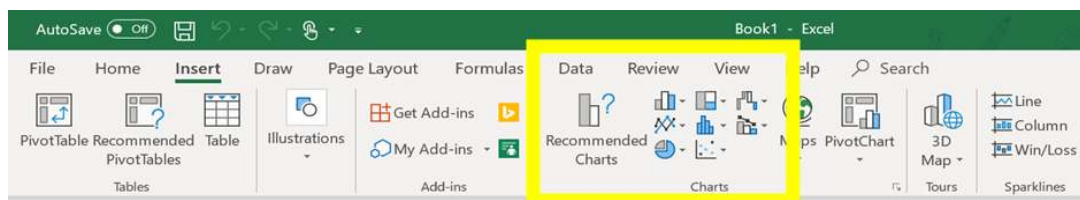
- The x values must be entered to the left of the y values in the spreadsheet. Remember that the independent variable (the one that you, as the experimenter, have control of) goes on the x-axis while the dependent variable (the measured data) goes on the y-axis.

3. Highlight the set of data (not the column labels) that you wish to plot (Figure 1). Click and drag your mouse from the top-left corner of the data set (e.g. cell A2) to the bottom right corner.

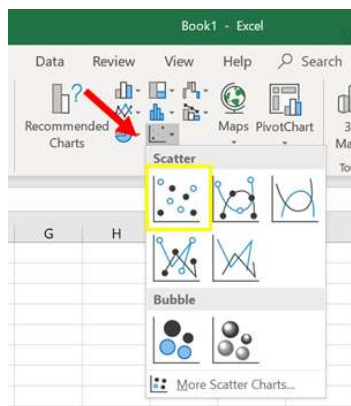


	A	B	C	D	E	F
1	mg Phosphate Absorbance					
2	4.984	0.097				
3	9.968	0.174				
4	19.935	0.303				
5	29.903	0.500				
6	49.838	0.774				
7						
8						

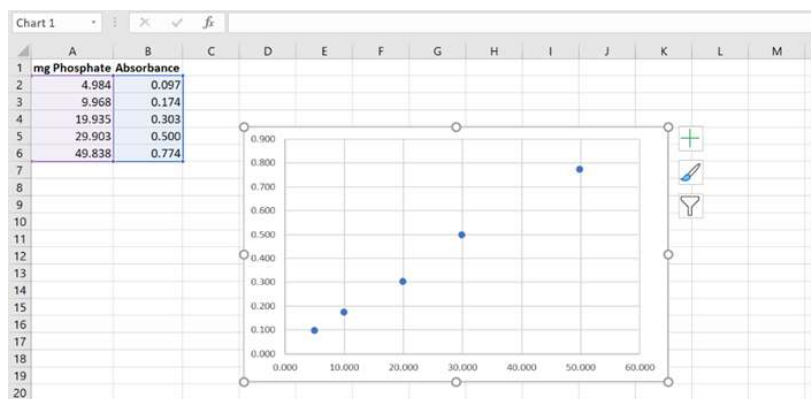
4. Click on 'Insert'. It's one of the tabs near the top of the Excel window. A toolbar below the 'Insert' tab should open (Figure 2).



In the Recommended Charts, select the 'Scatter' graph that shows data points only, with no connecting lines (Figure 3).



You should now see a scatter plot on your Excel workbook, which provides a preview of your graph (Figure 4).



5. If all looks well, you can now add title and label the axes of your graph.

a) To add title, click inside the chart; switch to the Design tab, and click Add Chart Element > Chart Title > Above Chart (Figure 5)

Note: The graph should be given a meaningful, explanatory title that starts out “Y versus X” followed by a description of your system.

b) To label the axes of your graph, click on Axis Titles > select Primary Horizontal Axis Title and > Primary Vertical Axis Title (Figure 6).

Note: It is important to label axes with both the measurement and the units used.

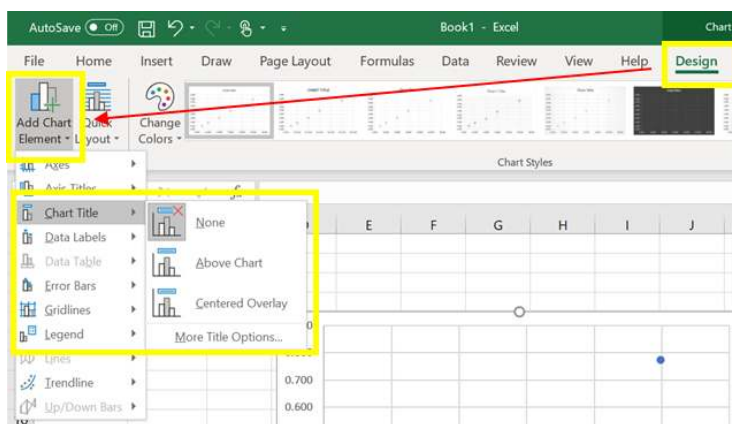


Figure 5

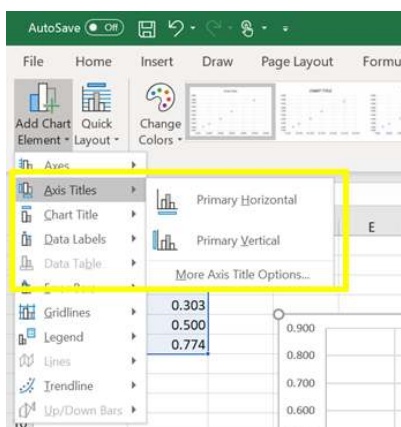


Figure 6

· To change the titles, click the text box for each title, highlight the text and type in your new title (Figure 7).

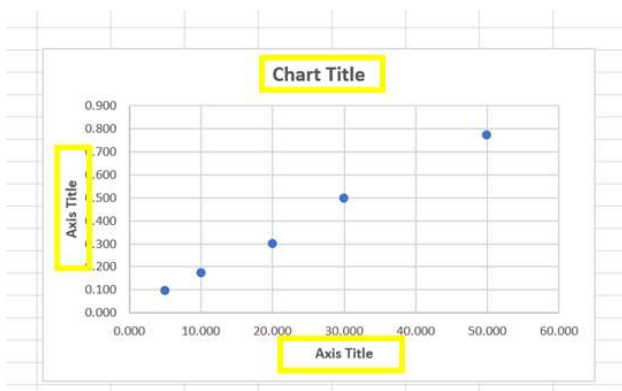
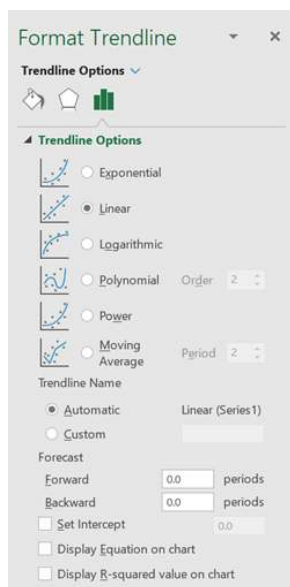


Figure 7

6. Your next step is to add a trendline to the plotted data points. A trendline represents the best possible linear fit to your data. To do this, "activate" the graph by clicking on any one of the data points. When you do this, all the data points will appear highlighted.



- Click the Chart Elements button next to the upper-right corner of the chart.
 - Check the 'Trendline' box.
 - Click 'More Options'. This will display the options shown in Figure 8. Notice that the 'Linear' button is already selected.
 - If you like to see the algebraic equation of the trendline, select the 'Display Equation' option. The equation (in the form $y = mx + b$) should appear on your graph.
 - If you like to know how well the data is fit by the equation, select the 'Display R-squared value' option. The closer the R^2 value is to 1, the better the fit.
 - Note that the program will always fit a trendline to the data no matter how good or awful the data is. You must judge the quality of the fit and the suitability of this type of fit to your data set.
7. Print a full-sized copy of your prepared graph and attach to your data recording sheet. Then, record the following information on the data recording sheet.
- The equation of the best-fit trendline to your data
 - The slope of the trendline
 - The y-intercept of the trendline
 - Whether the fit of the line to the data is good or bad, and why.

8. The graph contains a visual representation of the relationship (the graph) of the dependent and independent variables in your data and a mathematical expression of the relationship (the equation). It can now be used to make certain predictions.

For example, using the data above, you are asked to determine the phosphate concentration, in mg, in a different water sample. You used the same spectrometer and recorded an absorbance of 0.487. How can you find the concentration of phosphate in the water sample?

You can plug this value into the equation of the trendline and solve for the unknown concentration. Calculate and record your answer into your report. Note that this calculation will generally give a more precise value than inserting the data point (also called interpolating) and 'eyeballing' from the graph.

Part B. Two Data Sets with Overlay

Experimental Data: Two sets metal slugs of different sizes are measured. The masses and volumes are recorded, and the two sets of data collected are presented in the table below:

Metal A		Metal B	
Volume (cm ³)	Mass (g)	Volume (cm ³)	Mass (g)
1.00	11.87	1.20	7.64
2.50	21.66	4.00	8.71
4.25	32.27	5.00	15.72
5.00	41.69	6.00	19.93
7.00	54.47	7.00	15.75
8.00	59.46	8.50	23.88

You would like to see how these two sets of data relate to each other. To do this you will have to place both sets of data, as independent relationships, on the same graph. Note that this process only works when you have the same axis values and magnitudes.

1. Enter this new data on a new worksheet (Sheet 2) in Excel. Be sure to label your data columns A and B. Again, remember to enter the x values to the left of the y values.
2. First, plot data for Metal A only as an XY Scatter plot (the same way you did with the data in Part A). Fit a trendline to this data using linear regression and obtain the equation of this line.
3. Now you need to add data for Metal B to this graph.
 - Activate the graph by clicking on one of the data points, or anywhere on the chart.
 - Right-click the chart, and then choose 'Select Data' from the drop-down menu. The Select Data Source box appears on the worksheet with the source data of the chart.
 - Click the 'Add' tab and type "Metal B" for the Series Name.
 - Click on the box under Series X values, then highlight the x-axis values of Metal B. Press enter, then repeat this procedure for the Series Y Values, highlighting the y-axis values of Metal B. For each of these steps, you should see a display similar to what is shown in Figure 9. Note that slight differences may appear due to the version of Microsoft Excel installed on your computer.

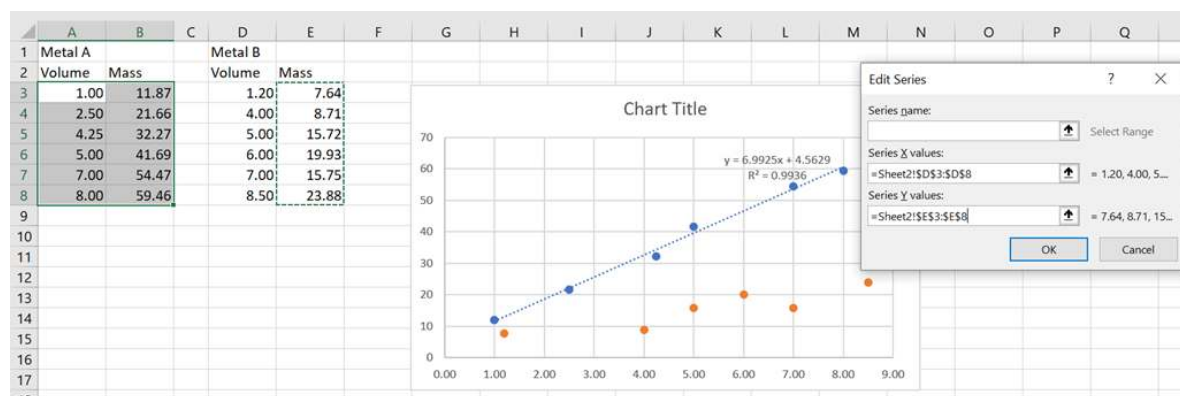


Figure 9

- Click 'OK' twice to return to the main Excel window.
- At this point, you should see the new data points as shown in Figure 10. You can now independently analyze the second dataset by inserting a trendline as before.

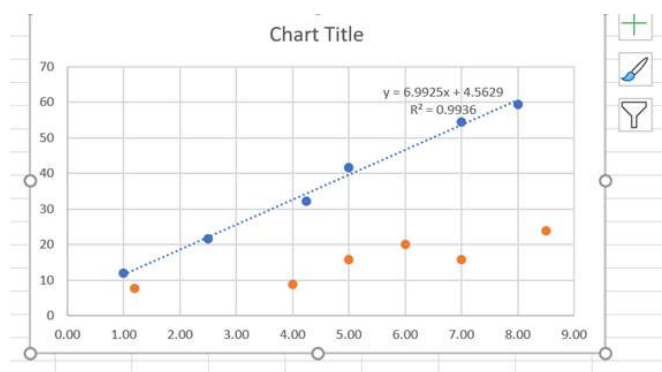


Figure 10

4. Print a full-sized copy of your prepared graph and attach to your data recording sheet. Then, record the following information on the data recording sheet.

- The equation of the best-fit trendline for Metal A
- The equation of the best-fit trendline for Metal B
- If these trendlines were extrapolated, they would intersect. Determine the values of x and y for the point of intersection using simultaneous equations.

Part C: Statistical Analysis

When many independent measurements are made for one variable, there is inevitably some scatter (noise) in the data. This is usually the result of random errors over which the experimenter has little control.

Experimental Data: Ten different students at two different colleges each measure the sulfate ion concentration in a sample of tap water:

College #1	35.9 ppm	43.2 ppm	33.5 ppm	35.1 ppm	32.8 ppm	37.6 ppm	31.9 ppm	36.6 ppm	35.0 ppm	32.0 ppm
College #2	45.1 ppm	34.2 ppm	36.8 ppm	31.0 ppm	40.7 ppm	29.6 ppm	35.4 ppm	32.5 ppm	43.5 ppm	38.8 ppm

Simple statistical analyses of these datasets might include calculations of the mean and median concentration, and the standard deviation. The mean () is simply the average value, defined as the sum () of each of the measurements () in a data set divided by the number of measurements ().

The median () is the midpoint value of a numerically ordered dataset, where half of the measurements are above the median and half are below. The median location of N measurements can be found using:



Standard deviation (s) is a measure of the variation in a dataset and is defined as the square root of the sum of squares divided by the number of measurements minus one.



While the mean, median and standard deviation can be calculated by hand using the above equations, it is often more convenient to use a calculator or computer to determine these values. Microsoft Excel[®] is particularly well suited for such statistical analyses, especially on large datasets.

1. Enter the data acquired by the students from College #1 (only) into a single column of cells (e.g. column A) on a new worksheet (Sheet 3) in Excel. Then, click on any empty cell (usually one close to the data cells) to enter the mean, median, or standard deviation for the data set.
2. Select the 'Formulas' tab on the top menu and click on 'Insert Function' button to perform the required functions on the data.
3. To compute the mean or average of the data entered in cells A1 through A10, on the box below 'Search for a function', type 'average' and then click 'Go'. Select 'average' under 'Select a function' and then click 'OK'.

Similarly, to compute the median, type 'median' and to obtain the standard deviation, type 'stdev'.

4. Record on your data recording sheet the mean, median, and standard deviation for the College #1 data set.

Rejecting Outliers

Do all the measurements in the College #1 data set look equally good to you, or are there any values that do not seem to fit with the others? If so, are you allowed to reject these measurements?

Outliers are data points which lie far outside the range defined by the rest of the measurements and may skew your results to a great extent. If you determine that an outlier resulted from an obvious experimental error (e.g., you incorrectly read an instrument or prepared a solution), you may reject the point without hesitation. If, however, none of these errors is evident, you must use caution in making your decision to keep or reject a point. One **rough criterion** for rejecting a data point is if it lies *beyond two standard deviations from the mean or average*.

5. Using the above criteria, determine if there are any outliers in the College #1 dataset.

- Record these outlier measurements (if any) on your data recording sheet.
- Then, excluding the outliers, re-calculate the mean, median and standard deviation of this data set.

6. Repeat steps 1 – 5 to analyze the data from College #2.

Important Note: Rejecting data points cannot be done just because you want your results to look better. If you choose to reject an outlier for any reason, you must always include documentation in your lab report which clearly states:

- that you did reject a point
- which point you rejected
- why you rejected it

Failure to disclose this could constitute scientific fraud.

3.0 DATA RECORDING SHEET

Attach the graphs you made for Parts A and B in this assignment. For each graph, make sure the following components are in the printout:

- Graph Title
- Labels for x and y axes (along with appropriate units when applicable)
- Best-fit line equation and R^2 when appropriate

Part A. Simple Linear Plot

1. Which set of data is plotted on the y-axis? _____
the x-axis? _____

2. From the graph:

Equation of the fitted trendline _____

Value of the slope of this line _____

Value of the y-intercept of this line _____

3. Is the fit of the trendline to your data good (circle one)? Yes / No

Explain why you think the line is a good fit to the data.

4. Determine the concentration of phosphate in the water sample with a recorded absorbance of 0.487 using

- Interpolation and "eyeballing" _____
- The equation of the trendline _____

Show your calculations for b) below.

Part B. Two Data Sets and Overlay

1. From the graph:

Equation of the trendline for Metal A data set _____

Equation of the trendline for Metal B data set _____

2. Perform a simultaneous equations calculation to determine the x and y values for the point of intersection between these lines. Show your work below.

Part C. Statistical Analysis

1. For the College #1 data set, record the following values:

the mean SO_4^{2-} concentration _____

the median SO_4^{2-} concentration _____

the standard deviation in the data set _____

2. Are there any outliers in the College #1 data set (circle one)? Yes / No

If yes, which measurements are the outliers? _____

Show the calculations you used to identify the outliers (or, if none, how you determined that there were none).

3. Re-calculate the following values (using Excel) excluding the outliers:

the mean SO_4^{2-} concentration _____

the median SO_4^{2-} concentration _____

the standard deviation in the data set _____

4. For the College #2 data set, record the following values:

the mean SO_4^{2-} concentration _____

the median SO_4^{2-} concentration _____

the standard deviation in the data set _____

5. Are there any outliers in the College #2 data set (circle one)? Yes / No

If yes, which measurements are the outliers? _____

Show the calculations you used to identify the outliers (or, if none, how you determined that there were none).

6. For College #2 data set, re-calculate the following values (using Excel) excluding the outliers:

the mean SO_4^{2-} concentration _____

the median SO_4^{2-} concentration _____

the standard deviation in the data set _____

7. Which data set give the more precise measurements? _____

How do you know?

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2502 Scientific Method

Scientific Method Laboratory

Last Name	First Name	Partner Name(s)	Date
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1.0 INTRODUCTION

While my children liked M&Ms, they always left the brown ones in the bowl, untouched. This happened whenever they had M&Ms to share. When I asked them why, they said the brown ones tasted different, tasted funny.

After a class discussion, enter the *hypothesis* that we developed here.

What predictions are we making about the results of the experiment we will do?

2.0 SAFETY PRECAUTIONS

Never eat or drink in the laboratory!
Some people are allergic to chocolate.

3.0 CHEMICALS AND SolutionS

4.0 GLASSWARE AND APPARATUS

5.0 PROCEDURE

1. _____
2. _____
3. _____
4. _____
5. _____

6.0 DATA RECORDING SHEET

Label the columns and rows, record your own data, and then collect the class data and add it to the table. (Add more columns and/or rows if you need them.)

7.0 CALCULATIONS AND ANALYSIS

What conclusion can you draw from the data collected?

Are there any biases in your work?

How could you improve the experiment?

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2503 Accuracy and Precision of Lab Glassware

Accuracy and Precision of Laboratory Glassware

(Adapted from *GCC CHM151LL Laboratory Manual*, *NSF-PACT Collection of Laboratory Activities*, & *TAMU Math Skills Review*)

1.0 INTRODUCTION

1.1 Objectives

After completing this experiment, the student will be able to:

- Use various volumetric glassware.
- Use the density formula.
- Determine which volumetric glassware contains or delivers the most accurate volume.
- Determine the percent error.

1.2 Background

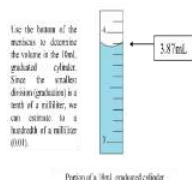
During the semester in the general chemistry lab, you will encounter various pieces of laboratory glassware.

Some of these pieces (e.g. beakers, Erlenmeyer Flasks) are used primarily to hold liquids during experiments. Upon closer inspection, you will also notice that they have graduations on the side to measure volumes. Both glassware mentioned above can measure volumes. Why do we have so many pieces of glassware if they all do the same basic job of measuring volumes?

Laboratory glassware can generally be divided into two main types based on how they measure volumes:

- those that are manufactured *to contain* certain volumes
- those that are manufactured *to deliver* certain volumes

The first thing to realize is that there is no such thing as a perfect measurement. Even when using expensive lab equipment there is some degree of uncertainty in measurement. The general rule of thumb is you can estimate one more digit past the smallest division on the measuring device. If you look at a 10 mL graduated cylinder, for example, the smallest graduation is tenth of a milliliter (0.1 mL). That means when you read the volume, you can estimate to the hundredths place (0.01 mL).



You must use the bottom of the meniscus to determine the volume in the 10 mL graduated cylinder. Since the smallest division (graduation) is a tenth of a milliliter, we can estimate to a hundredth of a milliliter (0.01).

Some glassware such as volumetric flasks and volumetric pipets only have a single line to indicate volume. They are made to measure just one specific volume. In the case of the glassware used in general chemistry lab, both the 10 mL volumetric pipet and 50 mL volumetric flask will have two significant figures (sig figs) after the decimal point (i.e. 10.00 mL and 50.00 mL), for a total of four significant figures each.

For the 150 mL beaker assume that 50. mL has two sig figs (it will not be obvious based on the volume markings).

1.3 Recording and Analyzing Lab Data

If a metal rod has a known length of 1.23 cm and you measure its length using three different measuring devices (A, B, and C), you obtain the following data.

Data Table: Measured Lengths of Device A, B, and C

	Device A (cm)	Device B (cm)	Device C (cm)
Trial 1	1.43	1.24	1.19
Trial 2	1.43	1.23	1.23
Trial 3	1.43	1.25	1.22
Trial 4	1.42	1.22	1.26

The first thing we want to do is to find an average from our multiple trials. As you may know, there are different types of averages. The types you may have heard about in your math classes are most likely *mean*, *median*, and *mode*. Many times, when we say “average” or “simple average” we are actually referring specifically to the *mean*. A *mean* is calculated as follows: add all of the values together and divide by the number of values. For example, for device A:

$$\frac{(1.43+1.43+1.43+1.42)}{4} = 1.\underline{4}275 \text{ or } \underline{1.43} \text{ cm (Based on sig. figs.)}$$

The “2” is underlined in the unrounded answer above because based on the addition and division rules, when the answer is rounded, it should have three significant figures. If I want to use the unrounded number for another calculation in the future, the underline will remind me of the significant figures that number actually has. The “4” that is in the denominator of the equation is an exact number and therefore has an infinite number of significant figures.

Using the same formula, we can then calculate a mean for the other measuring devices:

Data Table: Average Lengths of Device A, B, and C

	Device A (cm)	Device B (cm)	Device C (cm)
Average	1.43	1.24	1.23

How do we describe how “good” these measuring devices are based on our measurements and means? There are two ways we can describe these measurements – by their *accuracy* and *precision*.

Accuracy is a measure of how close your measured value is to the true value or other standard. Although sometimes we use *qualitative* (verbal) words to describe the accuracy (such as good accuracy and bad accuracy), a *quantitative* (numerical) measure is more useful when comparing measuring devices based on laboratory data. Our *quantitative* measure of accuracy is called percent error.

$$\text{Percent Error} = \left| \frac{(\text{Experimental Value} - \text{Theoretical Value})}{\text{Theoretical Value}} \right| \times 100$$

So, for each measuring device, using the averages calculated previously, we can calculate a percent error (remember the known length of the rod is 1.23 cm). If you use the values from Device A:

$$\text{Percent Error} = \left| \frac{1.4275 - 1.23}{1.23} \right| \times 100 = 16\%$$

The lower the percent error, the more accurate is the measuring device. In this case the mean for Device A had a 16% error from the true value of the length of the metal rod.

(Notice that when calculating percent error, we used some extra significant figures from our calculation of the mean. This is done to minimize rounding errors; errors that come from rounding and then using a rounded number in the next calculation. Remember it's always best

to round only when you need to report a number, as in a table. If you need to use a calculated number in another calculation, use the unrounded number.)

Data Table: Average Lengths of Device A, B, and C

	Device A (cm)	Device B (cm)	Device C (cm)
Percent Error	16%	0.5%	0.5%

The percent error for device A has two significant figures. Look at the formula for percent error. The first step is subtraction. Use the addition/subtraction rule for the numerator. For device A, take 1.4275 (unrounded average) and subtract 1.23 to calculate 0.1975 as a result. The numbers that are being subtracted each have two significant figures after the decimal point. Based on the addition/subtraction rule, the numerator can only have two sig figs after the decimal point: 0.1975. Once we divide the answer by 1.23 and multiply by 100 (an exact number), the result is a number with two sig figs, divided by a number with three sig figs. Based on the multiplication/division rule, the final answer can only have two sig figs.

The same logic yields only one significant figure for the percent error for devices A and B (do the calculations step by step to see for yourself).

Precision is a measure of how close repeated measurements are to each other.

Like accuracy, we can describe precision in qualitative terms (such as high precision and low precision) or quantitative terms. Precision is usually expressed in terms of the deviation of a set of results from the arithmetic mean of the set.

The *standard deviation* (S) is a measure of how precise the average is, that is, how well the individual numbers agree with each other.

$$\text{standard deviation, } S = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + (x_3 - \bar{x})^2 + \dots}{n - 1}}$$

The *relative standard deviation* (RSD) is often times more convenient. It is expressed in percent and is obtained by dividing the standard deviation by the mean, and multiplying by 100 to express as a percentage.

$$\text{RSD} = \frac{S}{\bar{x}} * 100$$

So, for each measuring device, using the averages calculated previously, we can calculate a standard deviation and relative standard deviation (remember the known length of the rod is 1.23 cm). If you use the values from Device A:

$$\text{standard deviation} = \sqrt{\frac{(1.43-1.4275)^2 + (1.43-1.4275)^2 + (1.43-1.4275)^2 + (1.42-1.4275)^2}{(4-1)}} = 0.005$$

$$\text{RSD} = \frac{100 \times 0.005}{1.4275} = 0.4\%$$

Our final result for Device A can be written as 1.43 ± 0.005 or $1.43 \pm 0.4\%$

The lower the standard deviation and RSD are, the more precise the measuring device is.

Data Table: Average Lengths of Device A, B, and C

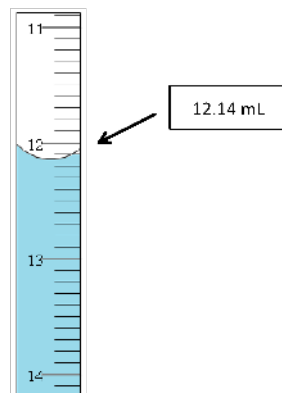
	Device A (cm)	Device B (cm)	Device C (cm)
RSD	0.4%	1%	2%

So, which measuring device is “best” to use? We want a device that is both accurate and precise. Based on our experimental data, using the percent error we calculated, we can see that Devices B and C have better accuracy than A. Therefore, we will need to look at the precision of these devices in order to make our final selection. Looking at the measurements, we can see that Device B has a higher degree of precision (lower standard deviation) than C. Therefore, Device B has the best combination of accuracy and precision.

Errors in general chemistry are classified as *systematic* (determinate) and *random* (indeterminate).

A *systematic error* is caused by a defect in the analytical method or by an improperly functioning instrument or analyst. A procedure that suffers from a systematic error is always going to give a mean value that is different from the true value. The measured value is described as being biased high or low when a systematic error is present. Systematic errors can be avoided. Sources of systematic errors include spectral interferences, chemical standards, glassware, and analytical balances where an improper calibration or use will result

in a systematic error. For example, a dirty glass pipet will always deliver less than the intended



volume of liquid.

Random errors are unavoidable. They are unavoidable due to the fact that every physical measurement has limitation, i.e., some uncertainty. Using utmost care, the chemist can only obtain a weight to the uncertainty of the balance or deliver a volume to the uncertainty of the glassware. For example, most four-place analytical balances are accurate to ± 0.0001 grams. Therefore, with care, a chemist can measure a 1.0000 gram weight (true value) to an accuracy of ± 0.0001 grams where a value of 0.9999 to 1.0001 grams would be within the random error of measurement.

A truly random error is just as likely to be positive as negative, making the average of several measurements more reliable than any single measurement.

1.4 Using Burets and Pipets

Using a buret

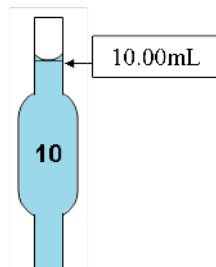
- Filling the buret: Close the stopcock. Use the beaker of water and a funnel to fill the buret about 1 mL above the “0” mark. Place a container under the buret tip and open the stopcock slowly. The buret tip should fill with solution, leaving no air bubbles. If the tip does not fill with solution, ask the instructor for help.

Continue to let out solution until the liquid level is at “0” or below.

- Reading the buret: Record the volume by noting the bottom of the meniscus. (Be sure that the meniscus is at eye level). If this reading is exactly “0,” record 0.00 mL. Otherwise, count the number of markings between each number, and estimate to the nearest 0.01 mL.

Using a pipet

- Identify and move a receiving container close to the pipet to minimize time and distance the pipet needs to travel.
- Do not pipet by mouth.
- Different pipet bulb styles are available. Some work differently. These instructions are for pipet bulbs that provide suction and do not form full seals on the top of the pipet.



- Place the pipet bulb loosely on the top of the pipet. Place it just far enough on the pipet to get an air-tight seal. Position the tip of the pipet below the liquid level in the beaker. Apply suction and fill the pipet well above the calibration line (etched or marked above the wide center section of the pipet). Take care not to get liquid into the pipet pump.
- Slide the bulb off the pipet while quickly sliding your index finger or thumb over the top of the pipet. Move your finger slightly and rotate the pipet to allow the liquid level to drop to the calibration line on the pipet. Stop the flow, and move the tip of the pipet into a position over the receiving container.
- Remove your finger and allow most of the liquid to drain out. Then hold the tip of the pipet against the inside of the container for about 10 seconds to allow more liquid to drain.
- Do not remove the small amount of liquid remaining in the tip. Pipets are calibrated to retain this amount.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

Goggles are not required for this experiment.

Be careful not to spill water on the balances.

Water can go down the drain.

If glass should break, do not pick it up with bare hands. Use a dust pan and broom and wear leather gloves.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate Amount	Notes
Laboratory water	Pure	100 mL	Be sure to use laboratory water

4.0 GLASSWARE AND APPARATUS

250 mL beaker	50 mL beaker
150 mL beaker	10 mL graduated cylinder
100 mL graduate cylinder	50 mL buret
50 mL volumetric flask	10 mL graduated pipet
Thermometer	Balance

5.0 PROCEDURE

5.1 GLASSWARE MEANT TO CONTAIN SPECIFIC VOLUMES

- Obtain approximately 100 mL of laboratory water in a 250 mL beaker. Place the beaker on a folded piece of paper towel on the bench. Place a thermometer in the water. Record the temperature after there is no change in temperature for at least ten minutes. This water will be used for all of the experiments. Look-up and record the following information using the CRC Handbook of Chemistry and Physics located in your laboratory or the excerpt available on Canvas.
 - The density of water at the exact temperature measured.
 - The density of water ten degrees higher than the temperature measured.
 - The temperature at which the density of water is the highest.

When using the CRC Handbook of Chemistry and Physics notice that the units are $\text{kg}\cdot\text{m}^{-3}$, which is the same as g/m^3 . You will need to convert to g/mL . Remember $1\text{mL} = 1\text{cm}^3$.

When using the excerpt available on Canvas, whole degrees are listed down the left hand side of the table, while tenths of a degree are listed across the top. First, find the whole degree by searching down the left hand column until you reach your desired whole number. Then slide across that row until you reach the column corresponding to your tenth of a degree.

- Obtain one of the pieces of glassware listed below. Use a top-loading balance as directed by the instructor (if using a balance with doors, make sure all of the doors of the balance are closed, as air currents in the lab can affect the readings). Ensure that the glassware is dry and determine its empty mass. Record its mass in the Data Recording Sheet. Remove the glassware from the balance.
 - 150 mL beaker
 - 25 mL graduated cylinder

- 50 mL volumetric flask
 - 50 mL beaker
 - 10 mL graduated cylinder
3. From the 250 mL beaker, transfer a little less than the volume of water listed for the glassware below, one piece of glassware at a time. Use a plastic disposable pipet to add the water drop-wise until the bottom of the meniscus is on the line that corresponds to the desired volume. If too much water was added, remove the extra water using the plastic disposable pipet. Wipe down the outside and bottom of the glassware in case any water was spilled during the transfer.
 - 150 mL beaker – 50 mL laboratory water
 - 25 mL graduated cylinder – 12 mL laboratory water
 - 50 mL volumetric flask – 50 mL laboratory water
 - 50 mL beaker – 10 mL laboratory water
 - 10 mL graduated cylinder – 10 mL laboratory water
 4. Using the same balance as before, record the mass of the glassware with the added water. Record its mass in the Data Recording Sheet.
 5. Pour the water from the glassware back into the 250 mL beaker and repeat steps 3-4 two more times with the same piece of glassware. You should have a total of three trials for each piece of glassware. You will use the empty mass from step 2 for all three trials.
 6. You will then repeat the process (three trials each) with each piece of glassware listed.

5.2 GLASSWARE MEANT TO DELIVER SPECIFIC VOLUMES

The proper use of a pipet is a skill that takes time and practice. Try at least three practice runs of filling the pipet and delivering the water back to the 250 mL beaker.

1. For the glassware listed below, the glassware will be used to transfer water to a beaker for weighing.
 - 50 mL buret
 - 10 mL volumetric or graduated pipet
2. Obtain a dry 50 mL beaker and determine its mass.
3. For each type of glassware, deliver 10.00 mL to the 50 mL beaker. For the buret, record initial and final buret readings in a table along with the amount of water delivered to the beaker.
4. Using the same balance as before, record the mass of the beaker with the added water. Record its mass.
5. Repeat the process so that you have three trials with both the buret and the pipet.

6.0 DATA RECORDING SHEET

Last Name	First Name		Partner Name(s)	Date

Table 1. Volumes and masses for 150 mL beaker

Mass of empty beaker: _____ Temperature of water: _____

Volume of water: _____ Theoretical density of water: _____

Trial	Mass of beaker with water (g)	Mass of water (g)	Calculated Density (g/mL)
1			
2			
3			

Table 2. Volumes and masses for 25 mL graduated cylinder

Mass of empty cylinder: _____ Temperature of water: _____

Volume of water: _____ Theoretical density of water: _____

Trial	Mass of cylinder with water (g)	Mass of water (g)	Calculated Density (g/mL)
1			
2			
3			

Table 3. Volumes and masses for 50 mL volumetric flask

Mass of empty flask: _____ Temperature of water: _____

Volume of water: _____ Theoretical density of water: _____

Trial	Mass of flask with water (g)	Mass of water (g)	Calculated Density (g/mL)
1			
2			
3			

Table 4. Volumes and masses for 50 mL beaker

Mass of empty beaker: _____ Temperature of water: _____

Volume of water: _____ Theoretical density of water: _____

Trial	Mass of beaker with water (g)	Mass of water (g)	Calculated Density (g/mL)
1			
2			
3			

Table 5. Volumes and masses for 10 mL graduated cylinder

Mass of empty cylinder: _____ Temperature of water: _____

Volume of water: _____ Theoretical density of water: _____

Trial	Mass of cylinder with water (g)	Mass of water (g)	Calculated Density (g/mL)
1			
2			
3			

Table 6. Volumes and masses for 50 mL buret

Mass of empty beaker: _____ Temperature of water: _____

Volume of water: _____ Theoretical density of water: _____

Trial	Initial buret reading (mL)	Final buret reading (mL)	Volume of water (mL)	Mass of beaker with water (g)	Mass of water (g)	Calculated Density (g/mL)
1						
2						
3						

Table 7. Volumes and masses for 10 mL pipet

Mass of empty beaker: _____ Temperature of water: _____

Volume of water: _____ Theoretical density of water: _____

Trial	Mass of beaker with water (g)	Mass of water (g)	Calculated Density (g/mL)
1			
2			
3			

7.0 CALCULATIONS

For each calculation show your work for at least one piece of glassware.

1. Calculate the mass of water for each trial by subtracting the mass of the empty glassware from the mass of the glassware with water. Remember to record the value in your data table with the appropriate number of significant figures and to retain all of the digits for further calculations.

2. Using the mass of water and volume of water for each trial, calculate the density of water in units of grams per milliliter (g/mL) for each trial using the formula for density (density = mass/volume). Remember to record the value in your data table with the appropriate number of significant figures and to retain all of the digits for further calculations.
3. Calculate the average density of water using the measured values for each piece of glassware. Record the value in the summary table with the appropriate number of significant figures and to retain all of the digits for further calculations.
4. Calculate the percent error for each piece of glassware using the average and theoretical values for the density of water with equation 1. Record the value in the summary table.
5. Calculate the standard deviation for each piece of glassware using the calculated densities for each trial, average density, and equation 2.
6. Calculate the relative standard deviation for each piece of glassware using the calculated densities for each trial, average density, and equation 3. Record the value in the summary table.

Table 8. Summary of glassware accuracy and precision

Glassware	Average density (g/mL)	Theoretical density (g/mL)	Percent Error	RSD
150 mL beaker				
25 mL grad. cylinder				
50 mL volumetric flask				
50 mL beaker				
10 mL grad. cylinder				
50 mL buret				
10 mL pipet				

8.0 POST-LAB QUESTIONS

1. List the glassware in order of the smallest percent error (best) to the largest percent error (worst).
2. Which piece of laboratory glassware is the most accurate? least accurate? Explain.
3. List the glassware in order of the smallest relative standard deviation (best) to the largest relative standard deviation (worst).
4. Which piece of laboratory glassware is the most precise? least precise? Explain.
5. What effects will a difference in temperature cause?

6. For each of the following situations, determine which type of glassware would be most appropriate. (150 mL beaker, 10 mL pipet, 50 mL buret, 100 mL volumetric flask, 100 mL graduated cylinder) Each will be used only once.

Note: When working with laboratory glassware, scientists choose the glassware that is appropriate while also efficient for the experiment. For example, if an experiment calls for using approximate volumes, it would be a waste of time to set up a buret.

1. A lab calls for adding enough water to dissolve 5 g of NaCl to make a total of 100.00 mL of solution.
2. A lab calls for adding very small amounts of liquid A to solution B until a color change is detected. The amount of liquid A added must be recorded.
3. A lab calls for adding approximately 50 mL of water to a solution.
4. A lab calls for adding 50.0 mL of water to a solution.
5. A lab calls for delivering 10.00 mL of solution Z to an Erlenmeyer flask.
7. For each of the following situations during the density of water experiment, determine if the mistake would give a false high, false low, or unchanged density for water. Also indicate if the error is systematic or random error. For each situation, assume everything else in the experiment is done correctly.

Note: It may be helpful to try some hypothetical calculations.

1. When weighing a volumetric flask full of water, a student does not notice that the outside of the flask is wet.

€ HIGH

€ LOW

€ UNCHANGED

◇ SYSTEMATIC

◇ RANDOM

Explanation:

2. Before placing an empty 10 mL graduated cylinder on the balance to get its mass, a student does not notice that the balance reads “-1.4444g”. When the student weighs the full 10 mL graduated cylinder later in the experiment, she correctly sets the balance to “0.0000g.”

€ HIGH

€ LOW

€ UNCHANGED

◇ SYSTEMATIC

◇ RANDOM

Explanation:

3. A student accidentally used the hot water faucet to deliver water to a piece of glassware. (Use the densities you recorded from the CRC Handbook during the lab in your explanation to received complete credit).

€ HIGH

€ LOW

€ UNCHANGED

◇ SYSTEMATIC

◇ RANDOM

Explanation:

8. In lab, your group used something called “laboratory” water to find the density of water. Explain what laboratory water is and why your group did not use tap water instead.

9. Would you expect the density of tap water to be higher, lower or the same as the density of laboratory water? Explain your answer.

10. Why is it important to know the accuracy and precision of glassware?

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2504 Mass and Density Measurements

Mass and Density Measurements

1.0 INTRODUCTION

You may be familiar with this riddle. “Which is heavier, a pound of feathers or a pound of lead?” The answer is, they are the same, they are both a pound. Heavy refers to the mass of an object. People are tempted to say “the lead” because there is a physical property that is very different between lead and feathers. The pound of lead would be a very small piece of lead and the pound of feathers would fill several pillows.

Different samples of any substance will have different masses and take up different volumes, but each sample will have the mass to volume ratio, known as density.

Density = mass/volume or $D = m/V$

The mass of an object can be determined by using a balance. In geometry classes you learned numerous formulas to calculate the volume of a regularly shaped object. For example, a cylinder is

$$V = \pi r^2 h$$

where $\pi = 3.14$, r = radius of the circle, and h = height of the cylinder.

In some cases, the volume of an object is difficult to determine because it has a very irregular shape. By Archimedes principle, you can measure the volume of that object by measuring the volume of water that it displaces.

In this experiment you will be examining the mass, the dimensions, and the volume of several objects and reviewing the relationships among these different properties.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

Let's develop good habits of safety by wearing goggles today and every day in lab.

Be careful not to break the glass graduated cylinders when inserting metal objects.

When you are finished with the experiment, return metal objects to the cart or stockroom.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate amount	Notes
Water	N/A	To partly fill graduated cylinders	Clean up any spillage with paper towels

4.0 GLASSWARE AND APPARATUS

Item	Use	Notes
6 aluminum cylinders (also called “slugs”) of various sizes	For analysis	Return to cart or stockroom (do not dispose!)
6 other metal cylinders (also called “slugs”) of various sizes – brass, copper, or iron	For analysis	Return to cart or stockroom (do not dispose!)
ruler	For measuring	Return to cart or stockroom (do not dispose!)
calipers	For measuring	
A graduated cylinder large enough to hold the largest metal “slug”. Plastic graduated cylinders are preferred.	For measuring	If you think the “slug” might get stuck in the graduated cylinder, use a larger graduated cylinder instead!
Balance	Measure mass	In room at back of laboratory

5.0 INVESTIGATION

Careful records of your observations are key to answering questions posed by your investigation. A bound laboratory notebook is one way to secure and maintain your records.

- Investigation: Measurement of various physical properties
- Data Collection

Your team will analyze 6 pieces of aluminum and 6 pieces of brass. It is best if the metal pieces are different sizes.

Each piece of metal is called a “slug”.

It is important to collect your data in an organized manner such that the measurement for one “slug” is not entered into the data table for a different “slug”. A data table is provided later in this document (you may use this one or draw your own data table in your notebook).

For each “slug”:

- Record any markings on the “slug” and note whether it is aluminum or brass.
- Measure the mass using an analytical balance. Record all digits for greater accuracy (the more significant figures, the better!)
- Measure the length using a ruler. Obtain as many significant figures as possible by estimating the distance between the 2 smallest marks of the ruler. For example, if your ruler measures millimeters and the length of the “slug” is exactly midway between 33

- mm and 34 mm, then record the length as 33.5 mm (If the length is closer to 34 mm, then you might state the length as 33.7 or 33.8 mm. Make your best estimate!)
4. Measure the diameter using a ruler or calipers. Again, obtain as many significant figures as possible by estimating the distance between the 2 smallest marks of the ruler.
 5. Measure the volume of the “slug” by using the water displacement method with a graduated cylinder. This method also works well for irregularly shaped objects (see the image shown below).
 1. Obtain a graduated cylinder large enough such that the slug will not get stuck.
 2. Partly fill the graduated cylinder with water. Gently tap the graduated cylinder to displace any air bubbles in the water.
 3. Measure and record the initial volume. Again, obtain as many significant figures as possible by estimating the distance between the 2 smallest marks of the graduated cylinder.
 4. Gently slide the “slug” into the graduated cylinder. Don’t let it fall quickly as it may crack the glass graduated cylinder! Gently tap the graduated cylinder to displace any air bubbles in the water.
 5. Measure and record the final volume. Again, obtain as many significant figures as possible by estimating the distance between the 2 smallest marks.
 6. The volume of the “slug” is the difference between final and initial volumes. Record this volume.



6.0 DATA RECORDING SHEET

Last Name	First Name	Partner Name(s)	Date
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6.1 BRASS DATA TABLE

(Add any special markings that appear on the sample) →	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6
Mass (g)						
Length (mm)						
Diameter (mm)						
Calculation of volume by cylinder formula(cm^3)						
Initial volume of water(mL)						
Final Volume of water(mL)						
Volume of water displaced (mL)						
Density using cylinder volume (g/cm^3)						
Density using displaced water volume (g/mL)						

6.2 ALUMINUM DATA TABLE

(Add any special markings that appear on the sample) →	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6
Mass (g)						
Length (mm)						
Diameter (mm)						
Calculation of volume by cylinder formula(cm^3)						
Initial volume of water(mL)						
Final Volume of water(mL)						
Volume of water displaced (mL)						
Density using cylinder volume (g/cm^3)						
Density using displaced water volume (g/mL)						

7.0 ANALYSIS

1. For the 6 aluminum “slugs”, construct a plot of volume determined by the water displacement on the x-axis and mass on the y-axis. Use a graphing program that calculates linear regression or the line of best fit (refer to Expt 2501).
2. On the same graph, plot the volume on the x-axis and mass on the y-axis for the 6 brass “slugs”. Also obtain the line of best fit. These 2 lines (one for aluminum and another for brass) should have different slopes.

3. Properly label your graph by including a title ("Mass vs. Volume of Metal Slugs"), y-axis label ("Mass in grams"), and x-axis label ("Volume in milliliters"). Also display a legend that clearly identifies which data points are for aluminum and which data points are for brass. Be sure to display both equations for the lines of best fit. Print a copy to include in your lab report.
4. Repeat the analysis for a different combination of properties such as 1) volume on the x-axis and length on the y-axis, 2) length on the x-axis and mass on the y-axis, or some other combination of properties of your choosing. Plot this second graph using data for both aluminum and brass slugs, displaying the equations for the lines of best fit for each type of metal. Be sure to properly label your graph as before.

Include in your lab report both of your graphs.

8.0 POST-LAB QUESTIONS

1. Comment on the quality of your data by inspecting how close the data points came to touching the line of best fit. Are there data points that you trust more than others?

2. From the plots of mass versus volume, obtain the slopes of the lines of best fit for aluminum and for brass. Because density is defined as mass divided by volume, the slopes of your lines are your experimentally determined values for the densities of aluminum and brass! Look up the densities for these metals on the internet and discuss how they compare to your slope values. (Because brass is an alloy of copper and zinc, the internet should mention that the density of brass can vary depending on the ratio of copper to zinc.)

3. For your second graph in which you choose to properties to plot, obtain the slopes of the lines of best fit and discuss whether the slopes have any meaning. Also compare the slopes between aluminum and brass: are they similar or different? Hypothesize about the reasons for any similarities or differences.

4. If you had to repeat this lab, would you have chosen different properties to graph? Why or why not?

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2505 Hydrates

HYDRATES

1.0 INTRODUCTION

Hydrates are ionic compounds that contain water molecules as part of their crystal structure. The bound water is called the water of hydration or water of crystallization. Some examples include minerals such as gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), borax ($\text{Na}_3\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$) and Epsom salts ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$).

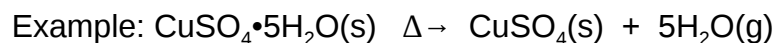
A hydrate contains a definite number of water molecules bound to each ionic compound (also called the anhydrous salt). The formula of the hydrate is represented by the formula of the anhydrous salt followed by a dot and $x\text{H}_2\text{O}$, where x is the number of moles of water per mole of the anhydrous salt. The hydrate is named by naming the anhydrous salt followed by the word 'hydrate' preceded by a Greek prefix to indicate the number of water molecules. For example, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ is named calcium sulfate dihydrate.

A newer naming convention is to explicitly state the component formula and name followed by '-water (x/y)' where x is the stoichiometric number of the parent compound and y is the stoichiometric number of the water molecules bound within the compound. Thus, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ would be named 'calcium sulfate-water (1/2)'. A second example would be the compound $\text{Co}_2\text{O}_3 \cdot n\text{H}_2\text{O}$ (the ' n ' indicates the number of water present is not fixed) which would be named 'cobalt(III) oxide-water (1/ n)'.

Although both naming conventions are valid, the older naming convention is more common and will be used here. The newer naming convention should be used if an unequivocal naming of the compound is needed.

1.1 Properties of Hydrates

In general, the water of hydration can be removed by heating a hydrate sample. After heating, the anhydrous salt remains as residue and the water escapes into the atmosphere as water vapor.



blue crystals *white ash*

The anhydrous salt will have a different structure and physical properties such as texture and color compared to the hydrate.

Most hydrates are stable at room temperature. However, some hydrates lose the water of hydration spontaneously and are said to be 'efflorescent'. On the other hand, other compounds spontaneously absorb water from the atmosphere and are said to be 'hygroscopic'. These hygroscopic substances are typically used to 'dry' other materials; they are referred to as

desiccants. An anhydrous salt (without water) can absorb so much water from the atmosphere that it will spontaneously dissolve in its own water of hydration and is said to be 'deliquescent'.

1.2 Formula of a Hydrate

As the formula indicates, water forms a definite percentage of the mass of any hydrate. For example, the mass of a mole of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ is 172 grams (this is the molar mass of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$). A mole of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ contains 2 moles of water (which corresponds to 36 grams of water) as part of its structure. Therefore, the compound $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ always consists of 36/172 or 21% water by mass.

All hydrated compounds may be dehydrated by heating. A solid sample will show a decrease in mass as water escapes into the atmosphere. As part of this experiment, the amount of water lost by an unknown hydrated compound on heating will be determined.

mass of water lost = mass of hydrate - mass of anhydrous salt

The ratio of the mass of water lost to the mass of the hydrate will indicate the percentage of the water in the hydrate.

% water of hydration = $\frac{\text{mass of water lost}}{\text{mass of hydrate}} \times 100$

References and further reading

Nomenclature of Inorganic Chemistry, IUPAC Recommendations 2005.
https://old.iupac.org/publications/b..._Book_2005.pdf (accessed 07/06/2020)

Technique F Use of Crucible

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

Use caution when heating the crucible and cover. A hot crucible looks like a cold one; avoid direct contact with the crucible, clay triangle and ring stand until you are sure they are cooled. A hot crucible will break if placed directly on a cold surface. Set hot crucibles on to wire screens to cool.

When you are finished with the experiment, discard solid waste in the container marked 'solid waste' in the fume hood.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate amount	Notes
$\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$	N/A	a few crystals	Do not leave the covers off these containers. Some of these compounds may react with water in the atmosphere (which you will observe).
Na_2SO_4	N/A	a few crystals	
CaCl_2	N/A	a few crystals	
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	N/A	a few crystals	
$\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$	N/A	a few crystals	
$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	N/A	a few crystals	
FeCl_3	N/A	a few crystals	
"Unknown" hydrates: $\text{CaCl}_2 \cdot x\text{H}_2\text{O}$ $\text{CaSO}_4 \cdot x\text{H}_2\text{O}$ $\text{AlK}(\text{SO}_4)_2 \cdot x\text{H}_2\text{O}$ $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$	N/A	2 – 4 grams	Determine the identity of the 'Unknown' by determining the value of 'x' and which of the four compounds you were issued.

4.0 GLASSWARE AND APPARATUS

Item	Use	Notes
Watch glass	To hold solids for observation	Several separate watch glasses are needed for Part A
Crucible with lid	To heat substances at high temperatures	Do not touch hot crucible – use tongs.
Wire triangle	To support a crucible while being heated	Do not touch hot wire triangle – use tongs
Crucible tong	To hold and transport crucible	
Wire gauze	To prevent hot crucibles from directly contacting cold surfaces	This reduces the risk of thermal shock and subsequent breakage of crucibles
Bunsen burner	To heat substances	
Mortar and pestle	To grind solid material	
Large test tube	To heat solid sample	
Test tube holder	To hold test tube while heating	

5.0 PROCEDURE

Part A. Hygroscopic and Efflorescent Solids

1. On an analytical balance, weigh a pea-sized sample of each of the compounds below (see 6.0 Data Recording Sheet) on separate, clean and dry watch glasses. Record the initial masses of watch glasses and samples.
2. Label and set them aside out of the way in one area on your work bench for observations. After one hour (or at the end of the lab period) note any changes in the physical appearance of each sample.
3. Weigh the samples and record the masses as final masses.
4. Calculate the change in mass for each sample. A substance is classified as efflorescent if its mass decreases by 0.005 g or more; and it is classified as hygroscopic if its mass increases by 0.005 g or more.

Part B. Reversibility of Hydration

1. Grind a few crystals of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ in a mortar. Gently heat the powder in a Pyrex test tube fastened in a horizontal position by a clamp on a ring stand. Observe whether any water can be seen (as a mist or droplets) condensing inside the test tube. Note the

appearance of the solid residue after heating. When the residue seems to be completely dehydrated, allow it to cool. Then add a few drops of water to the residue in the tube.

What does the water seem to have done to the solid? Record your observations.

2. Repeat with crystals of cobalt(II) chloride hexahydrate ($\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$). Do the same for iron(II) sulfate heptahydrate ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$). Record your observations.

Note: Because of the color changes associated with hydration and dehydration of anhydrous salts, some of these compounds can be used to detect small quantities of water.

Part C. Percent Water of Hydration

1. Place a clean, dry porcelain crucible and lid on a wire triangle, supported on a ring attached to a ring stand, with the lid slightly offset to allow an opening. Heat them strongly (high heat) for about 3 minutes to be sure they are perfectly dry. Allow the crucible and lid to stand in place on the triangle long enough to cool again, and then weigh them separately using an analytical balance. (Sometimes lids slip and break, so it is best to have a separate mass for each.) Record the exact mass of the crucible and lid.
2. Obtain a sample of one of the hydrated salts. Transfer a sample of the unknown hydrate weighing 2 to 4 grams into the crucible and weigh the crucible with salt and lid again as accurately as possible. Record the mass on your data recording sheet. Obtain the exact mass of the sample by subtraction of the mass of the empty crucible and lid.
3. Put the crucible, lid, and sample back on the wire triangle on the ring stand. Position the crucible such that it is at a slight angle on the triangle. With the lid slightly offset (allow an opening for the water to escape), heat the crucible gently for 5 minutes (slowly move the Bunsen burner back and forth across the bottom of the crucible). Then, gradually increase the heat. Continue heating for another 10-15 minutes. Do not let the bottom of the crucible turn red hot at any time.
4. Remove heat and allow the crucible, lid, and contents cool completely. Reweigh and record exact mass. (*Repeat this process, more than once if necessary, until the mass is within 0.01 g of the first mass in part 4.*) Obtain the exact mass of the heated sample by subtraction of the mass of the empty crucible and lid.
5. If time permits, perform a second determination using a different mass of hydrate sample.

6.0 DATA RECORDING SHEET

Last Name	First Name		Partner Name(s)	Date

1. Hygroscopic and Efflorescent Solids

	Compound:			
Measurement:	CaCl ₂	FeCl ₃	Na ₂ SO ₄ •10H ₂ O	Na ₂ SO ₄
Initial mass of watch glass				
Initial mass of watch glass and sample				
Final mass of watch glass and sample				
Initial mass of sample				
Final mass of sample				
Change in mass				
Observations				

2. Reversibility of Hydration

Compound	Observations		
	Before heating	After heating	After addition of laboratory water
CuSO ₄ •5H ₂ O			
CoCl ₂ •6H ₂ O			
FeSO ₄ •7H ₂ O			

3. Percent Water of Hydration

Identification code for "Unknown": _____

After cleaning and heating empty crucible

	(1)	(2), if time permits
Mass of crucible		
Mass of lid		
Mass of crucible, lid and sample		
Mass of sample (hydrate)		

After heating crucible with unknown

	(1)	(2), if time permits
Mass of crucible, lid and sample after 1 st heating		
Mass of crucible, lid and sample after 2 nd heating		
Mass of crucible, lid and sample after 3 rd heating, if needed		
Mass of heated sample (anhydrous salt)		
Mass of water lost (water of hydration)		
Percent (%) water of hydration		

Based on my results the formula for the unknown is: _____

7.0 CALCULATIONS AND DATA ANALYSIS

1. Calculate the percent water of hydration of your unknown hydrate sample from each determination and take the average, if you did two.

2. Obtain the identity of the unknown from your instructor. Calculate the percent water of hydration from the chemical formula. How does this compare with your experimental data?

8.0 POST-LAB QUESTIONS AND CONCLUSIONS

1. Write the chemical equations to illustrate the dehydration of hydrate samples used in Part B of this experiment.
2. Did the compound(s) that appeared wet in Part A lose or gain water? Explain what may have happened.
3. Determine the effect (too high, too low, or no effect) of each of the following procedural steps on the calculated percent of water in the hydrate sample. Explain your answer briefly.
 1. The hydrate sample was not heated sufficiently (for example, only 5 minutes of total heating time) to drive off all the water.
 2. The crucible lid was not slightly ajar while heating.
 3. The crucible was not cooled completely or was still warm when weighed.
 4. The hydrate was overheated so that may have led to the decomposition of the compound.

2506 Quantitative Reactions

QUANTITATIVE REACTIONS

1.0 INTRODUCTION

A limiting reagent (sometimes called limiting reactant) is the reagent that is completely consumed during a chemical reaction. Once this reagent is consumed the reaction stops. An excess reagent (sometimes called excess reactant) is the reactant that is left over once the limiting reagent is consumed. The maximum yield of a chemical reaction is dependent upon the limiting reagent thus the one that produces the least amount of product is the limiting reagent.

In class, stoichiometric calculations are used to identify the limiting and excess reagents. In lab, we can identify the limiting and excess reagents experimentally. In this experiment you will be create a mixture of two ionic solutions, $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 . When the solutions are mixed, they will react to form an insoluble salt, which can be separated from the liquid.

Precipitate is the term used for an insoluble solid which is produced by mixing two solutions. The liquid above the precipitate is called the supernatant. When the mixture is poured into a filter, the solid precipitate is trapped by the filter and the liquid supernatant passes through as the filtrate. You will then test the filtrate to find out the limiting reagent.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

When you are finished with the experiment, place any solid that contains cobalt in the proper container for solid waste.

Pour any solution that contains cobalt in the proper container for liquid waste.

Solutions without cobalt may be safely disposed of down the drain.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate amount	Notes
$\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	N/A	About the size of a pea (0.25 g)	
$\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}$	N/A	About the size of a pea (0.25 g)	
K_3PO_4	N/A	About the size of a pea or less	
$\text{CoCl}_2 \cdot 2\text{H}_2\text{O}$	N/A	About the size of a pea or less	
NaNO_3	N/A	About the size of a pea or less	
$\text{Co}(\text{NO}_3)_2$ solution	0.0183g/mL	90 mL	Record the exact concentration as printed on the bottle
Na_3PO_4 solution	0.0156g/mL	60 mL	Record the exact concentration as printed on the bottle

4.0 GLASSWARE AND APPARATUS

Item	Use	Notes
Several beakers of any size	Preparing and mixing solutions	Rinse well before reusing a beaker (they do not have to be dry).
4 funnels	For gravity filtration	
5 sheets(circles) of filter paper	For gravity filtration	
Buret or volumetric pipets of 5.0, 10.0, 15.0, 20.0 mL and bulb	For careful quantitative measurements of aqueous solutions	Either set of devices work, check stockroom for availability

5.0 PROCEDURE

Careful records of your observations are key to answering questions posed by your investigation. A bound laboratory notebook is one way to secure and maintain your records.

1. Investigation: Conduct and analyze the chemical reaction between $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 (Part 1).
1. First, let us explore the reaction without doing any careful measurements. In a beaker, dissolve about a pea-sized amount (about 0.25 g) of solid $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in approximately 20 mL of laboratory water. Label this beaker " $\text{Co}(\text{NO}_3)_2$ solution". Record the appearance of the solution (Clear? Cloudy? Colored?) in Data Table 1 in Section 6.
2. In a second beaker, dissolve about a pea-sized amount (about 0.25 g) of solid $\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}$ in approximately 20 mL of laboratory water. Label this beaker " Na_3PO_4 solution". Record the appearance of the solution (Clear? Cloudy? Colored?) in Data Table 1 in Section 6.
3. Start the chemical reaction by transferring about half of each solution into a third beaker and mixing well. Label this third beaker as "Reaction between $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 ". Record the appearance of the solution (Clear? Cloudy? Colored?) in Data Table 1 in Section 6.
4. Although you could use solubility rules (a topic discussed in class) to predict the products of today's reaction, let us instead conduct some additional experiments to determine the identity of the product. In Data Table 1 in Section 6, list the two ions formed when $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 dissolves.
5. Assuming that the reaction involves two ions bonding together, list all possible chemical formulas for the product that might result from the mixing of $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 in Data Table 1 in Section 6.
6. For each chemical formula that you listed, design and conduct a reaction to make the compound with that chemical formula. (Additional chemicals should be available in lab). For example, perhaps $\text{Co}_3(\text{PO}_4)_2$ is a product, so try reacting $\text{CoCl}_2 \cdot 2\text{H}_2\text{O}$ with K_3PO_4 to make $\text{Co}_3(\text{PO}_4)_2$. List the chemicals used, appearance of each solution and reaction mixture, as well as ions and products of each mixture in Data Table 2 in Section 6. How do these reaction mixtures compare with the contents of the beaker labeled "Reaction between $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 ".
7. Filter the mixture from the beaker labeled "Reaction between $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 ". The liquid obtained after filtration is called the supernatant, and it contains the product that is not a solid. However, we did not measure the reagents carefully, so there may be some leftover reagents also dissolved in the supernatant. Let us find out which reagent, if any, is still present in the supernatant.
8. Pour half of the supernatant into an empty beaker, and then pour the remainder of the supernatant into a second empty beaker. Pour some $\text{Co}(\text{NO}_3)_2$ solution into the first beaker, and pour some Na_3PO_4 solution into the second beaker.
9. In Data Table 3 in Section 6, record your observations and indicate the chemicals present in your supernatant. Did your supernatant contain $\text{Co}(\text{NO}_3)_2$ or Na_3PO_4 or neither? How do you know?

2. Investigation: Conduct and analyze the chemical reaction between $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 (Part 2).

1. Now let us repeat today's reaction by taking very careful measurements.

Obtain 4 clean beakers and label them #1, #2, #3, and #4. Into each beaker, place 20.0 mL of the 0.0183g/mL $\text{Co}(\text{NO}_3)_2$ solution from the stockroom. Record the actual volume measurements and concentration in Data Table 4 in Section 6.

Now transfer different amounts of the Na_3PO_4 solution into beakers #1, #2, #3, and #4. Suggested amounts are 5.0, 10.0, 15.0, and 20.0 mL. Record the the actual volume measurements and concentration in Data Table 4 in Section 6. Mix each solution completely and let sit for at least 5 minutes.

2. **Let us find out which reagent, if any, is still present in the supernatant.** Filter each reaction mixture and collect the supernatants into another set of 4 labeled beakers. You do not have to collect all of the supernatant from each reaction but collect at least 10 mL that is devoid of any solids.
3. Divide each supernatant into 2 portions: into one portion, add at least 5 drops of $\text{Co}(\text{NO}_3)_2$ solution; into the other portion, add at least 5 drops of Na_3PO_4 solution. Examine each of these new reactions for a precipitate (a solid). The appearance of a solid will help us identify if a reagent was still present in the supernatant. Complete Data Table 5 in Section 6 testing each filtrate.
4. Now that we know which reagent was present in the supernatant, then the other reagent must have been the limiting reagent! For example, if the supernatant from Beaker #1 produced a solid when Na_3PO_4 was added, then Na_3PO_4 must have been the limiting reagent because it was used up first, leaving behind some unreacted $\text{Co}(\text{NO}_3)_2$.

6.0 DATA RECORDING SHEET

Last Name	First Name	Partner Name(s)	Date
Data Table 1. Qualitative Data $\text{Co}(\text{NO}_3)_2 + \text{Na}_3\text{PO}_4$			
	Appearance	Ions When Dissolved	
$\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$			
$\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}$			
	Appearance	Product Chemical Formulas	
Mixture			

Data Table 2. Qualitative Data Reactions to Form Products

Product Being Formed: _____

Chemical Used	Appearance	Ions When Dissolved
	Appearance	Product Chemical Formulas
Mixture		

Product Being Formed: _____

Chemical Used	Appearance	Ions When Dissolved
	Appearance	Product Chemical Formulas
Mixture		

Data Table 3. Qualitative Data Supernatant Test

	Appearance
Addition of $\text{Co}(\text{NO}_3)_2$	

Addition of Na_3PO_4

Chemicals Present in Supernatant:

Reasoning:

Data Table 4. Quantitative Data $\text{Co}(\text{NO}_3)_2 + \text{Na}_3\text{PO}_4$

Concentration of Stock $\text{Co}(\text{NO}_3)_2$ _____ g/mL

Concentration of Stock Na_3PO_4 _____ g/mL

Beaker	Volume (mL) of $\text{Co}(\text{NO}_3)_2$	Volume (mL) of Na_3PO_4	Grams of $\text{Co}(\text{NO}_3)_2$	Grams of Na_3PO_4
#1				
#2				
#3				
#4				

Data Table 5. The Testing of the Supernatants

Supernatant from Beaker # 1		Supernatant from Beaker # 2		Supernatant from Beaker # 3		Supernatant from Beaker # 4	
To half of the supernatant, add...		To half of the supernatant, add...		To half of the supernatant, add...		To half of the supernatant, add...	
5 drops of $\text{Co}(\text{NO}_3)_2$	5 drops of Na_3PO_4	5 drops of $\text{Co}(\text{NO}_3)_2$	5 drops of Na_3PO_4	5 drops of $\text{Co}(\text{NO}_3)_2$	5 drops of Na_3PO_4	5 drops of $\text{Co}(\text{NO}_3)_2$	5 drops of Na_3PO_4
Did a solid form?	Did a solid form?	Did a solid form?	Did a solid form?	Did a solid form?	Did a solid form?	Did a solid form?	Did a solid form?
If yes, which reagent was still present in Beaker #1?	If yes, which reagent was still present in Beaker #1?	If yes, which reagent was still present in Beaker #2?	If yes, which reagent was still present in Beaker #2?	If yes, which reagent was still present in Beaker #3?	If yes, which reagent was still present in Beaker #3?	If yes, which reagent was still present in Beaker #4?	If yes, which reagent was still present in Beaker #4?

7.

CALCULATIONS AND DATA ANALYSIS

7.1 Investigation: Conduct and analyze the chemical reaction between $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 (Part 1).

1. How did the reaction mixtures in Data Table 2 compare with the contents of the beaker labeled “Reaction between $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 ” from Data Table 1? Use and discuss these experimental results to identify the solid formed in the beaker labeled “Reaction between $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 ”.
2. Now that you have identified the solid formed in the reaction between $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 , write the BALANCED chemical equation for this reaction. Include physical states in your balanced equation. Your balanced equation should contain two reagents and two products (one product is a solid and one product is aqueous).
2. Investigation: Conduct and analyze the chemical reaction between $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 (Part 2).
1. Review your results for the “The Testing of the Supernatants” in Data Table 4. Did all four supernatants give the same result? Discuss why or why not.

2. Now confirm your experimental results by calculating the limiting reagent for each reaction #1, #2, #3, #4. The following data tables will help you with this.

Beaker 1

Line 1	Number of grams of $\text{Co}(\text{NO}_3)_2$ present (refer to Quantitative Data Table 4)
Line 2	Molar mass of $\text{Co}(\text{NO}_3)_2$
Line 3	Number of moles of $\text{Co}(\text{NO}_3)_2$ present
Line 4	Using the balanced equation, how many moles of Na_3PO_4 are required to react with the Number of moles of $\text{Co}(\text{NO}_3)_2$ on Line 3?

Line 5	Number of grams of Na_3PO_4 present (refer to Quantitative Data Table 4)
Line 6	Molar mass of Na_3PO_4
Line 7	Number of moles of Na_3PO_4 present
Line 8	Using the balanced equation, how many moles of $\text{Co}(\text{NO}_3)_2$ are required to react with the Number of moles of Na_3PO_4 on Line 7?

If Line 8 is larger than Line 3 AND if Line 7 is larger than Line 4, then $\text{Co}(\text{NO}_3)_2$ is the limiting reagent.

However, if Line 4 is larger than Line 7 AND if Line 3 is larger than Line 8, then Na_3PO_4 is the limiting reagent.

Identify the limiting reagent for Beaker #1 _____

Beaker 2

Line 9	Number of grams of $\text{Co}(\text{NO}_3)_2$ present (refer to Quantitative Data Table 4)
Line 10	Molar mass of $\text{Co}(\text{NO}_3)_2$
Line 11	Number of moles of $\text{Co}(\text{NO}_3)_2$ present
Line 12	Using the balanced equation, how many moles of Na_3PO_4 are required to react with the Number of moles of $\text{Co}(\text{NO}_3)_2$ on Line 11?

Line 13	Number of grams of Na_3PO_4 present (refer to Quantitative Data Table 4)
Line 14	Molar mass of Na_3PO_4
Line 15	Number of moles of Na_3PO_4 present
Line 16	Using the balanced equation, how many moles of $\text{Co}(\text{NO}_3)_2$ are required to react with the Number of moles of Na_3PO_4 on Line 15?

If Line 16 is larger than Line 11 AND if Line 15 is larger than Line 12, then $\text{Co}(\text{NO}_3)_2$ is the limiting reagent.

However, if Line 12 is larger than Line 15 AND if Line 11 is larger than Line 16, then Na_3PO_4 is the limiting reagent.

Identify the limiting reagent for Beaker #2 _____

Beaker 3

Line 17	Number of grams of $\text{Co}(\text{NO}_3)_2$ present (refer to Quantitative Data Table 4)
Line 18	Molar mass of $\text{Co}(\text{NO}_3)_2$
Line 19	Number of moles of $\text{Co}(\text{NO}_3)_2$ present
Line 20	Using the balanced equation, how many moles of Na_3PO_4 are required to react with the Number of moles of $\text{Co}(\text{NO}_3)_2$ on Line 19?

Line 21	Number of grams of Na_3PO_4 present (refer to Quantitative Data Table)
Line 22	Molar mass of Na_3PO_4
Line 23	Number of moles of Na_3PO_4 present
Line 24	Using the balanced equation, how many moles of $\text{Co}(\text{NO}_3)_2$ are required to react with the Number of moles of Na_3PO_4 on Line 23?

If Line 24 is larger than Line 19 AND if Line 23 is larger than Line 20, then $\text{Co}(\text{NO}_3)_2$ is the limiting reagent.

However, if Line 20 is larger than Line 23 AND if Line 19 is larger than Line 24, then Na_3PO_4 is the limiting reagent.

Identify the limiting reagent for Beaker #3 _____

Beaker 4

Line 25	Number of grams of $\text{Co}(\text{NO}_3)_2$ present (refer to Quantitative Data Table)
Line 26	Molar mass of $\text{Co}(\text{NO}_3)_2$
Line 27	Number of moles of $\text{Co}(\text{NO}_3)_2$ present
Line 28	Using the balanced equation, how many moles of Na_3PO_4 are required to react with the Number of moles of $\text{Co}(\text{NO}_3)_2$ on Line 27?

Line 29	Number of grams of Na_3PO_4 present (refer to Quantitative Data Table)
Line 30	Molar mass of Na_3PO_4
Line 31	Number of moles of Na_3PO_4 present
Line 32	Using the balanced equation, how many moles of $\text{Co}(\text{NO}_3)_2$ are required to react with the Number of moles of Na_3PO_4 on Line 31?

If Line 32 is larger than Line 27 AND if Line 31 is larger than Line 28, then $\text{Co}(\text{NO}_3)_2$ is the limiting reagent.

However, if Line 28 is larger than Line 31 AND if Line 27 is larger than Line 32, then Na_3PO_4 is the limiting reagent.

Identify the limiting reagent for Beaker #4 _____

3. Record the identities of the limiting reagents in the table below.

Beaker	Limiting Reagent By Testing the Supernatants	Limiting Reagent By Calculations
#1		
#2		
#3		
#4		

Do your results make sense? Suggest reasons for any unusual result(s).

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2507 Stoichiometry of Reactions

STOICHIOMETRY OF REACTIONS

1.0 INTRODUCTION

Stoichiometry describes the ratio relationships between chemicals. These relationships are calculated mathematically using dimensional analysis by including units with quantities and canceling identical units to arrive at the unit of the answer. For example, grams can be converted to moles using the molar mass. If you were to weigh out 5.0 grams of calcium (which has chemical symbol Ca), it is possible to convert it to moles using the molar mass found on the periodic table.

$$(5.0 \text{ g Ca} / 1) \times (1.000 \text{ mol Ca} / 40.08 \text{ g Ca}) = 0.12 \text{ mol Ca}$$

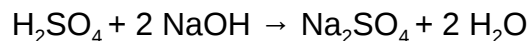
The units of 'grams of calcium' are in the numerator of the first term, and in the denominator of the second term. Those units cancel, and the remaining units are moles of calcium.

$$(5.0 \text{ g Ca} / 1) \times (1.000 \text{ mol Ca} / 40.08 \text{ g Ca}) = 0.12 \text{ mol Ca}$$

The result should have an appropriate number of significant figures. The molar mass is defined as the number of grams of a compound for exactly 1 mole of the substance. The molar mass therefore has the same number of significant figures in the numerator as the denominator. In the above example, the molar mass has four significant figures, but the quantity of 5.0 grams of calcium only has two significant figures. Thus there should only be two significant figures in the result.

The conversion between mol and the number of atoms or molecules can be done by using Avogadro's number (6.02×10^{23} molecules (or atoms) per mol). When a balanced chemical equation has been established, conversion between mol of one substance and mol of another is done using the coefficients in the balanced equation.

For example, the balanced equation between sulfuric acid, H_2SO_4 and sodium hydroxide, NaOH is:



This balanced equation indicates that 1 mol of H_2SO_4 reacts with 2 mol of NaOH to produce 1 mol of sodium sulfate, Na_2SO_4 and 2 mol of water, H_2O .

If there is 24.0 g of H_2SO_4 , we can determine how much NaOH is required. We will have to convert first to moles sulfuric acid using the molar mass of sulfuric acid, then to moles of sodium hydroxide using the coefficients in the balanced equation.

$$\text{g H}_2\text{SO}_4 \rightarrow \text{mol H}_2\text{SO}_4 \rightarrow \text{mol NaOH}$$

$$\frac{24.0 \text{ g H}_2\text{SO}_4}{98.09 \text{ g H}_2\text{SO}_4} \times \frac{1 \text{ mol H}_2\text{SO}_4}{1 \text{ mol H}_2\text{SO}_4} \times \frac{2 \text{ mol NaOH}}{1 \text{ mol H}_2\text{SO}_4} = 0.489 \text{ mol NaOH}$$

When setting up dimensional analysis problems, the units should be identified and retained in enough detail to avoid confusion. This is particularly important when using diluted solutions 15 ml of a solution of 1.5 molar NaOH should be written as "15 ml 1.5 M NaOH solution" to avoid confusion such as 'mol NaOH'. Similarly, "100 ml of 25% w/w H₂SO₄ solution" is different from "100 ml of 100% w/w H₂SO₄ solution"--both of these solutions have different densities which can easily lead to errors.

Figure 1 summarizes some stoichiometric relationships. These relationships are investigated in detail in this lab for reactions of calcium, water, hydrochloric acid and sodium hydroxide. Stoichiometric relationships can be reported graphically or numerically.

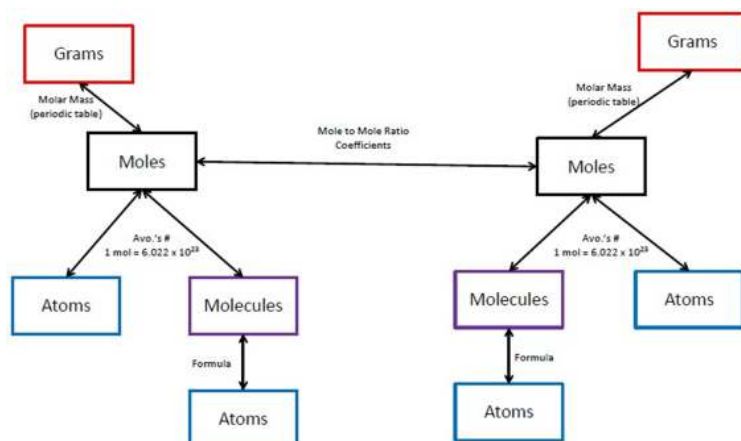


Figure 1 Stoichiometric Relationships

References and further reading

Los Medanos College Chemistry 25-26 Laboratory Manual, Technique B Significant Figures

Los Medanos College Chemistry 25-26 Laboratory Manual, Technique E, Volumetric Transfer of Reagent

Los Medanos College Chemistry 25-26 Laboratory Manual, Technique G, Buret Use

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

Sodium hydroxide and hydrochloric acid are both strong eye irritants. Be sure that the pathway to the eye wash station remains always unobstructed.

If any solution of acid or base used in this experiment is spilled on your hands, rinse thoroughly with plenty of water.

Wash your hands before leaving the laboratory. Do not eat or drink during this experiment.

Dispose of liquids down the drain

Once the reaction has been completed, the products are non-toxic. There are no hazardous wastes to be captured at the completion of this experiment.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate amount	Notes
HCl, hydrochloric acid	Approximately 1 M	50 ml	Record the concentration of the solution to 3 significant figures.
NaOH, sodium hydroxide solution	Approximately 0.5 M	100 ml	Record the concentration of the solution to 3 significant figures.
Methyl orange indicator solution	0.5%	Drops	Indicator
Calcium metal	N/A	0.2 g pieces	Calcium will react with moisture and the atmosphere. Keep stock containers sealed.
Laboratory water	N/A		
Litmus/pH paper	1-2 strips	pH measurement	
Matches			

4.0 GLASSWARE AND APPARATUS

Item	Use	Notes
1 small test tube	Mixing	
250 ml glass beaker		
400 ml glass beaker	Capturing waste for final disposal	
50 or 100 ml beakers	Weighing containers, solution containers	Start with clean, dry containers.
Spatula		Be sure to clean residues off spatula
Balance	Weighing	
2 flasks, 125-ml or 250-ml	Titration	
25-mL Pipette and bulb	Transfer liquid	
50-mL Buret	Transfer liquid	

5.0 PROCEDURE

Part A: Combining Calcium and Water

1. Fill a 250-mL beaker $\frac{1}{2}$ full of water.
2. Obtain 1 small piece of calcium. Observe the appearance of calcium metal.
3. Record your observations in the data section.
4. Obtain a small test tube, fill with water, cover its mouth with your finger, and invert the filled test tube into the beaker. Remove your finger when the mouth of the test tube is below the water level in the beaker. The objective is to invert the test tube without having any air bubbles.
5. Drop the small piece of calcium into the beaker.
6. Move the test tube so that it is directly over the piece of calcium. Capture gas formed (if any).
7. Record your observations.
8. Test the resulting solution with blue and red litmus paper, or pH paper.
9. Record your observations in the data section.
10. Use your observations to predict the reaction between calcium and water. Balance the reaction using correct chemical formulas.
11. If gas was formed and captured, test the gas with a lit match.
12. Discard the contents of the test tube(s) as directed by your instructor, clean all glassware used so far.

Part B:

1. Weigh out between 0.05 and 0.20 grams of calcium turnings. Record the mass in the data section to the nearest 0.0001 g on the analytical balance.
2. Transfer the calcium to a 125-mL flask.
3. Repeat for a second sample.
4. Return to your lab station with your two flasks.
5. Add 25.00 mL of 1.0 M HCl with the 25-mL pipette. Review Technique E on how to do this procedure with precision. Record the exact concentration of HCl.
6. Swirl the flask until all evidence of a reaction has ceased.
7. Add 1 drop of methyl orange to the test tube. Record the color in the data section.
 - a. A reddish orange color indicates the presence of acid (H^+).
 - b. Yellow indicates a more neutral solution.
 - c. You may need to put white paper under your beaker to clearly distinguish the color.
8. Prepare your buret by washing once with laboratory water and once with about 15 mLs of NaOH solution. Collect the waste in a beaker labelled "waste." Then fill your buret with NaOH solution and record the initial volume reading. Review Technique G on use of a buret. Record the exact concentration of NaOH.
9. Add the NaOH solution to your first flask slowly until one drop turns the color yellow. Record this final volume.
10. Refill your buret and titrate your second sample.
11. Calculate the number of moles of NaOH required to react with the excess reactant.
12. If there is time, do a third trial.
13. Report your results of the mass of calcium you used, the mol HCl added, and the mol NaOH added.
14. Record the class data. Make sure you calculate ALL missing data in the table. This will provide practice with stoichiometric calculations.

Last Name	First Name		Partner Name(s)	Date

6.0 DATA RECORDING SHEET

Part A

Action	Observation(s)
Inspection of piece of calcium metal.	
Interaction between calcium metal and water (if any)	
Test of reaction water with red litmus, blue litmus and pH paper.	
Hypothesis of reaction(s) occurred	
Evidence to confirm or refute hypothesis	
Other observations	
Notes:	

Part B: Titration

Exact Concentration of HCl _____

Exact Concentration of NaOH

	Trial 1	Trial 2	Trial 3 or Class Data	Class Data	Class Data
Gram of Ca					
Mole Ca					
Volume HCl added					
Mole HCl added					
Initial Volume NaOH					
Final Volume NaOH					
Volume NaOH used					
Moles NaOH used					
Moles HCl remaining in flask					
Moles of HCl that reacted with Ca					

7.0 CALCULATIONS AND ANALYSIS SHEET

Show your work here for calculations as needed.

Graph the moles of HCl that reacted with Ca versus the moles of Ca for your data. Attach your graph and give the equation for the line. What does this tell you about the stoichiometry of the reaction?

8.0 POST-LAB QUESTIONS AND CONCLUSIONS

1. Write out the balanced chemical reactions for part A and B.
2. Magnesium is in the same family of elements as calcium. If 0.0937 g of magnesium metal is reacted with 20.0 ml of 1.248 M hydrochloric acid solution, how many milliliters of 0.445 M sodium hydroxide solution would be expected to be needed to cause the resulting mixture to turn red litmus blue?
3. Aluminum is a Group 13 element. A sample of aluminum metal reacts with hydrochloric acid, much the same way that calcium metal does. What is the balanced chemical reaction for aluminum and hydrochloric acid?

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2508 Beer's Law

BEER'S LAW

1.0 INTRODUCTION

Many compounds are colored due to their absorption of visible light. Our eyes are naturally capable of detecting many different wavelengths or energies of light, and we perceive these different wavelengths as different colors.

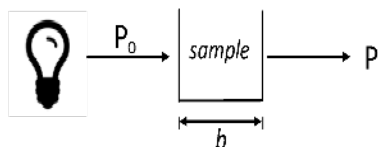
When white light passes through a colored liquid, some of the wavelengths of light are absorbed and some are transmitted or passes through the liquid. The observed color of the liquid is the sum of the wavelengths of light transmitted. For example, a substance that absorbs in the wavelength range 500 -750 nm will appear blue-purple to our eyes, because all of the red, orange, yellow and green light have been absorbed.

Wavelength Range (nm)	Color
380 – 450	Violet
450 – 495	Blue
495 – 570	Green
570 – 590	Yellow
590 – 620	Orange
620 – 750	Red

If a sample absorbs no light within the visible spectral range, then it will appear as a clear, colorless liquid. Water, for instance, absorbs such a small amount of visible light that our eyes generally cannot perceive the absorbance; unless you have a very large volume of water, it appears completely colorless.

Absorbance spectrophotometry is a commonly used laboratory technique for determining the concentration of substances in solutions.

A spectrophotometer is an instrument used to measure the amount of light absorbed by a sample. A light source directs light into the sample and the amount of light absorbed is measured. The power of the light source, P_0 , will be larger than P , the power of the light that is transmitted, because the molecules in the sample will absorb some of the light.



Most spectrophotometer provide measurements of % transmittance and/or absorbance, both are unitless (dimensionless). Percent transmittance, %T, a measurement of the light transmitted, is the ratio of the powers of the light source and transmitted light.

$$\% T = 100 P/P_0 \quad (\text{equation 1})$$

Absorbance, A, a measurement of the light absorbed, is the log of the reciprocal ratio.

$$A = \log_{10} P_0/P \quad (\text{equation 2})$$

If all the light passes through the sample without any absorption, then the absorbance reading on the spectrophotometer would be 'zero', and % transmittance is 100 and the sample appears colorless. Conversely, if all the light is absorbed, then the absorbance would be infinite and % transmittance would be 'zero'.

The following equation allows for interconversion between absorbance, A and % transmittance, %T.

$$A = -\log_{10} \%T/100 \quad (\text{equation 3})$$

A linear relationship exists between the absorbance of the solutes in solution and the concentration of the solution; the more solute molecules in the path of the light source, the greater will be the absorbance of the light. Low concentrations will transmit more light (and the color will appear lighter or less intense), and high concentrations will transmit less light (and the color will appear darker or more intense). This relationship is known as Beer's law and is given by the equation:

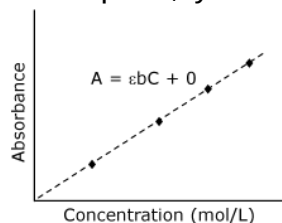
$$A = \epsilon bC \quad (\text{equation 4})$$

where A is the absorbance of the solution, ϵ is the molar absorptivity of the solute, b is the path length of light passing through the sample of solution, and C is the concentration of the solution in molarity (moles/L). The constant ϵ is specific to each solute at each wavelength and the units are $M^{-1}cm^{-1}$. If b and ϵ are constant, then the absorbance, A, is directly proportional to the concentration, C.

Since the molar absorptivity, ϵ , of the solute differs with wavelength, the optimal wavelength for each solution must be determined prior to absorbance spectrophotometry experiments. When working with solutions with only one solute absorbing visible light the optimal wavelength is the wavelength at which the solute is absorbing the strongest, λ_{max} . To find λ_{max} , a plot of absorbance vs. wavelength for a given solute is obtained. This plot is called an absorption spectrum.

At a given wavelength, plot of absorbance on the y-axis vs. concentration on the x-axis is predicted to be a linear plot, with a slope equal to ϵb , and an intercept of zero. In absorbance experiments, this is referred to as the standard (or calibration) curve. Once the calibration curve

is complete, you will be able to interconvert between absorbance, A and concentration, C .



Notice that the y-intercept of the line is zero – this makes sense, because if the concentration the light-absorbing compound is zero, then the absorbance should also be zero. In practice, the intercept will be a very small number, close to zero, but may not be exactly equal to zero.

In this experiment, students will prepare several dilute solutions from a standard stock solution of a dye. Using a spectrophotometer, students will determine the wavelength at which the solution is absorbing the strongest, λ_{max} . Using this wavelength, students will measure the absorbance of each of the dilute solutions and construct a standard curve. Students will be given a sports drink sample and measure its absorbance. Using this absorbance and the standard curve, students will determine the concentration of the dye in the sports drink.

Unfortunately, Beer's Law only holds true at absorbance values between 0.1 and 1. Above or below these absorbances readings no longer maintain a linear relationship. The original sports drink samples will be too concentrated. To ensure that the absorbance reading of the sports drink sample is within the usable range, the sample will need to be diluted. When determining which dilution to prepare, keep it simple such as a 1:2 or a 1:5 dilution. A 1:2 dilution indicates that the original solution has been diluted by adding enough solvent to double the volume. The diluted solution would therefore be $\frac{1}{2}$ the concentration of the original. For a 1:5 dilution, you would add water to 1 mL of sports drink sample to reach a total of 5 mL. The concentration of the dilute sports drink sample (determined from the absorbance and standard curve) can then be used to calculate the concentration of the original sports drink sample using the following equation.

$$M_1V_1 = M_2V_2 \quad (\text{equation 5})$$

References and further reading

Technique I: Use of Spectrophotometer of the laboratory manual

Experiment 2501 Using Excel for Graphical Analysis of Data of the laboratory manual

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

Spectrophotometers and cuvettes are expensive. Use them carefully.

Liquids can be disposed of in the sink.

Sports drinks are for experimentation only.

3.0 CHEMICALS AND SolutionS

Chemical	Formula	Approximate amount	Notes
Red No. 40 (496.43 g/mol)	$C_{18}H_{14}N_2Na_2O_8S_2$	25 mL dye solution	Synonyms: Allura Red, Allura Red, Food red 17
Yellow No. 5 (534.35 g/mol)	$C_{16}H_9N_4Na_3O_9S_2$	25 mL dye solution	Synonyms: Tartrazine, Acid yellow 23
Blue No.1 (791.84 g/mol)	$C_{37}H_{34}N_2Na_2O_9S_3$	25 mL dye solution	Synonyms: Brilliant Blue, Acid Blue 9
Commercial sports drinks			For experimentation only.

4.0 GLASSWARE AND APPARATUS

Item	Use	Notes
Beakers	Collecting and storage of samples	Start with clean, dry tubes. Label properly.
Test tubes	Solution dilutions	Start with clean, dry tubes. Label 1 – 4 for standard solutions and A – E for sports drink dilutions.
Three (3) 10-mL graduated pipets	Volume measurement	Label one 'water', one 'dye' and one 'sports drink'
Pipet bulb	Volume measurement	
Spectronic 20 &/or 200	% Transmittance &/or Absorbance measurement	Reference Technique I: Use of Spectrophotometer for proper use.
Square plastic cuvette or glass test tube cuvette	% Transmittance &/or Absorbance measurement	

5.0 PROCEDURE

Part A: Preparation of Standard Solutions

1.

Obtain about 25 mL of the assigned stock dye solution(s) in a clean, dry beaker. Please do not take more than what you need! Any excess will be wasted. Record the name, chemical formula, observed color and concentration of each stock dye solution (approximately 10 ppm) in Data Table 1 in Section 6.0.

2.

Prepare the following standard solutions using test tubes and graduated pipets to withdraw the specified volumes. Make sure to record the actual volumes to two (2) decimal places. For example, if you must withdraw 2 mL of water, the water level can go beyond the 2 mL mark but record the exact volume, such as 2.01 mL in Data Table 2 in Section 6.0.

Standard	Volume (mL) stock dye	Volume (mL) laboratory water
1	2.00	8.00
2	4.00	6.00
3	6.00	4.00
4	8.00	2.00

To use the graduated pipet, first use the bulb to draw solution into the pipet above the zero mark. Cap the end of the pipet with your finger. Slowly allow air into the top of the pipet until the meniscus is just touching the '0.00' calibration mark. Move the end of the pipet over to your receiving test tube. Drain the pipet until the meniscus is just touching the '2.00' mark. Remove the pipet and return the additional liquid back to the stock you have in your beaker. 2.00 mL has now been transferred. Adjust these instructions for other volumes as appropriate.

3. Cover and securely close each test tube with a small piece of parafilm. Shake each solution well to mix.

Part B: Dye Scan

Follow instructions on the preparation and proper use of a Spectronic 200 spectrophotometer (see Technique I: Use of Spectrophotometer of the laboratory manual for additional information). Turn on the spectrophotometer and allow it to warm up for 30 minutes.

1. Add laboratory water to a square plastic cuvette or glass test tube cuvette until there is about 4 cm of water from the bottom and place in the sample stage.
2. Blank the instrument by pressing the autozero button.
3. Empty the cuvette and shake out as much of the water as possible.
4. Run a scan of one of each of the standard dye solutions from 400 nm to 700 nm.
5. Record the maximum wavelength (λ_{max}) and absorbance at the peak in the spectrum in Data Table 1 in Section 6.0.

Part C: Absorbance Measurements

The following steps can be performed on a Spectronic 20 or Spectronic 200 spectrophotometer (see Technique I: Use of Spectrophotometer of the laboratory manual for additional information). If switching to a Spectronic 20 ensure that the instrument has warmed up, stabilized, and blanked with laboratory water. Absorbance measurements should all be done on the same day, using the same spectrophotometer.

Spectronic 20

1. Measure the % transmittance of the standard solutions at λ_{max} and record the data in Data Table 2 in Section 6.0. Note: Absorbance will be calculated in Section 7.0 Calculations and Data Analysis.
2. For each subsequent measurement, empty and rinse the cuvette with the next solution. Rinse the cuvette by adding a small amount of the next solution to be measured, swirling to coat the cuvette and empty before filling the cuvette until there is about 4 cm of the solution from the bottom.

Spectronic 200

1. Either in Scan mode, or in Live Display mode, measure the absorbance of the standard solutions at λ_{max} and record the data in Data Table 2 in Section 6.0.
2. For each subsequent measurement, empty and rinse the cuvette with the next solution. Rinse the cuvette by adding a small amount of the next solution to be measured, swirling to coat the cuvette and empty before filling the cuvette until there is about 4 cm of the solution from the bottom.

Part D: Analysis of Dye in Sports Drink

1. Obtain about 25 mL of the assigned sports drink sample in a clean, dry beaker.
2. Measure the absorbance or % transmittance of the undiluted sports drink sample at the appropriate wavelength(s) and record in Data Table 3 in Section 6.0.
3. Depending on how high the reading is, predict three dilutions to prepare to ensure that they all fall into the proper range. For example, if the undiluted sample has an absorbance of 1.200, and then you make a 1:1 dilution (5.00 mL undiluted + 5.00 mL water), you would expect the absorbance to be 0.600, which would be in the usable range.

4. Record how you prepared each dilution in Section 6.0 (be mindful of glassware and significant figures) as well as the % transmittance or absorbance of each of your three sports drink samples. Note: If % transmittance measured, absorbance will be calculated in Section 7.0 Calculations and Data Analysis.
5. If the absorbance reading of one of the dilutions does not fall within the usable range, prepare a fourth dilution and measure the % transmittance or absorbance.

6.0 DATA RECORDING SHEET

Last Name	First Name	Partner Name(s)	Date

Data Table 1. Stock Dye(s)

Name	Chemical Formula	Concentration
Observed Color	λ_{max}	Absorbance
Name	Chemical Formula	Concentration
Observed Color	λ_{max}	Absorbance
Name	Chemical Formula	Concentration
Observed Color	λ_{max}	Absorbance

Data Table 2. Standard Solutions

Dye						
Standard #	Volume of Stock (mL)	Volume of Dye	Volume of Laboratory Water (mL)	% T (if measured)	Absorbance	Concentration*
1						
2						
3						
4						
Dye						
Standard #	Volume of Stock (mL)	Volume of Dye	Volume of Laboratory Water (mL)	% T (if measured)	Absorbance	Concentration*
1						
2						
3						
4						
Dye						
Standard #	Volume of Stock (mL)	Volume of Dye	Volume of Laboratory Water (mL)	% T (if measured)	Absorbance	Concentration*
1						
2						
3						
4						

* See Section 7.0 Calculations and Data Analysis

Data Table 3: Sports Drink Sample Data

Data Table 3: Sports Drink Sample Data						
				Wavelength:	Wavelength:	
Sports Drink	Volume of	Volume of	% T	Abs	% T	Abs

Sample #	Sports Drink (mL)	Laboratory Water (mL)	(if measured)		(if measured)	
A (undiluted)						
B						
C						
D						
E (if needed)						

Briefly describe how the dilutions were prepared.

(e.g. 4.02 mL Gatorade + 6.00 mL water measure with graduated pipets)

Sports Drink Sample B:

Sports Drink Sample C:

Sports Drink Sample D:

Sports Drink Sample E (if needed):

7.0 CALCULATIONS AND DATA ANALYSIS

1. Calculate the molarity of the stock dye solution that was provided. Assume that the solution is dilute enough that its density is the same as that of water (1.00 g/mL). (Hint: Using the concentration provided and molar mass of the dye, treat this as a unit conversion problem. Convert g solute/g solution to moles solute/liter solution.)

2.

Using the dilution equation, $M_1V_1 = M_2V_2$, calculate the concentration of dye (M_2) in each of the diluted solutions, where M_1 is the stock concentration of dye calculated above, V_1 is the volume of stock dye recorded in Data Table 2, and V_2 is the total volume of the standard (sum of dye and water volumes). Record the values in Data Table 2. Remember to record the value in the Data Table with the appropriate number of significant figures and units. Show one sample calculation here.

3.

If a Spectronic 20 was used to measure the % transmittance, the absorbance of standard solutions and sports drink samples will need to be calculated. Using equation 3 ($A = -\log_{10} \%T_{100}$), calculate the absorbances. Remember to record the value in the Data Table with the appropriate number of significant figures. Show one sample calculation here.

4. After completing Data Table 2, make a graph of absorbance (y-axis) vs. the concentration of dye (x-axis). This is the standard curve. Notice that the data is linear. Obtain a best fit line or linear regression. Attach the graph when submitting and provide the equation here.

5.

Based on the absorbance of the sports drink samples in Data Table 3, determine the concentration in units of molarity of dye in each sports drink sample using the equation of the standard curve. Record the value in the Analysis Table below with the appropriate number of significant figures and units. Show one sample calculation here.

For example, if the equation of your standard curve is $y = 2.1033E4x + 0.0039$, and the measured absorbance for a sports drink sample is 0.3110. The concentration of dye is $1.46E-5$ M using the algebra shown.

$$(0.3110 - 0.0039) / 2.1033 \times 10^4 = [\text{dye}]$$

Analysis Table. Sports Drink Dye Concentration

Dye:		Wavelength:	
Sports Drink Sample #	Absorbance	Concentration of Diluted Sample	Concentration of Original Sample
A (undiluted)		N/A	
B			
C			
D			
E (if needed)			
Average:			
Dye:		Wavelength:	
Sports Drink Sample #	Absorbance	Concentration of Diluted Sample	Concentration of Original Sample
A (undiluted)		N/A	
B			
C			
D			
E (if needed)			
Average:			

6. Using the dilution equation, $M_1V_1 = M_2V_2$, calculate the concentration of dye (M_2) in the original sports drink sample, where M_1 is the concentration of dye calculated for each dilution, V_1 is the total volume of the dilution in Data Table 3 (sum of sports drink and water volumes), and V_2 is the volume of sports drink used in each dilution. These values should be very similar to each other as you are working with dilutions of the same original sports drink. Record the values in the Analysis Table. Remember to record the value in the Analysis Table with the appropriate number of significant figures and units. Show one sample calculation here.

7. Calculate the average concentration of dye in the original sports drink sample.

8.0 POST-LAB QUESTIONS AND CONCLUSIONS

1. Molar absorptivity, ϵ

a. Calculate ϵ based on Beer's Law (equation 4) and the absorbance and concentration of one of your standards. The pathlength of the cuvette used is 1.0 cm. Choose wisely. Do not choose a standard that might have an error.

b. The slope of the line of your standard curve is derived from the molar absorptivity (ϵ) of the dye and the pathlength (b) of the sample in the spectrophotometer. The pathlength of the cuvette used is 1.0 cm, calculate ϵ based on the trendline of your standard curve.

c. Compare the two molar absorptivity values (from Beer's Law and from the trendline). Which do you think is more accurate? Offer an explanation for your choice.

2. Locate and label the undiluted and diluted sports drink sample data points on your standard curve. Do all of these points fall within the usable range of the spectrophotometer? Do all of these points fall within the data points of the standard solutions?

3. How is the precision of your calculated concentrations of the original, undiluted sports drink sample? Calculate the standard deviation and relative standard deviation for the three dilutions. What does your relative standard deviation tell you about your precision?

4. Based on your average concentration of dye in the original sports drink sample and the molar mass of the dye, calculate how many grams of dye are present in 946 mL of sports drink.

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2509 Spectral Analysis of Cu²⁺

SPECTRAL ANALYSIS OF Cu²⁺

1.0 INTRODUCTION

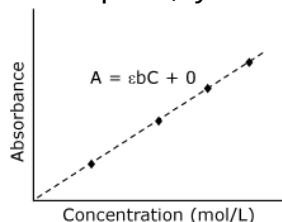
Reactions are often monitored by some form of spectroscopy. Many compounds are colored due to their absorption of visible light and absorbance spectrophotometry can be used to determine the concentration of these colored substances in solutions.

A linear relationship exists between the absorbance of the solutes in solution and the concentration of the solution; the more solute molecules in the path of the light source, the greater will be the absorbance of the light. Low concentrations will transmit more light (and the color will appear lighter or less intense), and high concentrations will transmit less light (and the color will appear darker or more intense). This relationship is known as Beer's law and is given by the equation:

$$A = \epsilon bC$$

where A is the absorbance of the solution, ϵ is the molar absorptivity of the solute, b is the path length of light passing through the sample of solution, and C is the concentration of the solution in molarity (moles/L). The constant ϵ is specific to each solute at each wavelength and the units are $M^{-1}cm^{-1}$. If b and ϵ are constant, then the absorbance, A , is directly proportional to the concentration, C .

At a given wavelength plot of absorbance on the y-axis vs. concentration on the x-axis is predicted to be a linear plot, with a slope equal to ϵb , and an intercept of zero. In absorbance experiments, this is referred to as the standard (or calibration) curve. Once the calibration curve is complete, you will be able to interconvert between absorbance, A and concentration, C .



Notice that the intercept of the y-intercept of the line is zero – this makes sense, because if the concentration the light-absorbing compound is zero, then the absorbance should also be zero. In practice, the intercept will be a very small number, close to zero, but may not be exactly equal to zero.

In this experiment, students will prepare several dilute solutions from a standard stock solution of Cu²⁺. Using a spectrophotometer, students will measure the absorbance of each of the dilute

solutions and construct a standard curve. Students will use the standard curve to determine the concentration of Cu^{2+} remaining in solution to investigate the reaction of copper (II) sulfate and potassium hydroxide.

References and further reading

Technique E: Volumetric Transfer Pipet Use of the laboratory manual

Technique I: Use of Spectrophotometer of the laboratory manual

Experiment 2501 Using Excel for Graphical Analysis of Data of the laboratory manual

Experiment 2508 Beer's Law of the laboratory manual

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

Spectrophotometers and cuvettes are expensive. Use them carefully.

Potassium hydroxide is a strong eye irritant. Be sure that the pathway to the eye wash station remains always unobstructed.

Solid potassium hydroxide is caustic! Do not handle the pellets!

Copper (II) solutions should be disposed of in the Inorganic Hazardous Waste container.

Dissolved copper salts are very toxic to marine and aquatic life.

3.0 CHEMICALS AND SolutionS

Chemical	Formula	Concentration	Approximate amount	Notes
copper sulfate pentahydrate (II)	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	N/A	About the size of a pea (0.25 g)	Very toxic to aquatic life with long lasting effects.
potassium hydroxide	KOH	N/A	1 g	Corrosive and hygroscopic.
copper sulfate (II)	CuSO_4	0.35 M	100 mL	Very toxic to aquatic life with long lasting effects.

4.0 GLASSWARE AND APPARATUS

Item	Use	Notes
Beakers	Collecting and storage of samples	Start with clean, dry beakers. Label properly.
Two (2) 25 mL volumetric transfer pipets	Volume measurement	Label one 'standard Cu^{2+} ' and one 'laboratory water'
Three (3) 10-mL graduated pipets	Volume measurement	Label one 'standard Cu^{2+} ' and one 'laboratory water'
Pipet bulb	Volume measurement	
Two (2) watch glasses	Cover reactions to prevent contamination	
Two (2) gravity filtration set-ups	Separation of precipitate from supernatant	Includes: filter funnel, filter paper, and Erlenmeyer flask. Label one 'Reaction A' and one 'Reaction B'
Test tubes	Solution dilutions	Start with clean, dry tubes. Label 1 – 4 for standard dilutions
Spectronic 20 &/or 200	% Transmittance &/or Absorbance measurement	Reference Technique I: Use of Spectrophotometer for proper use.
Square plastic cuvette or glass test tube cuvette	% Transmittance &/or Absorbance measurement	

5.0 PROCEDURE

Part A: Qualitative Investigation of Reaction

1. In a beaker add a small amount (1 cm^3) of solid copper (II) sulfate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) to about 25 mL of laboratory water. Mix to dissolve. Record the appearance of the solution in Data Table 1 in Section 6.
2. In a second beaker add a small amount (one or two pellets) of solid potassium hydroxide (KOH) to about 25 mL of laboratory water. Mix to dissolve. Record the appearance of the solution in Data Table 1 in Section 6.
3. Pour the solutions together and mix. Record the appearance of the mixture in Data Table 1 in Section 6.
4. Based on the observations of the mixture, provide a balanced chemical equation including state designations in Data Table 1 in Section 6 that represents the reaction.

Part B: Quantitative Investigation of Reaction

1. Obtain about 100 mL of the standard Cu^{2+} solution in a clean, dry beaker. Please do not take more than you need! Any excess will be wasted. Record the concentration of the standard Cu^{2+} solution (approximately 0.35 M) in Data Table 3 in Section 6.
2. Label two clean dry 100 ml or larger beakers "Reaction A" and "Reaction B". Into each mass 0.2 – 0.6 g of KOH using an analytical balance and record the exact mass in Data Table 2 in Section 6.
3. Using a volumetric transfer pipet, add 25.00 mL laboratory water to each beaker. Mix. Record the exact volume of laboratory water added in Data Table 2 in Section 6.
4. Once all the KOH has dissolved, add 25.00 mL standard Cu^{2+} solution to each beaker. Mix. Record the exact volume of standard Cu^{2+} solution added in Data Table 2 in Section 6. (You will use the rest of the standard Cu^{2+} solution in Part C.)
5. Allow the reactions to proceed for 10 minutes or more mixing periodically. (Feel free to complete Part C while you wait.) To prevent contamination, cover the beakers with watch glasses.
6. Prepare two filter papers for gravity filtration. The filter paper when folded should be a few mm below the rim of your funnel. Fold the filter paper into a cone by first folding it in half, and then in half again.
7. Place a funnel in the neck of an Erlenmeyer flask labeled "Reaction A". Open the cone filter paper for Reaction A so that three layers are on one side and one layer is on the opposite side. Support the filter paper in the funnel and pour a small amount of the reaction mixture to wet the filter paper. Do not wet with laboratory water as this will dilute the concentration of Cu^{2+} in the supernatant (liquid portion of the mixture).
8. Pour the remaining mixture to be filtered through the filter paper, in portions if necessary. It may be necessary to filter several times to separate the supernatant from the precipitate (solid portion of the mixture). If additional passes are necessary, the same filter paper can be used. The filter paper and precipitate can be discarded according to your instructor's directions. Record the appearance of the supernatant in Data Table 2 in Section 6, and retain the supernatant for analysis in Part D.
9. Repeat the filtration process for Reaction B using an Erlenmeyer flask labeled "Reaction B" and the second filter paper cone.

Part C: Preparation of Standard Dilutions

1. Prepare the following standard dilutions using test tubes and graduated pipets to withdraw the specified volumes of the standard Cu^{2+} solution from Part B and laboratory water. Make sure to record the actual volumes to two (2) decimal places. For example, if you must withdraw 2 mL of water, the water level can go beyond the 2 mL mark but record the exact volume, such as 2.01 mL in Data Table 3 in Section 6.

Dilution	Volume (mL) stock dye	Volume (mL) laboratory water
1	2.00	8.00
2	4.00	6.00
3	6.00	4.00
4	8.00	2.00

To use the graduated pipet, first use the bulb to draw solution into the pipet above the zero mark. Cap the end of the pipet with your finger. Slowly allow air into the top of the pipet until the meniscus is just touching the '0.00' calibration mark. Move the end of the pipet over to your receiving test tube. Drain the pipet until the meniscus is just touching the '2.00' mark. Remove the pipet and return the additional liquid back to the stock you have in your beaker. 2.00 mL has now been transferred. Adjust these instructions for other volumes as appropriate.

2. Cover and securely close each test tube with a small piece of parafilm. Shake each solution well to mix.

Part D: Absorbance Measurements

The following steps can be performed on a Spectronic 20 or Spectronic 200 spectrophotometer (see Technique I: Use of Spectrophotometer of the laboratory manual for additional information). Ensure that the instrument has warmed up, stabilized, and blanked with laboratory water. Absorbance measurements should all be done on the same day, using the same spectrophotometer.

Spectronic 20

1. Measure the % transmittance of the standard Cu^{2+} solution and standard dilutions at 620 nm and record the data in Data Table 3 in Section 6.0. Note: Absorbance will be calculated in Section 7.0 Calculations and Data Analysis.

2. For each subsequent measurement, empty and rinse the cuvette with the next solution. Rinse the cuvette by adding a small amount of the next solution to be measured, swirling to coat the cuvette and empty before filling the cuvette until there is about 4 cm of the solution in the bottom.

3. With the same techniques measure the % transmittance of the filtered reaction supernatants at the same wavelength and record the data in Data Table 2 in Section 6.0.

Spectronic 200

1. Either in Scan mode, or in Live Display mode, measure the absorbance of the standard Cu^{2+} solution and standard dilutions at 620 nm and record the data in Data Table 3 in Section 6.0.
2. For each subsequent measurement, empty and rinse the cuvette with the next solution. Rinse the cuvette by adding a small amount of the next solution to be measured, swirling to coat the cuvette and empty before filling the cuvette until there is about 4 cm of the solution in the bottom.
3. With the same techniques measure the absorbance of the filtered reaction supernatants at the same wavelength and record the data in Data Table 2 in Section 6.0.

6.0 DATA RECORDING SHEET

Last Name	First Name	Partner Name(s)	Date

Data Table 1. CuSO_4 +
KOH

	Appearance	
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$		
KOH		
Mixture		

Balanced Chemical Equation:

Data Table 2. Reaction Solutions

Reaction	A	B
Mass KOH (g)		
Laboratory water (mL)		
Standard Cu^{2+} solution (mL)		
Total volume (mL)		
Supernatant Appearance		
Absorbance		

Data Table 3. Cu^{2+} Standard Dilutions

Concentration of Standard Cu^{2+} Solution (mol/L): _____ Absorbance: _____

Dilution #	Volume of Standard Cu^{2+} (mL)	Volume of Laboratory Water (mL)	% T (if measured)	Absorbance	Concentration*
1					
2					
3					
4					

* See Section 7.0 Calculations and Data Analysis below

7.0 CALCULATIONS AND DATA ANALYSIS

1. Using the dilution equation, $M_1V_1 = M_2V_2$, calculate the concentration of Cu^{2+} (M_2) in each of the diluted solutions, where M_1 is the concentration of standard Cu^{2+} solution, V_1 is the volume of standard Cu^{2+} solution recorded in Data Table 3, and V_2 is the total volume of the dilution (sum of standard Cu^{2+} solution and water volumes). Record the values in Data Table 3. Remember to record the values in the Data Table with the appropriate number of significant figures and units. Show one sample calculation here.

2. If Spectronic 20 was used to measure the % transmittance, the absorbance of standard solution, dilutions, and sports drink samples will need to be calculated. Using the equation, $A = \log_{10} 100\%T$, calculate the absorbances. Remember to record the values in the Data Table with the appropriate number of significant figures. Show one sample calculation here.

3. After completing the Data Table 3, make a graph of absorbance (y-axis) vs. the concentration of Cu^{2+} (x-axis). This is the standard curve. Notice that the data is linear. Obtain a best fit line or linear regression. Attach the graph when submitting and provide the equation here.

4. Based on the absorbances of the reaction supernatants in Data Table 2, determine the concentration in units of molarity of Cu^{2+} of each supernatant using the equation of the standard curve. Record the value in the Analysis Table below with the appropriate number of significant figures and units. Show one sample calculation here.

For example, if the equation of your standard curve is $y = 1.2439x + 0.004637$, and the measured absorbance for a reaction supernatant is 0.303. The concentration of Cu^{2+} in the supernatant is 0.240 M using the algebra shown.

$$(0.303 - 0.004637) / 1.2439 = [\text{Cu}^{2+}]$$

Analysis Table

	Reaction A	Reaction B
Supernatant Cu^{2+} concentration		
Moles of Cu^{2+} in supernatant		
Moles of Cu^{2+} added		
Actual moles of Cu^{2+} consumed		
Moles of KOH added		
Theoretical moles of Cu^{2+} consumed		
% error		

5. Calculate the number of moles of Cu^{2+} remaining in the filtered supernatant of each reaction mixture using the supernatant concentration of Cu^{2+} in the Analysis Table and the total volume of the reaction mixture in Data Table 2. Record the values in the Analysis Table. Remember to

record the value in the Analysis Table with the appropriate number of significant figures and units. Show one sample calculation here.

6. Calculate the number of moles of Cu^{2+} added to each reaction mixture using the concentration of standard Cu^{2+} solution in Data Table 3 and the volume of standard Cu^{2+} solution of each reaction mixture in Data Table 2. Record the values in the Analysis Table. Remember to record the value in the Analysis Table with the appropriate number of significant figures and units. Show one sample calculation here.

7. Calculate the number of moles of Cu^{2+} consumed in the reaction by subtracting the number of moles of Cu^{2+} remaining in the supernatant from the number of moles of Cu^{2+} added to each reaction mixture. Record the values in the Analysis Table. Remember to record the value in the Analysis Table with the appropriate number of significant figures and units. Show one sample calculation here.

8. Calculate the moles of KOH added to each reaction using the molar mass of KOH and mass of KOH added to each reaction in Data Table 2. Record the values in the Analysis Table. Remember to record the value in the Analysis Table with the appropriate number of significant figures and units. Show one sample calculation here.

9. Calculate the theoretical moles of Cu^{2+} consumed in each reaction using the moles of KOH added to each reaction and the balanced chemical equation in Data Table 1. Record the values in the Analysis Table. Remember to record the value in the Analysis Table with the appropriate number of significant figures and units. Show one sample calculation here.

10. Using the actual and theoretical moles of Cu^{2+} consumed in each reaction calculate the percent error of each reaction. Record the values in the Analysis Table. Remember to record the value in the Analysis Table with the appropriate number of significant figures and units. Show one sample calculation here.

$$\% \text{ error} = \frac{\text{theoretical mol Cu}^{2+} \text{ consumed} - \text{actual mol Cu}^{2+} \text{ consumed}}{\text{theoretical mol Cu}^{2+} \text{ consumed}} \times 100\%$$

8.0 POST-LAB QUESTIONS AND CONCLUSIONS

1. Locate and label the reaction mixture data points on your standard curve. Thinking about the appearance of the supernatants (Data Table 2), how do the qualitative observations correlate to the quantitate data? Provide a brief explanation to illustrate your reasoning.

2. In Data Table 1 you provided a balanced chemical equation for the reaction between copper (II) sulfate and potassium hydroxide, using the data in the Analysis Table, does your data support the chemical equation? Summarize data for and against your balanced chemical equation. Explain your reasoning using your data.

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2510 Thermochemistry

THERMOCHEMISTRY: HEATS OF REACTIONS AND HESS'S LAW

1.0 INTRODUCTION

Thermochemistry is a study of energy and heat changes which accompany chemical reactions. Heat given off or absorbed by a chemical reaction at constant pressure is called the *enthalpy change*, ΔH . A reaction which loses heat to the surrounding is said to be *exothermic*. Reactions which absorb heat from the surrounding are said to be *endothermic*.

The sign of ΔH is relative to the surroundings. Exothermic reactions have negative enthalpy changes ($\Delta H < 0$). Conversely, endothermic reactions have positive enthalpies ($\Delta H > 0$). A way to remember this is that if heat is subtracted from the reaction final products ($\Delta H < 0$), it must go to the surroundings and is therefore exothermic.

Thermochemical reactions follow Hess's law, named after Germain Hess in the year 1840. It can be summarized as:

"If a process proceeds through one or several steps, the enthalpy change for the overall process can be calculated as the sum of the enthalpy changes for the individual steps".

This is a useful law. It allows the measurement of enthalpy changes for chemical reactions that are difficult or even impossible to determine directly. It can be written as an algebraic formula:

$$\Delta H (\text{Reaction 1} + \text{Reaction 2}) = \Delta H (\text{Reaction 1}) + \Delta H (\text{Reaction 2})$$

This experiment measures the following exothermic reactions:

- Enthalpy of the reaction of dissolving solid sodium hydroxide in water
- Enthalpy of neutralization reaction of aqueous hydrochloric acid by aqueous sodium hydroxide
- Enthalpy of neutralization reaction of solid sodium hydroxide by an aqueous solution of hydrochloric acid

The overall analysis of results should permit verification of Hess's Law.

Heat and temperature measurements are carried out in equipment called a calorimeter. A calorimeter is designed to insulate a container to minimize loss or gain of heat from the surroundings. A simple one consists of two Styrofoam™ cups nested together. The cups are covered by a corrugated cardboard square with a center hole for a thermometer. No calorimeter is perfectly insulated from its surrounding. Measurements must be corrected due to losses (or gains) from the calorimeter.

The immediate surroundings for the reaction are the reaction solution plus the calorimeter. The measured heat of the reaction, ΔH_{rxn} is equal to the heat gained by the surroundings:

$$\Delta H_{\text{rxn}} = -q_{\text{surr}} = -(q_{\text{soln}} + q_{\text{cal}})$$

Remember that the three reactions that are being studied in this experiment are all exothermic, hence ΔH_{rxn} will be negative.

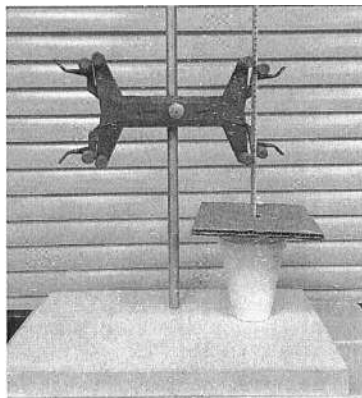
The heat of the solution, q_{soln} is given by:

$$q_{\text{soln}} = m_{\text{soln}} \times C_{\text{soln}} \times \Delta T_{\text{soln}}$$

where m_{soln} is the mass of the solution and ΔT_{soln} is the temperature change of the solution. The specific heat, C_{soln} , is the quantity of heat required to raise the temperature of one gram of a material by one degree Celsius. The total heat capacity of the double nested foam cups and corrugated board, q_{cal} , is calculated by the formula:

$$q_{\text{cal}} = C_{\text{cal}} \times \Delta T_{\text{cal}}$$

The heat capacity of calorimeter, C_{cal} , is the heat required to raise the temperature of an empty calorimeter by one degree Celsius, and therefore is measured in Joules per degree C. Unless you measure C_{cal} separately, assume that the heat capacity of the double-nested foam cup shown in the picture is 15 J/°C.



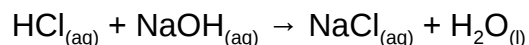
Double-Nested Foam Cup Calorimeter for Measuring Heats of Reactions

At least three factors, make it difficult to measure temperature changes:

- Heat is lost from the calorimeter to the surroundings. This causes the temperature to drop by about 0.1 °C/min
- It takes time for the calorimeter to reach thermal equilibrium. Usually this is at least 1 minute.
- It takes for sodium hydroxide to dissolve, sometimes as much as 4 min.

After correction is made for the heat absorbed by the calorimeter, the standard enthalpy of the reaction, ΔH , is calculated by dividing the negative of the corrected heat by the number of moles of a reactant and/or product.

Standard enthalpy of neutralization is calculated per one mole of water formed in the course of neutralization reaction. To find the number of moles of water formed, use the stoichiometry of the neutralization reaction:



A table may help organize the reactions and mixtures present in the parts of this experiment.

Summary Table

	Target mixture	Total volume of solution	M of HCl, NaOH in solution after mixing	Moles of product
Part A:	50.0 ml of 1.00 M HCl added to 50.0 ml of 1.00 M NaOH	100 ml	0.500	0.0500
Part B:	2.00 g $\text{NaOH}_{(\text{s})}$ are added to 50.0 ml of laboratory H_2O	100 ml	0.500	0.0500
Part C:	2.00 g of $\text{NaOH}_{(\text{s})}$ are added to 50.0 ml of 1.00 M $\text{HCl}_{(\text{aq})}$ and 50.0 ml of laboratory H_2O	100 ml	0.500	0.0500

The concentrations and amounts listed in the above table are ideal quantities. You will be determining the mass of your $\text{NaOH}_{(\text{s})}$ which is to be close to 2.00 grams but may not be exactly 2.00 grams. You will therefore have to calculate the number of moles of $\text{NaOH}_{(\text{s})}$ for your grams and it may therefore be a little more or a little less than 0.0500 mol. Similar adjustments might be needed for different concentrations of HCl. Remember to record your values.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

Sodium hydroxide and hydrochloric acid are both strong eye irritants. Be sure that the pathway to the eye wash station remains always unobstructed.

Solid sodium hydroxide is caustic! Do not handle the pellets!

If any solution of acid or base used in this experiment is spilled on your hands, rinse thoroughly with plenty of water.

Rinse the Styrofoam™ cups well with laboratory water prior to returning them. The cups are reusable.

Wash your hands before leaving the laboratory. Do not eat or drink during this experiment.

This experiment uses digital thermometers that should not be completely submerged in liquids. Only the measuring probe should be submerged. If glass thermometers are used, they are fragile. Mercury-containing thermometers have mostly been replaced due to problems with cleanup in the event of a break. If a mercury thermometer breaks, notify your instructor and the stockroom for appropriate and safe cleanup procedures.

Dispose of liquids down the drain

Once the reaction has been completed, the products are non-toxic (salty water). There are no hazardous wastes to be captured at the completion of this experiment.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate amount	Notes
HCl, hydrochloric acid	1.0 M	100 ml	
NaOH, sodium hydroxide solid	NA	4 g	Hydroscopic! Solid sodium hydroxide absorbs moisture directly from the air. Weigh quickly and seal the container.

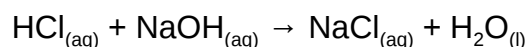
4.0 GLASSWARE AND APPARATUS

Item	Use	Notes
400 ml glass beaker	Store and use 1.00 M $\text{NaOH}_{(\text{aq})}$	If different concentration is used, label it properly.
250 ml glass beaker	Store and use 1.00 M $\text{HCl}_{(\text{aq})}$	If different concentration is used, label it properly.
50-ml graduated cylinders	Volume measurement	Keep separate cylinders for NaOH and HCl solutions
Stopwatch or phone	Time measurement	
50 or 100 ml beaker	Weighing container for solid NaOH	
Spatula		Be sure to clean residues off spatula
Thermometer	Temperature measurement	Do not submerge electronic portion
2 Styrofoam™ cups (8 oz)	Calorimeter	
Cardboard square	Insulation	Cut to fit if needed

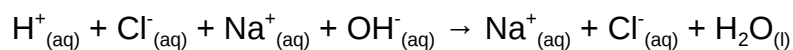
5.0 PROCEDURE

Part A: The Enthalpy of Neutralization of $\text{HCl}_{(\text{aq})}$ and $\text{NaOH}_{(\text{aq})}$

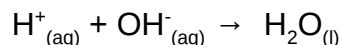
1. Prepare a double nested foam-cup calorimeter as shown in the picture.
2. Place 50.0 ml (measured by a 50-ml graduated cylinder) of 1.0 M $\text{NaOH}_{(\text{aq})}$ into calorimeter.
3. Use another 50-ml graduated cylinder to measure 50.0 ml of 1.0 M $\text{HCl}_{(\text{aq})}$. Leave the acid in the cylinder.
4. Measure the temperature of sodium hydroxide solution inside the calorimeter (with the accuracy of $\pm 0.1^\circ\text{C}$) for 3 minutes at 1-minute intervals.
5. At the 4th minute do not read temperature, but instead quickly mix $\text{HCl}_{(\text{aq})}$ into the calorimeter. Immediately close the lid to prevent the loss of heat. Gently swirl the calorimeter for 1 minute to mix the solutions.
6. Measure the temperature every minute after mixing. Continue until the temperature starts dropping at a steady rate for at least 5 minutes.
7. For this part, the complete reaction being studied is:



It can also be written as:



There are several species on both sides of the above equation. Because those species are not relevant to the reaction (they are called *spectator species* because they witness the reaction but do not chemically react), they can be canceled from both sides of the chemical equation. The result is a *net ionic* reaction:



This is a neutralization reaction. The enthalpy of this reaction is called the *enthalpy of neutralization*.

Part B: The Enthalpy of dissolving $\text{NaOH}_{(\text{s})}$

1. Because it is difficult to measure exactly 2.00 grams of sodium hydroxide, use between 1.9 and 2.1 grams. Precisely measure the amount of $\text{NaOH}_{(\text{s})}$ to at least 3 significant figures. Record the exact mass in your report.
2. Place 50 ml of laboratory water in your rinsed and dried calorimeter.
3. Record the temperature of the water for 3 minutes at one-minute intervals.
4. At the 4th minute, add the solid NaOH . Gently swirl the calorimeter.
5. Measure the temperature every minute after mixing.
6. It will take several minutes for the NaOH to dissolve, so the temperature will continue rising. Continue until the temperature starts dropping at a steady rate for at least 5 minutes.
7. For this part, the complete reaction being studied is:



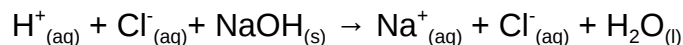
There are no spectator species present, so the reaction cannot be further simplified. This reaction is a dissolution (or solvation) of a chemical into a solvent. The enthalpy of this reaction is called the *enthalpy of dissolving* (sometimes called the *enthalpy of solvation*).

C. The Enthalpy of the Neutralization of $\text{HCl}_{(\text{aq})}$ with $\text{NaOH}_{(\text{s})}$

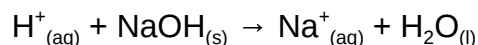
1. Because it is difficult to measure exactly 2.00 grams of sodium hydroxide, use between 1.9 and 2.1 grams. Precisely measure the amount of $\text{NaOH}_{(\text{s})}$ to at least 3 significant figures. Record the exact mass in your report.
2. Place 50.0 ml of 1.00 M $\text{HCl}_{(\text{aq})}$ in your rinsed and dried calorimeter.
3. Record the temperature of the solution for 3 minutes at one-minute intervals.
4. At the 4th minute, add the solid NaOH . Gently swirl the calorimeter.
5. Measure the temperature every minute after mixing.

6. It will take several minutes for the NaOH to dissolve, so the temperature will continue rising. Continue until the temperature starts dropping at a steady rate for at least 5 minutes.

7. For this part, the complete reaction being studied is:



The $\text{Cl}^-_{(\text{aq})}$ is a spectator ion that does not participate in the reaction. So it can be subtracted from both sides of the equation.



Note that if you combine the reaction in C with the reverse reaction in B (and after eliminating common ions) get the reaction in A. The enthalpy of this reaction will be the combination of the enthalpy of dissolving and the enthalpy of neutralization. This enthalpy does not have a specific name.

6.0 DATA RECORDING SHEET

Last Name	First Name		Partner Name(s)	Date

Temperature Readings ($^{\circ}\text{C}$)

	Time (minutes)	Part A	Part B	Part C	
	0.00 (Start)				
	1				
	2				
	3				
	4 Mix reagents				
	5				
	6				
	7				
	8				
	9				
	10				
	11				
	12				
	13				
	14				
	15				
	16				
	17				
	18				
	19				
	20				

Note: Continue to measure the temperature until it has stabilized.

7.0 CALCULATIONS AND ANALYSIS SHEET

	Part A	Part B	Part C
Amount of NaOH added	Conc: Volume:	Grams:	Grams:
Maximum temperature reached, $T_{\max}$, °C			
Temperature before mixing, T_o , °C			
Temperature change, $\Delta T = T_{\max} - T_o$			
Moles of NaOH present in the calorimeter (based on ml of NaOH in Part A or grams of solid NaOH in parts B and C)			
Heat gained by the solution: $q_{\text{soln}} = m_{\text{soln}} \times c_{\text{soln}} \times \Delta T$	*	**	*
Heat gained by the calorimeter, $q_{\text{cal}}^{***} = C_{\text{cal}} \times \Delta T$			
Heat gained by both calorimeter and $\text{NaCl}_{(\text{aq})}$, q_{surr} , in Joules			
Total heat of the reaction, in kJ $\Delta H_{\text{rxn}} = -q_{\text{surr}}$			
Enthalpy of the reaction in kJ per mole $\Delta H_{\text{rxn}} / \text{moles of reaction} = \text{standard } \Delta H$			

* The density of 0.5 M $\text{NaCl}_{(\text{aq})}$ is 1.02 g/mL. 100 mL of this solution is therefore 1.02 g/mL X 100 mL = 102 grams. The specific heat of this solution is approximately 4.19 J/g·°C.

** The density of water is 1.00 g/mL. Assume that the specific heat of this solution is approximately 4.184 J/g·°C.

*** Unless you measure C_{cal} separately, assume that the heat capacity of the double-nested foam cup shown in the picture is 15 J/°C.

8.0 POST-LAB QUESTIONS AND CONCLUSIONS

1. Write out the reactions for parts A, B, and C showing how A and B combine to give C.

2. Use the enthalpies you calculated for parts A, B, and C. Combine them in the same way as you did the reactions to see if your data is consistent with Hess's Law.

3. Does your data support Hess' Law? Why or why not? What sources of error could be present that might cause differences between the theoretical value and the observed value of the enthalpy of this reaction? How might you change the experiment to minimize sources of these errors?

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2511 Atomic Hydrogen and Line Emission Spectra

ATOMIC HYDROGEN AND LINE EMISSION SPECTRA

Last Name	First Name	Partner Name(s)	Date

1.0 OVERVIEW

One of the most tantalizing puzzles at the beginning of the 1900s for scientists was to describe the make-up of the atom. In this activity, you will look at spectra predicted by different models of the atom. The models you will test include:

- Niels Bohr's Shell model
- Louis de Broglie's Electron Wave model
- Erwin Schrödinger's Quantum Mechanical model

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

Simulation. No safety precautions.

3.0 CHEMICALS AND SolutionS

Simulation. No required chemicals.

4.0 GLASSWARE AND APPARATUS

The [simulation](#) can be downloaded and run offline on a computer that has [Java](#) installed or a smartphone with internet connection. Unfortunately, the simulation does not run on iPads.

5.0 PROCEDURE AND ANALYSIS

1. Access the **PhET** website.
2. Search for **Models of the Hydrogen Atom** and run it.
3. Turn on the *White* light gun with **Experiment** highlighted in the top left corner. The white light is shining into a transparent box with a "?" containing hydrogen gas molecules.

Why, with white light, the light photons passing up through the box have different colors?

4. Check the **Show Spectrometer** box. Notice that the color of the photons coming from the box corresponds to a wavelength of UV, visible, or IR radiation.
5. In the **Light Controls**, click on *Monochromatic*. Notice that the incoming photons are now all the same color. A spectrum slider appears that allows you to change the energy of the incoming photons. Move the slider across the spectrum from ultraviolet (UV) down to the infra-red (IR). Notice the color of the lamp and the photons moving up the screen.

Describe how you can distinguish between UV and IR photons. Record that information below. (You may want to move the *Slow...Fast* slider all the way over to *Slow*.)

6. Switch the **Light Controls** back to *White* light
7. Move the *Slow...Fast* slider all the way over to *Fast*, **Reset the Spectrometer**, and let the simulation run for 1 minute.
8. After 1 minute, click on the **Spectrometer** camera to take a snapshot of the **Experiment**.
9. Describe what is happening to the spectrum. Include in your description the *colors*, estimated *wavelengths*, and *relative numbers* of stacked colored balls. These colored balls correspond to photons emitted by the ? box.

10. Slide the snapshot off to the right of the screen for later comparison with the models.
11. In the top left corner, switch from **Experiment** to **Predict** and highlight the **Bohr** model, click on **Show electron energy level** in the upper right-hand corner, and set the speed to *Slow*. *NOTE: These three next steps are critical to your understanding of the Bohr atom model.*
 1. Describe what you see in the atom diagram.

 2. Describe what you see in the energy level diagram

 3. Describe how the atom diagram and energy level diagrams are related.

4. Notice that the electron decays in the energy level diagram are colored. What is the meaning of the colors?

12. Move the *Slow...Fast* slider all the way over to *Fast*, reset the **Spectrometer**, and let the simulation run for 1 minute.
13. Click on the **Spectrometer** camera to take a snapshot of the **Bohr** model. Compare it to the **Experiment**. Slide the snapshot off to the right for later comparisons.
14. *In the **Bohr** model, electrons can exist only at certain energy levels (also called shells), not at any energy levels between them. Shells in this energy diagram go from 1 to 6. An increase in energy level (say from 1→6) can only occur if a photon of incoming light with enough energy is absorbed. A decrease in energy level (say from 3→1) is accompanied by the emission of a photon as the excited electron releases its excess energy.*
 1. Calculate the wavelength of incoming light that must be absorbed to make a 1 → 6 transition.

 2. Calculate the energy of the photon emitted when an electron makes a 3 → 1 transition.

15. In the *Help* menu, select *Transitions* to see all possible transitions. Does this match your calculations?

16. Move the *Slow...Fast* slider all the way over to *Slow*.
17. Set the **Light Controls** to **Monochromatic**.
18. Set the monochromatic light source to 122 nm by clicking on the slider box and type in "122." This provides photons of just the right energy to raise (excite) the electron from $n=1$ to $n=$ _____ shell. Describe the atom diagram and the energy level diagrams.

19. Set the light source to 103 nm. This provides photons of just the right energy to raise (excite) the electron from $n=1$ to $n=$ _____ shell. Describe the atom diagram and the energy level diagrams.
20. Just to make sure you understand this model, set the light source to 97 nm. This provides photons of just the right energy to raise (excite) the electron from $n=1$ to $n=$ _____ shell. Describe the atom diagram and the energy level diagrams. How do the transitions differ from the two above?
21. Set the light source to 656 nm. This corresponds to a transition from $2 \rightarrow 3$. What happens? *Explain why.*
22. *Louis de Broglie was the first atomic theorist to incorporate the ideas of Planck and Einstein that electrons can be both waves and particles. He developed the de Broglie hypothesis stating that any moving particle or object had an associated wave (wave-particle duality). De Broglie thus created a new field in physics called wave mechanics, uniting the physics of light and matter. For this he won the Nobel Prize in Physics in 1929.*
23. Set the **Light Controls** back to **White**. Switch to the **de Broglie** Electron Wave model. Describe what is the same and what is different about the de Broglie and Bohr models of the hydrogen atom.
24. Move the *Slow...Fast* slider all the way over to *Fast*, reset the **Spectrometer**, set **Light Controls** to **White**, and let the simulation run for 1 minute.
25. Click on the **Spectrometer** camera to take a snapshot of the **de Broglie** model. Compare it to the **Experiment**. Slide the snapshot off to the right for later comparison with other models.
26. *Erwin Schrödinger began thinking about wave mechanics in 1925. His interest was sparked by a footnote in a paper by Albert Einstein. Like de Broglie, he began to think about explaining the movement of an electron in an atom as a wave. By 1926 he published his work, providing a theoretical basis for the atomic model that Niels Bohr*

had proposed based on laboratory evidence. The equation at the heart of his publication became known as *Schrödinger's wave equation*.

27. Move the *Slow...Fast* slider all the way over to *Fast*, reset the **Spectrometer**, set **Light Controls** to **White**, and let the simulation run for 1 minute.
28. Click on the **Spectrometer** camera to take a snapshot of the **Schrödinger** model. Slide the snapshot off to the right for later comparison with other models.
29. Switch to the **Schrödinger** model with **White** light and speed set to **Fast**. Describe what you see in the atom diagram and the energy level diagram.

1. The l value in the top left-hand corner corresponds to the l value we covered in lecture. What is this value telling us in the *Electron Energy Level Diagram*?
2. Describe what you see in the atom diagram when $n=1$ and $l=0$.
3. Describe what you see in the atom diagram when $n=2$ and $l=1$.
4. Compare the atom diagrams when $n=2$ & $l=1$ and $n=3$ & $l=1$.

8.0 CONCLUSIONS

Spread the **Experimental** snapshot and the model snapshots across the screen. Now that all of the models are side-by-side, describe **in detail** how each model's spectrum compares to the other models and most importantly to the real hydrogen spectrum. Compare the number of photons in each region of the spectra. What types of transitions are best predicted by which model(s)? Why?

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2512 Periodic Properties

PERIODIC PROPERTIES

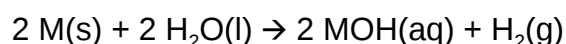
1.0 INTRODUCTION

Elements in the periodic table are arranged in order of increasing atomic number into horizontal rows whose lengths are such that elements with similar properties recur periodically. The elements within a given column of the periodic table, while having similar electron configuration, tend to have similar chemical properties. Because of this, columns of elements are often referred to as “groups” or “families” of elements. These families include the most familiar alkali metals, alkaline earth metals, halogens, and noble gases. Other families are the pnictogens (nitrogen family) and the chalcogens (oxygen family).

The physical and chemical properties of the elements within a given family tend to change gradually as one goes from one element in the column down to the next. In this experiment, the some of the properties of the alkali and alkaline earth metals will be observed and compared. The solubility properties of various ionic compounds containing alkaline earth metal cations will be observed. The properties of the halogen family, as neutral molecules and as anions, will be studied including the relative reactivity between halogens and halides.

1.1 Group 1 (or 1A) – The Alkali Metals

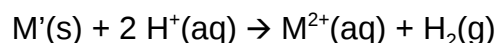
Alkali metals are some of the most reactive metals in the periodic table. Extreme caution should be observed in handling these metals. Lithium, sodium, potassium, rubidium, and cesium all react vigorously or even violently in cold water. The reaction produces alkali metal hydroxide and hydrogen gas.



Alkali metals lose an electron to make ions with a +1 charge. They are easily oxidized and typically used in organic reactions as reducing agents, species that tend to reduce (give electrons to) other species.

1.2 Group 2 (or 2A) – The Alkaline Earth Metals

The alkaline earth metals— beryllium, magnesium, calcium, strontium, barium, and radium—are all moderately reactive. They are less metallic in character than the alkali metals. Alkaline earth metals lose two electrons to make ions with a +2 charge. Their reaction with acids (H^+), produces the metal cation and generates hydrogen gas.



When solutions of Group 2 metal cations are mixed with solutions containing anions such as CO_3^{2-} or SO_4^{2-} , ionic compounds of the general form MX will precipitate if the compound MX is insoluble under the reaction conditions used, as shown in the net ionic equation below.

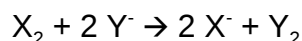


No precipitate will be observed if the compound MX is soluble.

When the solubilities of compounds containing various cations combined with a given anion are compared, a solubility trend that follows the order in the periodic table is expected. For example, for the solubilities of the sulfate salts, the solubility is expected either to increase or decrease as we go down the alkaline earth family.

1.3 Group 17 (or 7A) – The Halogens

The halogens include fluorine, chlorine, bromine, iodine, and astatine. They are also relatively reactive. Unlike the alkali and alkaline earth metals, the halogens tend to gain electrons to form anions, such as Cl^- , Br^- , and I^- . Since they are reduced when this occurs, the halogens are oxidizing agents, species that tend to oxidize (remove electrons from) other species. Thus, it is possible for some halogens (Cl_2 , Br_2 , I_2) to react with halide ions (Cl^- , Br^- , I^-) in a single replacement reaction. Taking X_2 to be a halogen, and Y^- to be a halide ion, the reaction would be as follows:

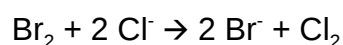


The reaction will only occur if X_2 is a better oxidizing agent than Y_2 , since X_2 has to remove electrons from the Y^- ions. If Y_2 is a better oxidizing agent than X_2 then no reaction will occur. Solutions of halogens and halide ions will be combined to determine the relative oxidizing abilities of the halogens. A trend can be shown as one goes from one halogen to the next in the Periodic Table.

Halogens have characteristic colors in non-polar organic solvents, such as cyclohexane, while the halide ions are all colorless in polar solvents like water. In this case, we can use the color changes that occur (or don't occur) to determine whether or not one halogen has displaced another in the above reaction. The relative oxidizing abilities of the halogens will be determined from the reactivity patterns.

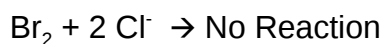
Example. Suppose that an aqueous solution of bromine or bromine water ($\text{Br}_2 \cdot \text{H}_2\text{O}$) is mixed with cyclohexane. Cyclohexane is less dense than and insoluble in water. The cyclohexane layer will be the top layer and water will be the bottom layer. The bromine is much more soluble in cyclohexane than in water and goes into the cyclohexane layer if you shake the mixture well, giving the cyclohexane layer an orange color. Now suppose we add a solution containing chloride ion (Cl^-), to the bromine mixture and mix well. There are two possible results:

a. If bromine (Br_2) is a better oxidizing agent (more easily reduced) than chlorine (Cl_2), it will take electrons from the chloride (Cl^-) ions to be converted to bromide (Br^-) ions:



If this reaction occurs the color of the cyclohexane layer will change from orange (the color of Br_2 in cyclohexane) to that of a solution of Cl_2 in cyclohexane.

b. On the other hand, if no reaction occurs:



There will be no color change; the cyclohexane layer will remain orange. This implies that bromine is not a better oxidizing agent than chlorine or, in other words, that chlorine is a better oxidizing agent than bromine.

In summary, no color change means that no reaction occurred, and a color change means that a reaction occurred. In either event, the halogen that remains after the halogen and halide are mixed is the less active halogen.

NOTE: It is important to keep in mind the difference between the halogen elements and the halide ions:

Halogens Halide Ions

Bromine, Br_2 Bromide ion, Br^-

Chlorine, Cl_2 Chloride ion, Cl^-

Iodine, I_2 Iodide ion, I^-

The halogens are molecular substances, oxidizing agents, have odors, and are distinct colors. They are only slightly soluble in water and much more soluble in cyclohexane.

The halide ions are soluble only in water, have no color or odor, and are not oxidizing agents. They do not dissolve in cyclohexane.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

Handle all metals with forceps, tongs, or a spatula, **not with your fingers.**

Use plastic containers when adding small samples of Na and K to water, along with all normal safety precautions.

Na and K metals must be kept separate. They form an air-reactive liquid metal alloy at room temperature and require special precautions if they are to be used. NaK is much more inherently dangerous and requires special techniques to be safely used.

Cyclohexane is flammable. Do not use it anywhere near an open flame.

Avoid breathing the halogen vapors.

Don't use your finger to stopper the tubes. Notify your instructor if you spill any bromine solution; it can be cleaned up with sodium thiosulfate solution.

All reaction mixtures should be placed in the waste container when you are done with them.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate amount	Notes
Na metal	N/A	One small piece	These metals are soaked/stored in mineral oil. Consult your instructor for proper handling.
K metal	N/A	One small piece	
Ca metal	N/A	One small piece	
Mg metal	N/A	One small piece	
Phenolphthalein	N/A	A few drops	
HCl	6 M	10 mL	
Ba(NO ₃) ₂	0.1 M	10 mL	
Ca(NO ₃) ₂	0.1 M	10 mL	
Mg(NO ₃) ₂	0.1 M	10 mL	
Sr(NO ₃) ₂	0.1 M	10 mL	
H ₂ SO ₄	1 M	10 mL	
Na ₂ CO ₃	1 M	10 mL	
(NH ₄) ₂ C ₂ O ₄	0.25 M	10 mL	
K ₂ CrO ₄	1 M	10 mL	
CH ₃ COOH	1 M	10 mL	
Cl ₂ •H ₂ O	N/A	5 mL	
Br ₂ •H ₂ O	N/A	5 mL	
I ₂ •H ₂ O	N/A	5 mL	
NaCl(aq)	0.1 M	5 mL	
NaBr(aq)	0.1 M	5 mL	
NaI(aq)	0.1 M	5 mL	
Cyclohexane	N/A	10 mL	

4.0 GLASSWARE AND APPARATUS

Item	Use	Notes
Test tubes	To hold reaction mixtures for observation	
Cork or rubber stopper	To contain liquid sample in a test tube	
Test tube rack	To place test tube in while observing reaction	
Stirring rod	To stir reaction mixture	
250-mL beaker	To hold reaction mixtures for observation	
Watch glass	To cover beaker	
Crucible tong or forceps	To pick up or grasp small metal pieces	

5.0 PROCEDURE

Part A. Reactivities of Some Alkali and Alkaline Earth Metals

1. Metallic sodium and potassium (Optional Instructor Demo)

a) For safety, these reactions must be conducted under the fume hood and performed separately.

2. Fill a small, plastic, rectangular bin about halfway with laboratory water. Place 2 to 3 drops of phenolphthalein solution into the water.
3. Using large forceps, obtain a small piece of Na metal from the reagent bottle and soak in the rinsing solution provided. Dry the metal piece completely in a paper towel.
4. Carefully place the metal into the bin. Lower the glass shield of the fume hood and allow the reaction to proceed. Record your observations.
5. Make sure that the metal has completely reacted before discarding the solution into the sink.
6. Repeat steps 2-5, this time using K metal.
2. Metallic magnesium and calcium

Place two clean, dry medium test tubes in a test tube rack. Using crucible tongs, put a small piece of Mg in one test tube and a small piece of Ca into another. Add 5 mL of laboratory water into each test tube. Record your observations in the data recording sheet.

Add 5 drops of 6M HCl to each test tube. If no change is evident, add an additional 5 mL of 6 M HCl. Record your observations.

Part B. Solubilities of Some Salts of Alkaline Earth Metal Cations

Four tests will be performed on solutions containing these ions: Mg^{2+} , Ca^{2+} , Sr^{2+} , and Ba^{2+} . You will be given aqueous solutions of $\text{Mg}(\text{NO}_3)_2$, $\text{Ca}(\text{NO}_3)_2$, $\text{Sr}(\text{NO}_3)_2$, and $\text{Ba}(\text{NO}_3)_2$. Nitrate is a spectator ion and will not interfere with any of the tests.

Between tests, the test tubes should be washed, rinsed with tap water, and then rinsed at least twice with small amounts of laboratory water. The test tubes do not have to be dried. The stirring rod, if you must use one, should be rinsed in a beaker of laboratory water between uses.

1. Add about 1 mL (approximately 12 to 15 drops) of the 0.1 M solutions of $\text{Mg}(\text{NO}_3)_2$, $\text{Ca}(\text{NO}_3)_2$, $\text{Sr}(\text{NO}_3)_2$, and $\text{Ba}(\text{NO}_3)_2$ to separate test tubes. To each test tube, add 1 mL of 1 M H_2SO_4 . Swirl the test tube vigorously or use a stirring rod to mix the contents. Observe whether or not an insoluble precipitate forms.
2. Record your observations in the data recording sheet, noting whether precipitate forms, and any characteristics—as color, odor, amount, size of particles, and settling tendencies—that might distinguish it.
3. Discard reaction mixture into a waste beaker.
4. Clean the test tubes as directed above and add 1 mL of each of the metal nitrate solution once again. To each test tube, add 1 mL of 1 M Na_2CO_3 . Mix the contents as before and record observations.

5.

Repeat the procedure two more times, once with 1 mL of 0.25 M $(\text{NH}_4)_2\text{C}_2\text{O}_4$ added to each of the four metal nitrate solutions, and finally with a mixture of 1 mL of 1 M K_2CrO_4 and 1 mL of 1 M CH_3COOH added to the four metal nitrate solutions.

Part C. Reactivities of Halogens and Halides

Obtain solutions containing Cl_2 , Br_2 , I_2 , Cl^- , Br^- , and I^- . For the halogens, you will be given halogen water solutions: $\text{Cl}_2 \cdot \text{H}_2\text{O}$, $\text{Br}_2 \cdot \text{H}_2\text{O}$, and $\text{I}_2 \cdot \text{H}_2\text{O}$. For the halides, you will be given aqueous solutions of NaCl , NaBr , and NaI . The sodium ion (Na^+) is a spectator ion and will not interfere with any of the tests.

1. Determine the color of the bromine in cyclohexane by placing 1 mL of bromine water ($\text{Br}_2 \cdot \text{H}_2\text{O}$) in a small test tube, adding 1 mL of cyclohexane, and mixing the contents well until the bromine color is mostly in the cyclohexane layer. Note the color of the

cyclohexane layer. Although the colors of the two layers may be different, it is the color in the cyclohexane layer that will be observed throughout this experiment.

- Determine the colors of chlorine and iodine in cyclohexane the same way as for bromine. Combine chlorine water ($\text{Cl}_2 \cdot \text{H}_2\text{O}$) with cyclohexane, and then iodine water ($\text{I}_2 \cdot \text{H}_2\text{O}$) with cyclohexane. Record the color of the cyclohexane layer in each case.
- Prepare each of the following mixtures in a small test tube by combining 1 mL of cyclohexane with 1 mL of each of the two reagents listed. Cover the test tube with a cork stopper and shake the contents vigorously. Note and record the color of the (upper) cyclohexane layer in your data recording sheet. The test tubes must be washed and then rinsed at least twice with laboratory water between tests.

Mixture #	1 mL of halogen + 1 mL of cyclohexane; shake	Then add 1 mL of halide; shake
1	Br_2	NaCl
2	Br_2	NaI
3	Cl_2	NaBr
4	Cl_2	NaI
5	I_2	NaCl
6	I_2	NaBr

6.0 DATA RECORDING SHEET

Last Name	First Name	Partner Name(s)	Date
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1. Reactivities of Some Alkali and Alkaline Earth Metals

Metal	Behavior in Cold Water
Sodium	
Potassium	

Write balanced chemical equations consistent with your observations above.

Metal	Behavior in Cold Water	6 M HCl Added
Magnesium		
Calcium		

Write balanced chemical equations consistent with your observations above.

2. Solubilities of Some Salts of Alkaline Earth Metal Cations

Indicate whether precipitate forms ("ppt") or there is no precipitate ("no ppt"). Note any distinguishing characteristics of the precipitates.

	1 M H_2SO_4	1 M Na_2CO_3	0.25 M $(\text{NH}_4)_2\text{C}_2\text{O}_4$	1 M K_2CrO_4 + 1 M acetic acid
$\text{Ba}(\text{NO}_3)_2$				
$\text{Ca}(\text{NO}_3)_2$				
$\text{Mg}(\text{NO}_3)_2$				
$\text{Sr}(\text{NO}_3)_2$				

3. Reactivities of Halogens and Halides

Indicate the color of the cyclohexane layer after mixing cyclohexane and each halogen.

Halogen	Color of Cyclohexane Layer
Br_2	
Cl_2	
I_2	

Indicate (a) the color of the cyclohexane layer after addition of each halide reagent and (b) the corresponding halogen that is present. If a reaction occurred, write a net ionic equation. If no reaction occurred, write "NR".

Mixture	Color of cyclohexane layer after mixing	Halogen present in the cyclohexane layer	Reaction
$\text{Br}_2 + \text{Cl}^-$			
$\text{Br}_2 + \text{I}^-$			
$\text{Cl}_2 + \text{Br}^-$			
$\text{Cl}_2 + \text{I}^-$			
$\text{I}_2 + \text{Cl}^-$			
$\text{I}_2 + \text{Br}^-$			

7.0 POST-LAB QUESTIONS AND CONCLUSIONS

- Which is more reactive with water, Na or K? Explain your answer based on your observations.
- Based on your answer in #1, what would be the trend for reactivity with water for alkali metals as atomic number increases.
- Which is more reactive, Mg or Ca? Explain your answer based on your observations.
- Will you expect a similar reactivity trend for alkaline earth metals as you observed for alkali metals? Why or why not?
- Consider the relative solubilities of the Group 2 cations in the various precipitating reagents. On the basis of the trends you observed, list the four alkaline earth metal ions

in order based on the solubility of the salts formed—the one that forms the greatest number of soluble compounds is the most soluble and the one that forms the greatest number of precipitates is the least soluble

_____ > _____ > _____ > _____

most soluble least soluble

6. Is this order consistent with the data from all of the precipitation reagents? If not, why did you order them as you did?

7. Is this order consistent with a periodic trend? If no, how is it different? If yes, does solubility increase or decrease down the column?

8. Rank the halogens in order of their decreasing oxidizing ability:

_____ > _____ > _____

most reactive least reactive

9. Does the order above follow a periodic trend? If no, how is it different? If yes, does oxidizing ability (“activity”) increase or decrease down the column?

2513 Molecular Structures

2513: MOLECULAR STRUCTURES

Last Name	First Name	Partner Name(s)	Date
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1.0 INTRODUCTION

1.1 Lewis Structures

Lewis structures are two-dimensional representations of the relative position of atoms in a molecule or polyatomic ion. All valence electrons are shown distributed around the bonded atoms as either shared electron pairs (bonding pairs) or unshared electron pairs (nonbonding or lone pairs). A shared pair of electrons is represented as a short line (a single bond). Sometimes atoms can share two pairs of electrons, represented by two short lines (a double bond). Atoms can even share three pairs of electrons, represented by three short lines (a triple bond). Examples are shown for SO_2 and HCN below.



1.2 Resonance Structures

Resonance refers to bonding in molecules or ions that cannot be correctly represented by a single Lewis structure. These structures are often equivalent, meaning that they contain the same number of bonds at different locations. The molecule SO_2 has two such resonance forms.



Resonance structures can also be non-equivalent, in which case they will have different numbers and/or locations of bonds. Note that any valid resonance structure of a molecule can be used to determine its shape and polarity.

1.3 VSEPR Theory

The Valence Shell Electron Pair Repulsion (VSEPR) theory is used to predict the three-dimensional shape of molecules (or polyatomic ions). It is based on the number of electron groups around a central atom. Electron groups can be a single bond, a double bond, a triple bond, a lone pair of electrons, or a single unpaired electron. Based on VSEPR theory, the molecular shape is primarily determined by minimizing the repulsion between these electron groups. Lone pairs of electrons tend to take more space than the bonded pairs leading to somewhat distorted structures.

The difference in electronegativity of the atoms in a bond determines whether a bond is polar or nonpolar. Depending on the arrangement of the polar bonds in a three-dimensional molecular structure, the molecule can be classified as polar or nonpolar. A molecule will be nonpolar if it contains all nonpolar bonds, OR it contains polar bonds that are arranged in a symmetrical manner around the central atom. A molecule will be polar if it contains polar bonds that are arranged in a nonsymmetrical manner resulting from the presence of lone pairs on the central atom, or when there are different atoms surrounding the central atom. $\longleftrightarrow$



linear, nonpolar linear, polar

In this experiment, students will practice drawing Lewis structures of covalent compounds and polyatomic ions. Using a molecular model kit, students will build these molecules and determine their shapes and bond angles. Students will also determine the polarity of the molecule based on the structure that they have drawn and modeled.

2.0 PRELAB ASSIGNMENT

Review (in your textbook or notes)

- the steps and rules for drawing Lewis structures, including rules for structures that follow the octet rule, and those that involve odd electrons, electron deficiencies, and expanded octets.
- the different geometries and bond angles that can be formed by different electron group arrangements around the central atom.

3.0 MATERIALS

Molecular Model Kits

Quick Guide to using your molecular model kit:

- Each colored ball corresponds to a specific atom. Refer to the color key provided in the kit.
- Use the medium length gray links for single bonds.
- Use the short white links for lone pairs.
- Use the flexible long gray links for making double or triple bonds.

4.0 INSTRUCTIONS

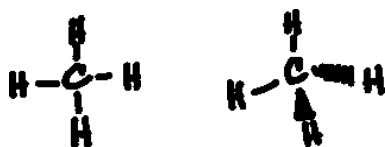
Complete the following for each of the molecules and ions on your data recording sheet.

1. Calculate the total number of valence electrons of all atoms in the molecule or ion. For a polyatomic ion, add one electron for each negative charge, and subtract one for each positive charge.
2. Draw the Lewis structure of the molecule or ion, including all resonance structures, if applicable.

3. Build an exact geometric model of the molecule using the balls and links provided in your kit. These 3D models will serve as a visual aid to help you in drawing the perspective structure of the molecule.
4. Look at your model and draw the perspective structure of the molecule or ion. Use the following guidelines:
 - For bonds lying in the plane of the paper, use a regular solid line.
 - For bonds that project down into the paper away from you, use a hatched wedge-shaped line. 
 - For bonds that project up out of the paper towards you, use a solid wedge-shaped line. 

Example: CH₄

Lewis Structure Perspective Diagram



5. Determine the electron group geometry around the central atom (or each central atom, if there is more than one), and the ideal bond angle(s) associated with this geometry.
6. Determine the shape of the molecule based on the bonded atoms.
7. Determine the polarity of the molecule. Use an arrow to show the direction of the overall dipole on the perspective drawing.
8. **Disassemble the models and return all balls and sticks to the model kit when finished.**

5.0 DATA RECORDING SHEET

Last Name	First Name		Partner Name(s)	Date

1. **HCN** Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry	Molecular shape
Bond angle(s)	Polarity

2. **CH₃OH** Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry	Molecular shape
Bond angle(s)	Polarity

3. **SeF₆** Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry	Molecular shape
Bond angle(s)	Polarity

4. **NO_2^-** Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry	Molecular shape
Bond angle(s)	

5. **AsF_5** Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry	Molecular shape
Bond angle(s)	Polarity

6. **XeF_2** Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry	Molecular shape
Bond angle(s)	Polarity

7. **TeF₅⁻** Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry Bond angle(s)	Molecular shape

8. **CH₂O** Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry Bond angle(s)	Molecular shape Polarity

9. **SF₄** Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry	Molecular shape
Bond angle(s)	Polarity

10. **ICl₂⁺** Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry	Molecular shape
Bond angle(s)	

11. PO_4^{3-} Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry	Molecular shape
Bond angle(s)	

12. CH_3CO_2^- Total number of valence electrons: _____

Lewis structure	Perspective diagram
Electron group geometry	Molecular shape
Bond angle(s)	

6.0 POST-LAB QUESTIONS AND CONCLUSIONS

1. The tetrahedral shape is one of the most fundamental shapes in chemical compounds. How would you describe it in words to someone who has not seen it?
2. The “octet” rule appears to be a very important rule governing the structures of molecules. Considering your work with models, provide a simple explanation for the importance of 8 electrons.

3. Explain in your own words why nonbonded electron pairs help determine the shapes of molecules.
4. Do all of the assigned molecules obey the octet rule? If not, why (or in what way) did the octet rule fail?
5. As a test of what you have learned, predict the shapes of (a) TeCl_4 (b) H_2S (c) Cl_2O .
6. Models do not necessarily have to be physical objects. They can be 2-dimensional drawings or even mental constructs. Cite one or more examples of such models encountered outside of chemistry. Can you think of models that are used in your own field of study or that you will use in your future career?

2514 Calculating R

DETERMINATION OF THE UNIVERSAL GAS CONSTANT

1.0 INTRODUCTION

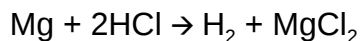
A gas is the state of matter that is characterized by having neither a fixed shape nor a fixed volume. Gases exert pressure, are compressible, have low densities and diffuse rapidly when mixed with other gases. On a microscopic level, the molecules (or atoms) in a gas are separated by large distances and are in constant, random motion.

Four measurable properties can be used to describe a gas quantitatively: pressure (P), volume (V), temperature (T) and mole quantity (n). The relationship among these properties is summarized by the Ideal Gas Law:

$$PV = nRT$$

Here, P represents as the gas pressure (in atmospheres or torr); V is the gas volume (in liters); n is the number of moles of gas in the sample; T is the gas temperature (in Kelvin). R is a proportionality constant called the Universal Gas Constant, and has a theoretical value of 0.08206 L•atm/K•mol or 62.363 L•torr/mol•K, depending on the units for pressure. Note that the units of R will allow the units of P, V, n and T in the Ideal Gas Law to cancel correctly.

In this lab, students will measure various properties of a sample of hydrogen gas to experimentally determine the value of the Universal Gas Constant, R. The single displacement reaction between magnesium metal and hydrochloric acid will be used to generate the hydrogen gas (with magnesium chloride as a byproduct).



The hydrogen gas will be collected in a eudiometer, a tube closed at one end and marked in milliliter volume units. The gas will be collected in the closed end of the tube over a water bath via the technique of water displacement (see figures shown later in this document).

Students will then obtain the following values for the collected sample of hydrogen gas: (1) Volume, (2) Temperature, (3) Moles, and (4) Pressure. The hydrogen volume will be directly measured from the eudiometer scale. The hydrogen temperature will also be directly measured using a thermometer. However, the mole quantity and pressure of the hydrogen gas must be determined indirectly. The mole quantity of the collected hydrogen can be easily calculated from the measured mass of the magnesium reactant using stoichiometry. But the hydrogen pressure is a little more difficult to obtain. Since hydrogen is collected over a water bath, a small amount of water vapor is mixed with the hydrogen in the eudiometer. The combined pressure of the H₂ and H₂O gases (after adjustments) will be equal to the external atmospheric pressure:

$$P_{\text{atm}} = P_{\text{hydrogen}} + P_{\text{water vapor}}$$

P_{atm} (atmospheric pressure) will be measured using a barometer. $P_{\text{water vapor}}$ (the partial pressure of water vapor) depends on the temperature of the water bath, and can be obtained from the table supplied below. By substituting these values in the above equation, the pressure of hydrogen (P_{hydrogen}) will be determined.

Finally, to determine the value of the Gas Constant (R), the quantities V , T , n and P obtained for the hydrogen gas must simply be substituted into the Ideal Gas Equation. Students can then evaluate their accuracy in this experiment by comparing their experimental result to the true theoretical value of R , and by calculating their percent error.

Temperature ($^{\circ}\text{C}$)	$P_{\text{water vapor}}$ (mm Hg or torr)
16	13.5
17	14.5
18	15.5
19	16.5
20	17.5
21	18.7
22	19.3
23	21.1
24	22.4
25	23.8
26	25.2
27	26.7
28	28.3
29	30.0

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

Concentrated HCl is corrosive and can burn your skin! If any spills occur, inform your instructor immediately. Wash under running water (sink or shower) and use the neutralizing sodium bicarbonate solution supplied at the sinks if necessary. Confirm that access to an eyewash station is available and know where it is. 6 M HCl is an eye hazard. Also note that hydrogen gas is flammable, so be sure to have no open flames nearby when you perform this experiment.

At the end of the experiment, the HCl solution will be much more dilute. The water, diluted HCl, and MgCl_2 mixture may be rinsed down the sink with plenty of water.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate amount	Notes
Magnesium (ribbon)	N/A	4 cm	
HCl solution	6 M	10 mL	Corrosive!

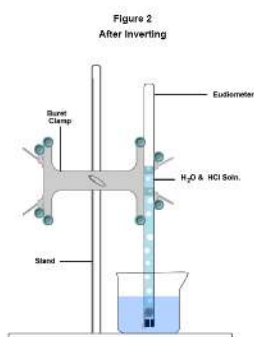
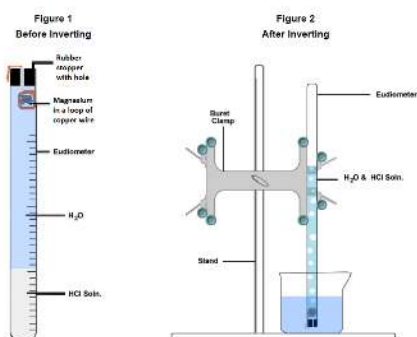
4.0 GLASSWARE AND APPARATUS

Item	Use	Notes
Copper wire	For holding magnesium ribbon	Return to cart or stockroom (do not dispose!)
Eudiometer	For analysis	50mL or 100mL
Rubber stopper with one hole	For eudiometer	
Ruler	For taking measurements	
Buret clamp	For supporting eudiometer	
Several beakers	For holding laboratory H_2O , HCl solution, water bath	
Thermometer	For temperature measurement	
10 mL graduated cylinder	For measuring HCl solution	Rinse well after use
Funnel	For transferring HCl solution	Rinse well after use
Sandpaper or steel wool	For cleaning magnesium ribbon	Return to cart or stockroom (do not dispose!)

5.0 PROCEDURE

1. Setup a ring stand with a buret clamp. Attach the empty eudiometer to the buret clamp with the open-end facing up. (If there is water in the eudiometer, pour it out before attaching it to the clamp.) Label a large beaker "Laboratory H_2O " and fill it $\frac{3}{4}$ full with laboratory H_2O .

2. Obtain a strip of Mg ribbon no greater than 0.0400 g and a piece of sandpaper (or steel wool) *if you are using the 50-mL eudiometer*. *If you have the 100-mL eudiometer, obtain a ribbon no greater than 0.0800 g*. Take the Mg to your bench and lay it down on a couple of clean paper towels. Use the sandpaper to remove the black oxide on the Mg please take care not to scratch the surface of the workbench. Wipe the Mg with a damp paper towel. Dry Mg completely. Hold the Mg with a paper towel and take the Mg to the balance; precisely record its mass (Data table, page 7).
3. Make a metal loop out of copper wire leaving a tail that will fit through the hole in a rubber stopper that fits into a eudiometer. Take the strip of Mg and place it through the metal loop attached to the rubber stopper so that it is folded in half and the ends are hanging down (Figure 1).
4. Label a 100 mL beaker "6M HCl". Obtain about 35 mL of 6M HCl from the reagent cart and return to your bench. Note the exact concentration of the HCl (Data table, page 7).
5. Use your 10 mL graduated cylinder to measure approximately 10 mL of the 6M HCl (you do not need exactly 10 mL, but you need to know exactly how much you have) and record the volume of 6M HCl (Data table, page 7). Place the funnel in the eudiometer and pour the 10 mL of HCl into the eudiometer.
6. Remove the eudiometer from the buret clamp. Hold it at a 45° angle. Slowly add the laboratory H₂O from the large beaker to the HCl in the eudiometer; try not to mix the HCl and the water as you pour. The goal is to layer the water over the acid. Completely fill the eudiometer with laboratory water; no space or air should remain.
7. Insert the rubber stopper with the magnesium into the opening of the eudiometer- push it all the way in and refill the hole with laboratory water (Figure 1).



8. Place your finger over the hole in the rubber stopper. Invert the tube and place it in the large beaker full of laboratory water. Once the hole is submerged in the water, you can take your finger off it. The reaction will begin shortly.

9. Measure the temperature of the water in the graduated cylinder at the end of the reaction. (Data table, page 7)
10. Gently tap the sides of the eudiometer to dislodge any stuck gas bubbles.
11. Measure the volume of hydrogen gas produced in the eudiometer. Without removing the eudiometer from the water, recover the hole in the rubber stopper with your finger. Carefully take the eudiometer to the large graduated cylinder at the sink. Again, once the hole is submerged in the water, you can take your finger off of it. Raise or lower the eudiometer so that the solution inside is level with the solution in the graduated cylinder. Once this is accomplished, you can read the volume of gas in the eudiometer. (Data table, page 7).

If you cannot get the two water levels equal, simply measure the difference in height in mm between the level of water in the eudiometer and the height in the graduated cylinder. This height can be converted to a pressure and used to correct the total pressure.

$$P(\text{total}) = P(\text{H}_2) + P(\text{H}_2\text{O}) + P(\text{water level difference})$$

$$P(\text{water level difference}) = \text{height}/13.59$$

12. Remove the eudiometer, invert it so that the rubber stopper is facing up, and dispose of the liquid into the sink. Rinse the eudiometer out with a small amount of tap water and final rinse with laboratory water.
13. Repeat the entire procedure two more times for a total of three trials.
14. Determine the barometer pressure and temperature of the room (Data table, page 7). Be mindful to include units.
15. Obtain the data for 3 additional trials by sharing with another group (Data table, page 7). Choose data that is consistent- don't include points in which there is an obvious error.

Last Name	First Name	Partner Name(s)	Date
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6.0 DATA AND CALCULATIONS

	Your group's data			Another group's data		
	Trial #1	Trial #2	Trial #3	Trial #1	Trial #2	Trial #3
Mass of Mg (g)						
Concentration of HCl (M)						
Moles Mg						
Moles H ₂ produced						
Volume of gas collected (mL)						
Temp. of water/gas (°C)						
Barometric Pressure (torr)						
VP of H ₂ O (torr)						
Height difference						
Pressure correction (0 if heights same)						
Partial pressure of H ₂ (torr)						
R (L•torr/mol• K)						
Average R value						

1. Calculate the number of moles of Mg from the mass of the magnesium ribbon in each trial, record your data above.
2. Assume that all of the magnesium reacts and that none of the gas escapes during the experiment. Magnesium is the limiting reactant; therefore, the number of moles of hydrogen gas produced (theoretical yield) is determined from the number of moles of magnesium reacted. Use the balanced chemical equation and moles of Mg to calculate the moles of H₂ produced in each trial. *This is a stoichiometry calculation not an ideal gas law calculation.* Record your answers in the data table on page 7.
3. Determine the vapor pressure of water using the temperature of the water/gas measured in step 10 of the procedure and record the vapor pressures in the data table on page 7. The vapor pressure for water at various temperatures is available on page 2.

- When the reaction was complete, the eudiometer was adjusted so that water in the graduated cylinder was at the same level as the water in the eudiometer. In this case, the atmospheric pressure (P_T) is approximately equal to the pressure inside the eudiometer. Dalton's Law of Partial Pressures can be used to calculate the hydrogen partial pressure (P_{H_2}) in units of torr with the atmospheric pressure (P_T) and vapor pressure for water (P_{H_2O}). Record the calculated partial pressures of H_2 in the data table on page 7.

$$P_T = P_{H_2} + P_{H_2O} (+ P_{\text{water level}})$$

- Calculate the value of R . You now have the four variables needed to calculate the Universal Gas Constant, R .

$$R = \frac{PV}{nT}$$

$$PV = nRT \text{ or } ,$$

where n is the number of moles of H_2 produced, T is the temperature of the gas in Kelvin (temperature of the water in the graduated cylinder at the end of the reaction), P is the partial pressure of hydrogen in torr, and V is the volume of gas produced in Liters.

7.0 ANALYSIS

Calculate your percent error using the average R as your experimental value and 62.363 L·torr/mol·K as the accepted value of R .

$$\text{Percentage error} = \frac{\text{experimental value} - \text{accepted value}}{\text{accepted value}} \times 100$$

8.0 POST-LAB QUESTIONS

- Explain why the vapor pressure of water must be subtracted from the atmospheric pressure to obtain the partial pressure of hydrogen gas.
- The experiment called for about 0.040 g of Mg and 6M HCl. What volume of 6.0 M HCl is required to react with 0.040 g of Mg? What was the limiting reactant?

3. Using the percent error, discuss the accuracy and precision, respectively, of the experiment.

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2515 Molar Mass of a Liquid

MOLAR MASS OF A LIQUID

1.0 INTRODUCTION

One of the most important applications of the Ideal Gas Law is the experimental determination of the molar masses (or "molecular weights") of gases. In fact, the method used in this experiment, the Dumas method, was the first widely used method for determining molar masses and was extremely important in the development of chemical theory.

The density of a gas is given by the ideal gas equation as

$$D_{\text{gas}} = \frac{m}{V} = \frac{PM}{RT}$$

where M is the molar mass of the gas. Rearranging this equation to obtain molar mass gives

$$M = \frac{mRT}{PV}$$

Thus, the molar mass of a gas can be determined by measuring the **temperature**, **pressure**, **mass**, and **volume** of a substance in its gaseous phase.

The 'unknown' liquids used in this experiment have boiling points less than 100°C. The liquid sample will be placed in a flask and then vaporized by heating the flask in a water bath. We will assume that the **temperature** of the vapor produced is the same as the temperature of the warm water bath that surrounds the flask (thermal equilibrium). An excess initial quantity of liquid sample is used so that its vapor will be more than enough to fill the flask. As a result, the vaporizing sample forces the air out of the flask as the vapor itself fills the flask. Any excess vapor also leaves the flask. Thus, after a few minutes, the flask contains only the vapor. The **pressure** of the vapor in the flask must equal the pressure of the atmosphere at the time (obtained from the barometer reading in the lab), since the flask is not closed off against atmospheric air. In addition, the **volume** of the vapor equals the volume of the flask. The flask is then removed from the warm water bath and the sample is allowed to cool and condense back into the liquid phase. The **mass** of the vapor is measured by weighing the condensate remaining in the flask.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

Use tongs or mitts to handle hot flasks and beakers.

Dispose of organic liquids in organic waste

Most of the liquids that are used in this experiment will be evaporated into the atmosphere. Any excess organic liquids need to be disposed of in the organic waste container.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate amount	Notes
"Unknown" liquid sample	N/A	7 mL	
Dithizone dye crystal	N/A	1 small piece	

4.0 GLASSWARE AND APPARATUS

Item	Use	Notes
250-mL Florence flask	To hold liquid sample during heating	
Aluminum foil	To cover the Florence flask	
Copper wire	To secure the aluminum foil cover on the flask	
Ring stand with iron ring		
Crucible tong	To hold and transport heated glassware	
Wire gauze	To support beaker on an iron ring	
Bunsen burner	To heat substances	
Utility clamps (2)	To secure flask while being heated; to secure thermometer	
Thermometer	To measure temperature of water	
Rubber stopper with hole	To support thermometer during use	
500-mL Graduated cylinder	To measure total volume of the 250-mL Florence flask	

5.0 PROCEDURE

Liquid sample in this experiment may be any of the following: acetone, isopropyl alcohol, ethyl acetate, or cyclopentane. Additional liquids may also be added at the request of the instructor.

1. Obtain a liquid sample from your instructor. Your instructor may choose to issue the liquid as an unknown sample. Record the name of the liquid sample in your data recording sheet, or the identification code of the unknown.
2. Obtain a clean, dry 250-mL Florence flask, put one small crystal of dithizone dye and loosely cover the flask with a square of aluminum foil. Secure the aluminum foil by tying a copper wire. Weigh the flask with the cover and copper wire on it. In your data recording sheet, record mass to at least the nearest 1 milligram (0.001 grams).
3. Carefully remove the copper wire and cover from the flask and add a small sample of liquid unknown - about 3 mL - to the flask. Swirl to dissolve the dithizone dye in the unknown liquid. Replace the cover and crimp it tightly around the neck of the flask. Secure the foil with the copper wire. Avoid leaving any loose foil or plastic wrap hanging down that might collect moisture from the water bath during the experiment.
4. Use a pin to make a tiny pinhole in the center of the foil cover. This is done to allow the air and excess vapor to leave the flask, maintaining a pressure

equal to atmospheric pressure.

5. Prepare a water bath by filling a 1-L beaker with about 600 mL of tap water. Add a few boiling chips. Place it on a wire gauze on an iron ring attached to a ring stand. Adjust the height of the ring to make sure that the hottest part of the flame of the Bunsen burner touches the bottom of the beaker. The beaker can be secured with an oversize ring to prevent it from falling over. (If the beaker of water is too far away from the flame, it will take a very long time for the water to heat up!)
6. Clamp the flask at its very top and suspend it in the water bath on the ring stand. Without allowing any water to get into the flask, push the flask as deep into the bath as possible, since the entire vaporized sample should be surrounded by water.
7. Begin heating the bath with Bunsen burner. Mount the thermometer in the bath so that it is as close to the unknown as possible. Leave the thermometer in the bath throughout the warming time.
8. Occasionally stir the water in the water bath with a stirring rod. Monitor the temperature of the water bath with a thermometer. Maintain the temperature of the water at around 95°C, or just below boiling throughout this experiment. Add additional water as needed to maintain the level of water.

Alternatively, a different setup may be used with hot plates instead of Bunsen burners. Check with your instructor as to the specific procedure to use.

9. Continue heating and watch the green color of the dithizone disappear – this indicates that the liquid sample has completely vaporized. If the water starts to boil while the liquid is evaporating, record this observation in your data recording sheet.

It is very important that you keep the sample submerged during the vaporization process - do not take the flask out and put it back in the bath before the vaporization is complete, or it may

jeopardize the accuracy of your results.

10. Keep the flask in the boiling water for three minutes after the last of the unknown liquid vaporizes to ensure that the vapor is in complete thermal equilibrium with the boiling water in the bath. Record the temperature of the boiling water indicated on your thermometer. Turn off the Bunsen burner.
11. Carefully remove the flask from the water bath by raising the clamp on the ring stand. The clamp may be hot; use thermal gloves or paper towels to protect from heat, if necessary. Let the flask hang on the ring stand until it has cooled to room temperature. During the wait, read the barometric pressure in the lab and record it.
12. When the flask has cooled, dry the outside of the flask thoroughly using paper towels. Make sure there are no water droplets on the outside of the aluminum foil cover. Weigh the flask (with the condensed unknown liquid inside) with the aluminum foil cover and copper wire still in place. Record this mass in your data recording sheet.
13. For a second trial, add about 2 - 3 mL of the same unknown to the flask

Replace the cover (make sure it is the same cover!), and repeat steps 5-12. Use the same water bath as before - since the water is already warm, it will take less time to heat up.

14. You have now recorded temperature, pressure, and weight of the sample for two separate trials. Volume must be measured next. This is NOT the same as the "250 mL" marked on the flask! To determine the volume, remove the cover and dump out the sample into the organic waste. Then, rinse out the flask and fill it to the brim with tap water. Pour this water into a 500-mL graduated cylinder and measure the volume to the nearest milliliter. You need to determine the volume of the expanded vapor in the flask, not just recording the volume marking on the flask.

6.0 DATA RECORDING SHEET

Last Name	First Name		Partner Name(s)	Date
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6.1 Data

Identification Code of Unknown	
Mass of the dry flask, dithizone crystal, foil, and copper wire	
Atmospheric Pressure (from barometer)	
Volume of flask (determined from your measured volume of water used to fill flask)	

	Trial 1	Trial 2
Temperature of water bath		
Mass of flask, dithizone, foil, copper wire, and condensate		
Mass of condensate		

6.2 Observations

7. CALCULATIONS AND DATA ANALYSIS

1. Calculate a separate value of molar mass for each trial. Calculate the average value and the percent difference between the two values. If you trust one result more than the other, state which one and explain your reasoning.
2. Obtain the identity of the unknown from your instructor. How does this compare with your experimental data? Calculate the percent error.

8.0 POST-LAB QUESTIONS AND CONCLUSIONS

1. Knowing the exact volume of the unknown liquid sample initially placed in the flask is not important. Why?
2. Why do you have to immerse the flask as deep into the water bath as possible?

3. When doing a second trial, why should you not dump out the condensate remaining in the flask before adding more sample?

4. There are some experimental errors that could lead to a higher or lower calculated molar mass compared to the true value. In each case below, determine whether the experimental value would be higher or lower than the true value of the molar mass of the compound analyzed. Justify each of your answers with supporting rationale or equations.
 1. Air was not completely swept out of the flask by the unknown liquid after it had completely vaporized, so that the gas in the flask was a mixture of air and unknown.
 2. The volume of the flask at 100°C was greater than that at room temperature because of expansion of the glass. (The expansion of the glass is $1.4 \times 10^{-5} \text{ deg}^{-1}$. That is, the volume of the flask increases by 1.4×10^{-3} percent per degree rise in temperature.)

3. The vapor was not an ideal gas. (Hint: Remember that the temperature of the vapor was not far above the boiling point of the unknown liquid. This fact should tell you the direction of the deviation from ideality.)

4. Although the unknown liquid completely vaporized, the gas in the flask never reached the temperature of the boiling water.

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2516 Intermolecular Forces of Liquids

INTERMOLECULAR FORCES OF LIQUIDS

1.0 INTRODUCTION

The attractive forces between molecules, the intermolecular forces, are responsible for many of physical properties of a substance. Forces *between* molecules in a liquid are called *intermolecular*. Forces *within* the molecule that keep the molecule together (such as the bonds between atoms) are called *intramolecular*.

Evaporation is vaporization that occurs at the surface of a liquid. Evaporation is an endothermic process and the extent of the temperature change is related to the strength of the intermolecular forces of the substance.

The vapor pressure of a liquid is the pressure of the vapor from evaporation of a liquid above a sample of the liquid in a closed container. The vapor pressure of a liquid depends on the intermolecular forces of the liquid as well as temperature and external pressure.

The surface tension of a liquid occurs at the interface between the liquid and a gas (or sometimes, another liquid). Surface tension is affected by the intermolecular forces of the material at the interface.

This experiment measures the evaporation temperature changes for several liquids and surface tension. Differences among these surface tensions and temperature change during evaporation illustrate different intermolecular forces that are present, which are in turn dependent upon the chemical structure of the liquid.

References and further reading

OpenStax College. (2015, March 17). *Chemistry*. OpenStax College. Retrieved from <https://openstax.org/details/books/chemistry>

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

Several liquids will be evaporated during this experiment. Avoid breathing in vapors. Use fume hoods if available as much as possible. Avoid eye and skin contact with liquids.

This experiment uses digital thermometers that should not be completely submerged in liquids. Only the measuring probe should be submerged. If glass thermometers are used, they are fragile. Mercury-containing thermometers have mostly been replaced due to problems with cleanup in the event of a break. If a mercury thermometer breaks, notify your instructor and the stockroom for appropriate and safe cleanup procedures.

Dispose of organic liquids in organic waste

Most of the liquids that are used in this experiment will be evaporated into the atmosphere. Any excess organic liquids need to be disposed of in the organic waste container.

Solid filter papers, if dry, can be disposed of as regular trash.

3.0 CHEMICALS AND SolutionS

Chemical	Formula	Approximate amount	Notes
Water	H ₂ O	Drops	Use laboratory water
Hexane	CH ₃ (CH ₂) ₄ CH ₃	Drops + 2 ml	Flammable liquid and vapor
Acetone	CH ₃ COCH ₃	Drops + 2 ml	Flammable liquid and vapor
Isopropanol (Rubbing alcohol, also called 2-propanol)	CH ₃ C(OH)HCH ₃	Drops + 2 ml	Flammable liquid and vapor
Methanol (methyl alcohol)	CH ₃ OH	2 ml	Flammable liquid and vapor
Ethanol (ethyl alcohol)	CH ₃ CH ₂ OH	2 ml	Flammable liquid and vapor
1-Butanol	CH ₃ CH ₂ CH ₂ OH	2 ml	Flammable liquid and vapor

4.0 GLASSWARE AND APPARATUS

Item	Use	Notes
Test tubes	Collecting and storage of samples	Start with clean, dry tubes. Label properly.
Pennies	Thermal sink	Pennies prior to 1982 were 100% copper. Pennies after 1982 are copper-coated zinc. Differences may be present.
50-ml graduated cylinders	Volume measurement	Keep separate cylinders for NaOH and HCl solutions
Stopwatch or phone	Time measurement	
Molecular model kits	Chemical models	
Filter paper	Solvent dispersal	Use approximately 2.5 cm x 2.5 cm
Thermometer	Temperature measurement	Do not submerge electronic portion

5.0 PROCEDURE

Part A: Penny Drops

1. Clean and dry several pennies. Using a dropper, drop one of the liquids slowly onto the top of a penny laying down on the top of the lab bench. Count the number of drops a penny can hold before the liquid starts to run off the side of the penny for each of the solvents. Be consistent with the size of each drop.
2. Repeat the process for each liquid on separate pennies.
3. Complete Data Table 1 in Section 6.0 with other students' data for comparison.

Part B: Evaporation Temperature Change

1. In clean, labeled, small test tubes, obtain approximately 2 mL of the following liquids:

methanol (methyl alcohol)

ethanol (ethyl alcohol)

isopropanol

1-butanol

hexane

acetone

2. Wrap a digital thermometer probe tip with square pieces of filter paper (2.5 cm × 2.5 cm) in a cylinder or test tube. Secured the cylinder with a small rubber band. The paper should be even with the probe end.

3. Submerge the filter paper in methanol for 45 seconds to soak the filter paper. Monitor the temperature while submerged for 15 seconds to establish the initial temperature. Record the initial temperature in Data Table 2.
4. Remove the thermometer from the liquid and use tape to secure the thermometer to the lab bench. The thermometer probe tip should extend about 5 cm over the edge of the bench. Continue monitoring the temperature until it reaches a minimum and begins to increase. Readings should be every 15 - 30 seconds. Record data in the table in Section 6.
5. Subtract the minimum temperature from the initial temperature to determine ΔT , the temperature change during evaporation.
6. Roll the rubber band up the probe and dispose of the filter paper.
7. Replace the filter paper and repeat the evaporation with the other liquids. Record their data in the data table.

Last Name	First Name	Partner Name(s)	Date
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Note: Continue to measure the temperature until it has begun to increase.

7.0 Data Analysis

Part A: Penny Drops

1. Group the liquids used in the penny drop portion on the experiment according to intermolecular forces. Does your data support your grouping? Explain.

Part B: Evaporation Temperature Change

2. Plot a graph of ΔT values of the four alcohols versus their respective molar masses. Plot molar mass on the x-axis and ΔT on the y-axis. Use a linear regression to find an algebraic expression to express the data.

8.0 POST-LAB QUESTIONS AND CONCLUSIONS

1. Provide a sketch of four water molecules interacting. Label the intermolecular and intramolecular forces.

2. Provide a sketch of four acetone molecules interacting. Label the intermolecular and intramolecular forces.

3. Provide a sketch of four hexane molecules interacting. Label the intermolecular and intramolecular forces.

4. Two of the liquids, hexane and 1-butanol, had nearly the same molar masses, but significantly different ΔT values during evaporation. Explain the difference in ΔT values of these liquids, based on their intermolecular forces.

5. Which of the alcohols studied has the strongest intermolecular forces of attraction? The weakest intermolecular forces? Explain using the results of this experiment.

6. Using the language of intermolecular forces, explain the order of the evaporation temperature changes and surface tensions you observed.

CHAPTER OVERVIEW

Part 2 Techniques

- A Use of Laboratory Notebook
- B Significant Figures
- C Use of Balances
- D Setup of vacuum filtration
- E Volumetric Transfer Pipet Use
- F Crucible use
- G Buret use
- H Use of the Bunsen burner
- I Use of Spec 20/Spec 200
- J: Use of Centrifuge

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A Use of Laboratory Notebook

Technique A: Use of Laboratory Notebook

SECTION 1: PURPOSE OF TECHNIQUE

The laboratory notebook is a permanent record of research activities. It preserves experimental plans, study designs or protocols, procedures, observations, results, conclusions, and recommendations. Information must be documented in sufficient detail so another scientist of equivalent training and experience can replicate the results.

Supplemental quality control activities and practices related to equipment maintenance and calibration are recorded on appropriate forms and maintained in log books as required, often under laboratory Standard Operating Procedures. Unless there is a critical operation, it is generally not necessary to record supplemental quality control data twice, though the activities should be referred to and cross-referenced in all appropriate laboratory notebooks. For example, if the laboratory water supply is monitored, measured and recorded in a separate logbook, then a second record of that monitoring result is not needed.

The laboratory notebook and its contents may be owned by a business, government agency, or a private individual. Laboratory notebooks are often involved as supporting documentation in courts of law. Ownership of the laboratory notebook and its contents should be clearly defined.

SECTION 2: HOW TO START YOUR LABORATORY NOTEBOOK

Requirements for a Laboratory Notebook

The laboratory notebook is usually bound with sequentially numbered pages. Several styles are available. Some organizations require acid-free paper for archival purposes. Others require a carbon-copy for immediate archival at the end of the day. Pages can be ruled or graph-ruled. Some examples are provided below:



For most educational settings a bound composition book is sufficient. If the pages are not sequentially numbered, then sequences can be established by diligent recording of entry dates:



Laboratory notebooks should be used with either blue or black permanent ink. Many non-black or blue inks fade with time.

SECTION 3: COMPONENTS OF A LABORATORY NOTEBOOK

Part 1: Owner's Name, Dates of Use, and Course Title

1. Put your name on the outside cover, inside cover, or on the first page of your notebook to identify that you own this laboratory notebook.
2. Specify when this laboratory notebook is in service. Write a start date on the cover, followed by an end date when the notebook is completed.
3. If used in an educational course, record the course title, professor, and section number for reference.
4. Provide an email address and contact phone number to help reunite it with you if it



becomes lost.

Part 2: Table of Contents and Numbered Pages

1. A Table of Contents is optional, but useful.
2. The Table of Contents should list page numbers, titles of experiment, and dates of when started and finished. This makes it easier to find information.
3. Make sure to include enough pages in the front for the Table of Contents.

Table of Contents		
Experiment	Page #	
2/6-2/9	1	Lesand Trace Evidence Lab
2/16-2/19	13	Physical Fit Lab: Glass
2/19-2/24	17	Soil Microscopy Lab
2/24-3/5	20	Questioned Documents Lab
3/4-3/4	35	Biometric Analysis Lab
3/4-3/4	41	Recreation Anthropometry Lab
3/4-3/10	45	Skull ID Lab

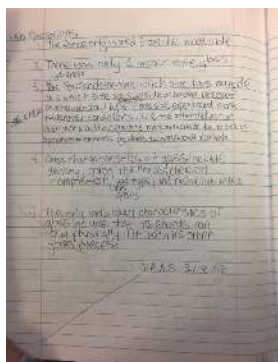
4. If your notebook is not sequentially numbered, make sure to do so using black or blue ink. Date and initial all information collected, recorded, and calculated in your notebook at the end of each day of working in the laboratory.

Part 3: Experiment Entries

1. For each experiment, include the day you started it and the title of the experiment at the beginning of the page.
2. Document the daily plan of the experiment before starting it. Include the purpose of the experiment, background information, techniques to use, equations used for calculations, and what reagents and materials are needed.
3. Start a section for recording observations. Record clear, concise, and detailed descriptions. Pre-laboratory research and background information may provide predicted results. Include all that was expected and any unexpected or unusual observations. Unusual observations should be recorded—these observations may lead to further experimentation. Entries should be grammatically correct, legible and honest.
4. Include experimental raw data. Minor details may be important, so record data as feasible. Organize data in tables. Tape or permanently glue any data printouts or separate forms. For computer programs, record the program name and version number if relevant.
5. After observations, prepare a data analysis section. This will include any calculations needed to properly interpret data.
6. Include a conclusions section of what went well, what could have been done better, and what future scientists can do to improve the experiment.
7. If another person did some parts of the experiment, note have them date and initial the relevant parts of the laboratory notebook.
8. If an experiment is too long for one page, show that the experiment continues on to the next page. If pages are skipped because of concurrent experiments, include references back to the original experiment. So if Experiment 12 ends on page 23 and is continuing on page 35, a note on page 23 should read 'continued on page 35' and a note on page 35 should read 'Experiment 12 (continued from page 23).'

Keep A Legally Defensible Document

All data goes in the notebook, even outliers, or “bad” data points. A failed experiment should still be documented in the laboratory notebook.



Do not remove any pages, or data. Do not skip any pages. Cross out any unused portions of your laboratory notebook and mark it with the date and signature.

Cross out any mistakes with a single line. Add corrections with initials and a date the correction was made. To correct a large section, block out with one diagonal line from corner to corner followed by a date, signature, and reason for crossing it out.

REFERENCES:

“Standard Operating Procedure for Use and Maintenance of Laboratory Notebooks and Project Binders”, U. S. Environmental Protection Agency Office of Pesticide Programs, Microbiology Laboratory, Environmental Science Center, Ft. Meade, MD. SOP Number: ADM-05-04; Date Revised: 06-20-17. <https://www.epa.gov/sites/production.../adm-05-04.pdf> (Accessed 10/29/2018).

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B Significant Figures

Technique B: Significant Figures

SECTION 1: PURPOSE OF TECHNIQUE

The number of *significant figures* of a quantity is the count of digits required to maintain precision. The number of significant figures is very important. Proper use of significant figures gives an immediate estimate of how precisely the quantity has been measured.

For example, a bathroom scale shows a value of 201.3 pounds for a person's weight. When converting from pounds to kilograms, you divide by a conversion factor of 2.20462 pounds per kilogram. Modern calculators show a result of 91.30825267 kg, but not all of these digits are needed (or desired). You should keep only the digits necessary to describe the precision of the original measurement but in the new units. Using the rules for significant figures and mathematical operations that follow, the result should be reported as 91.31 kg.

SECTION 2: RULES FOR SIGNIFICANT FIGURES

Determine the number of significant figures in a measurement:

1. All nonzero digits are significant.

Examples: 4.79 has three significant figures.

2779 has four significant figures.

856.429 has six significant figures.

2. Zeros in the middle of a number are significant.

Examples: 1.049 has four significant figures.

4.50809 has six significant figures.

780.12 has five significant figures.

3. Leading zeros (at the very beginning of the number) are not significant.

0.032 has two significant figures. (It may be written as 3.2×10^{-2}).

0.0000411 has three significant figures. (It is also 4.11×10^{-5} .)

If you are not sure of how many significant figures are present, convert the number to scientific notation. Numbers written in scientific notation are in the form of a coefficient multiplied by 10 raised to an exponent. The number of significant figures is the number of digits in the coefficient.

Examples:

0.00579 has three significant figures (may be written as 5.79×10^{-3})

0.048023 has five significant figures (may be written as 4.8023×10^{-2})

0.000024 has two significant figures (may be written as 2.4×10^{-5})

4. Zeros at the end of a number are significant if the number contains a decimal point.

Examples: 1.30 has three significant figures.

4.00 has three significant figures.

350. has three significant figures. (The decimal point at the end is the indication that the zero is significant.)

5. Zeros at the end of a number which does not contain a decimal point may or may not be significant (it is unclear). When in doubt, assume they are not significant.

The number 2500 could have been measured to the nearest ± 100 or to the nearest ± 10 or to the nearest ± 1 . Since it is not indicated, assume the worst case: the greatest uncertainty. 2500 has two significant figures. (2.5×10^3)

140 has two significant figures. (1.4×10^2)

If you are recording a number such as "2500", it is best to include the error estimate so there is no ambiguity.

SECTION 3: RULES FOR ROUNDING

If a calculated number has too many digits, it needs to be rounded to the appropriate number of significant figures. The rules for rounding are as follows:

1. Increase the last retained digit by one if the next digit is greater than or equal to 5.

Examples:

4.248 rounded to three significant figures. It rounds to 4.25

347.9 rounded to three significant figures. It rounds to 348

7815.246 rounded to six significant figures. It rounds to 7815.25

2. Decrease the last retained digit by one if the next digit is less than 5.

Examples:

94.24571 rounded to three significant figures. It rounds to 94.2

1.50802 rounded to five significant figures. It rounds to 1.5080

624804 rounded to two significant figures. It rounds to 620000

NOTE:

There are several ways of dealing with rounding when the next digit after the last retained digit is 5. The convention we are using is covered by Rule 1 (above). However, your instructor may want you to use a different one.

SECTION 4: DETERMINING THE NUMBER OF SIGNIFICANT FIGURES IN A CALCULATION

A calculation cannot be more precise than the least precise measurement used to do the calculation. (The least precise measurement limits the precision of the result.)

1. Multiplication and Division:

The number of significant figures in the result should be the same as that in the number with the fewest significant figures.

$$3.42 \div 1.3 = 2.630769 \quad \text{Rounded result} = 2.6$$

The reported result of 2.6 has two significant figures. This is because the number “1.3” in the original calculation has 2 significant figures (the fewest in this calculation).

$$43.10 \times 9.22713 = 397.6893 \quad \text{Rounded result} = 397.7$$

The reported result of 397.7 has four significant figures. This is because the number “43.10” in the original calculation has 4 significant figures (the fewest in this calculation).

$$7.58432 * 18.54 * 6.542 \div 42.5 = 21.64452 \quad \text{Rounded result} = 21.6$$

The reported result of 21.6 has three significant figures. This is because the number “42.5” in the original calculation has 3 significant figures (the fewest in this calculation)

2. Addition and Subtraction:

The number of decimal places in the result should be the same as in the number with the fewest decimal places. The digits in **bold** are the important ones to look at.

$$\begin{array}{r} 4.381 \quad 31.5563 \\ +131.4 \quad -1.410 \\ \hline 135.781 \quad 30.1463 \end{array}$$

Round to nearest 0.1 Round to nearest 0.001

Report result as 135.8 Report result as 30.146

3. Logarithms and Exponents:

A logarithm is made of two parts, the *characteristic* and the *mantissa*. The characteristic is everything before the decimal point in a logarithm. The mantissa is everything after the decimal point in a logarithm. Unless the base of the log is specified differently, it is assumed to be the common log of base 10.

Example 1:

$$\log (345.2) = 2.5381 \quad \text{which means that } 10^{2.5381} = 345.2$$

In this example, the number “2” in the logarithm is the characteristic and “.5381” is the mantissa.

Example 2:

$$\log (0.0024) = -2.62 \text{ or, } 10^{-2.62} = 0.0024$$

In this example, the number “-2” is the characteristic and “.62” is the mantissa.

With regard to significant figures, the number of significant figures in the mantissa should be the same as the number of significant figures of the original number. In Example 1, “345.2” and “.5381” have the same count of four significant figures. In Example 2, “0.0024” and “0.62” have the same count of two significant figures.

Converting quantities into logarithms is commonly found when working with pH and concentrations.

The mathematical definition of pH is:

$\text{pH} = -\log(\text{concentration of } \text{H}^+ \text{ in moles per liter}).$

Moles per liter is abbreviated as ‘M’.

Example: Determine the pH of a solution containing a concentration of H^+ of $2.56 \times 10^{-4} \text{ M}$, the calculation is:

$\text{pH} = -\log(2.56 \times 10^{-4} \text{ M}) = -(-3.591760035)$ which rounds to 3.591

In this example, the original concentration (**$2.56 \times 10^{-4} \text{ M}$**) has 3 significant figures (shown in **bold**). The mantissa of “3.**591**” has 3 significant figures.

4. Physical Constants, Discrete Numbers, and Mathematical Constants.

Physical Constants

Final results should reflect the uncertainty due to experimental error, not from calculations. Many physical constants have been measured to great precision so you have many more significant figures than you require. For example, Avogadro's Number (6.0221409×10^{23}) or the speed of light (2.99792×10^8 meters/second) have been measured to great precision. If you are using these constants in a calculation, and are hand-entering the numbers in your calculating device, then include at least two more significant figures in the constant than the quantity with the fewest significant figures in your calculation.

Example: If you are calculating how many atoms are present in a mass of 1.25×10^{-5} grams of lead (Pb), using the atomic mass of 207.19 grams/mole, the calculation should be set up as:

$1.25 \times 10^{-5} \text{ grams Pb} \times \frac{1 \text{ mole Pb}}{207.19 \text{ grams Pb}} \times 6.0221409 \times 10^{23} \text{ atoms Pb} \times \frac{1 \text{ mole Pb}}{1 \text{ mole Pb}}$
 $= 3.63321 \times 10^{16} \text{ atoms Pb}$

Note that the units cancel except for *atoms Pb*. The rounded result should have three significant figures, so the final result is $3.63 \times 10^{16} \text{ atoms Pb}$.

In this example, the atomic mass of Pb is a reference value presented with five significant figures, two more than the minimum of three significant figures. Avogadro's Number of 6.0221409×10^{23} should be entered in the calculation with at least five significant figures (more significant figures will not affect the calculated result after rounding).

Discrete Numbers

Measurements of discrete objects can be considered to have infinite precision. For example, there are precisely 12 eggs in a dozen (i.e., 12.00000...).

Example: If you are calculating a molecular weight that contains 2 copper atoms, the mass of the copper atom (63.55 g/mol) can be multiplied by 2 with no loss of significant figures.

Mathematical Constants

It generally does no harm to use too many significant figures in your constants, but it is not necessary. If your calculator has a constant such as π or e , then just use the button on your calculator and the number provided.

Example: The area of a circle is given by $A = \pi r^2$. For a radius of 4.247 cm, find the area.

$A = 3.14159 \times 4.247 \text{ cm} \times 4.247 \text{ cm} = 56.6648 \text{ cm}^2$. This rounds to 56.66 cm^2 .

Note that the value of π used has six significant figures, only two more than is needed for the quantity of "4.247 cm" which has four significant figures.

The use of 3.14 as a value of π in the above calculation would be incorrect because it has too few significant figures. The following calculation is provided to show what happens when you use too few significant figures.

Example of *incorrect* significant figure usage:

$A = 3.14 \times 4.247 \text{ cm} \times 4.247 \text{ cm} = 56.6362 \text{ cm}^2$. This rounds to 56.64 cm^2 .

Although it seems like a minor difference (0.02 cm^2) between the two numbers, these differences can propagate through many calculations and result with real differences in results.

SECTION 5: DETERMINING SIGNIFICANT FIGURES FOR MEASURED NUMBERS

Laboratory measuring equipment such as balances, scales, glassware marks, rulers, and calipers have limits on precision. The number of significant figures in measured quantities should imply the precision in the measurement. Equipment with digital readouts can be usually read as-is. All other equipment may have precision limits declared on them, or may be derived from the precision of the smallest graduation.

Refer to the specific users' manual, experiment or technique for additional guidance, or ask your instructor. Some general examples are provided below for common laboratory equipment.

For balances:

A balance that is precise to the nearest 0.01 grams (or 10 milligrams) will have a number precise to the nearest 0.01 grams. A mass measured on this balance as 5.24 g has three significant figures and is assumed to have an error of ± 0.01 grams.

A balance that is precise to the nearest 0.1 milligrams will have a number precise to the nearest 0.0001 grams. A mass measured on this balance as 5.2400 has five significant figures and is assumed to have an error of ± 0.0001 grams.

For rulers:

A ruler that is graduated to 0.1 centimeter (or 1 millimeter) will be measured to the nearest 0.1 cm or 1 mm. The length should be estimated to the nearest graduation. A length of 4.2 cm or 42 mm has two significant figures.

For graduated cylinders:

A 10 ml graduated cylinder that is graduated to 0.1 milliliter has a tolerance of 0.01 ml. It should be measured to the nearest 0.01 ml by estimation. A volume measured by this cylinder as 5.62 ml has three significant figures.

For burets:

A 50 ml buret that is calibrated to 0.1 milliliter can be estimated to the nearest 0.01 milliliter if sufficient distance exists between calibration marks. A volume measured as 15.24 ml has four significant figures.

For transfer pipets:

A 25 ml "Class A" transfer pipet has a specification of ± 0.03 ml. Properly used, this transfer pipet will deliver 25.00 ml of a solution, which should be recorded with four significant figures. Note: A more formal error analysis may use the specific declared error limits and is beyond the scope of this document.

For volumetric flasks:

A 100 ml Class A volumetric flask has a tolerance of ± 0.08 ml. Properly used, this flask will contain 100.0 ml of a solution, which should be recorded with 4 significant figures. Note: A more formal error analysis may use the specific declared error limits and is beyond the scope of this document.

SECTION 6: OTHER WAYS OF REPORTING SIGNIFICANCE

Using significant figures is the most common way to present quantities and imply the uncertainty in the last digit. It is a little more complicated to calculate statistically valid error range estimates.

However, sometimes specific error estimates are presented along with the number. Here are some results from metals analysis of a National Institute of Standards and Technology (NIST) standard reference material (Buffalo River Sediment, Standard Reference Material 2704):

Selected element	Weight %	Error
Aluminum	6.11	± 0.16
Calcium	2.60	± 0.03
Carbon	3.348	± 0.016
Iron	4.11	± 0.10
Magnesium	1.20	± 0.02

The declared numbers on the standard reference material have a minimum number of significant digits, and the error estimates are explicitly stated. This way of presenting the limits of uncertainty with data contains more information and is generally preferred for precision work.

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C Use of Balances

Technique C: Use of Balances

SECTION 1: PURPOSE OF TECHNIQUE

This technique describes how to use a laboratory balance to accurately measure the mass of a sample.

SECTION 2: OPERATIONS OF BALANCES

There are two general types of balances in the chemistry laboratory. There are precision balances, which usually measure to the nearest 0.01 grams (g), and the analytical balances, which usually measure to the nearest 0.1 milligrams (mg) (0.0001 g). The visibly distinguishing factor between the two is the presence of sliding doors on the analytical balance to protect from air currents. An example precision balance in the laboratory is model PB 602-S, or equivalent. The analytical balance is model AL 204, or equivalent. Both models are produced by Mettler-Toledo, LLC of Columbus, Ohio. Other manufacturers supply balances which have fundamentally similar basic operations.



Analytical Balance Precision Balance

Part A: Normal Weighing Procedure

1. Tare your scale before use, making sure the reading is 0. (See Part B for instructions on weighing material directly into a container).
2. Open one/two doors of the balance isolation cabinet (if using the analytical balance).
3. Place your sample on the balance pan.
4. Close side door(s) of the balance.
5. Wait until the stabilizing circle disappears.
6. Record the number as the mass of your sample.

When the number displayed no longer has a circle displayed next to it, the mass and balance has stabilized. Some balances have other ways to indicate that the balance is now stable. This means weighing analysis is complete and the number can be recorded.

Some balances are sufficiently sensitive to measure water as it is evaporating from aqueous solutions, or atmospheric water as it is hydrating the surface of reagents and equipment. In this situation, the balance may be stable and accurately displaying the correct mass, but the mass of the material may not be constant. In this situation, select the reading that will most closely represent the true value of the material being weighed.

A balance has a limit to how much weight it can read. There is normally a maximum capacity displayed on the front of the balance. **Do not go over this range—most balance pans are balanced on a small point and exceeding the capacity of the balance may cause damage.**

AL 204 (analytical balance)	PB 602-S (precision balance)
Max: 210 g	Max: 610 g
Min: 0.01 g	Min: 0.5 g

Part B: Weighing Procedure Using Tare Function

If you intend to use a container to carry your sample, you can tare (zero) the container on the balance. You will not have to subtract the initial weight of the container from the weight of the sample and the container together. However, review your procedure to see if you need the weight of the container and be sure to record the container weight if it will be needed in the future.



1. Set the empty container on the balance pan and close the doors if there is an isolation chamber. Press the tare button.
2. The balance should now display "0". This allows the balance to ignore the weight of the container and accurately measure only the sample.
3. Add your sample to the container and record the weight that the balance shows.
Remember to close the isolation chamber doors for an analytical balance.

SECTION 3: MAINTENANCE OF BALANCE ACCURACY

Modern balances are marvelous devices and are precision instruments. However, any balance will not give an accurate weight if it is mishandled. Depending on balance precision, a balance may need to be adjusted to match the gravitational acceleration of its location. Moving the

balance and level may affect balance measurements. Avoid moving the balance in order to preserve its reading accuracy.

Balances should be left powered on in order to establish stable voltages and thermal gradients within electronic components. Some balances may enter a standby mode after a period of time, but should remain powered.

If a leveling bubble is present, the feet of the balance should be adjusted until the leveling bubble is centered as shown:



Centered Leveling Bubble

Precision

The precision of a balance can be monitored by recording the weight of a stable object over time. Most commercial and government laboratories monitor balances weekly, although some places require a check prior to each use. The calibration of the balance may often be adjusted to a standard mass. Usually this adjustment is done by an independent calibration company.

Static Charge

Contact between two materials and then separating them may generate a static charge. This can introduce unknown forces (either up or down) on the balance pan and cause an error. The static can be lessened by using a balance pan with doors to insulate it. This helps with the accuracy of an analytical balance.

Thermal Effects

Weighing an object that is not at constant temperature can affect the accuracy of a balance reading. If the object is either too hot or too cold, the weight could drift due to air currents and thermal expansion of the weighing pan. A constant temperature results with increased stability and accuracy. Laboratories are generally kept at a constant temperature to prevent measurement drifting due to thermal effects.

Buoyancy Error

When weighing is carried out in air, this can create buoyancy error, which may or may not be significant. The balance is subjected to buoyancy due to being submerged in air. Taring can

prevent a drift due to buoyancy to help cancel this effect, or the buoyancy of air can be subtracted by calculation.

Refer to the following for more discussion and calculation:

https://chem.libretexts.org/Ancillary_Materials/Reference/Reference_Tables/Analytic_References/Appendix_09%3A_Correcting_Mass_for_the_Buoyancy_of_Air

SECTION 4: USER'S MANUAL FOR BALANCES

For further information on the balance in your laboratory, search online for the maker and model number. This manual can explain what features that specific balance can do which can vary between balance instruments.

<https://www.labcompare.com/191234-Eliminating-Inaccuracy-in-Precision-Weighing-Caused-by-Static-Charge/>

<https://bitesizebio.com/33245/drift-measurements-analytical-balances/>

https://www.mt.com/dam/mt_ext_files/Editorial/Generic/6/Weigh_Uncertain_Number5_0x0003d6750003db670009174c_files/measure_mass_force.pdf

https://www.mt.com/dam/product_organizations/laboratory_weighing/basic_weighing/Products/AL_and_PL/manuals/en/AL-PL-PL-S_OI_en_11780906C.pdf

<https://academic.oup.com/labmed/article/35/1/48/2504342>

All websites in this page were in service as of 06/26/2019.

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D Setup of vacuum filtration

Technique D: Setup of Vacuum Filtration

SECTION 1: PURPOSE OF TECHNIQUE

Vacuum filtration is a technique for separating a solid product from a reaction mixture. The mixture is passed through filter paper in a Buchner funnel. The solid is trapped by the filter and the liquid is drawn through the funnel into the flask below. A vacuum system is used to make the filtration faster.

Only a moderate vacuum is typically needed. **Fluid is NOT to enter the vacuum system.** Care should be taken to avoid liquid getting into the vacuum system. After vacuum filtration, the solid may not be completely dry. Filtered solid material is damp or wet on the surface and might still have to be air-dried or oven-dried.

Buchner funnels are typically used for stable solids and liquids (i.e., air and water unreactive material). Special techniques for dealing with air and water reactive materials are beyond the scope of this technique.

SECTION 2: COMPONENTS OF VACUUM FILTRATION

Filter flask

Buchner funnel

Vacuum adapter

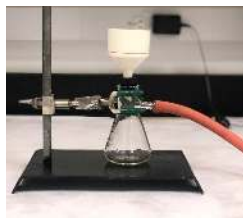
Vacuum tubing (thick tubing that resists compression)

Vacuum system (provided by the building services or a vacuum pump)

Filter paper

Stand and clamp

Set up the vacuum equipment similar to as the photo below. Your setup may not exactly look like the photo due to available equipment, but it should be functional.



Setup of Vacuum Equipment

SECTION 3: SETUP OF VACUUM FILTRATION

Part 1: Inspect the equipment

1. Check the equipment (filter flask, Buchner funnel, and vacuum tubing) for broken parts or cracks before use.
2. Make sure the filter paper fits in the Buchner funnel. (Folding of the filter paper is fine if it is bigger than the Buchner funnel; do not use filter paper if it is smaller than the Buchner funnel).
3. Make sure the Buchner funnel fits snugly on the filter flask.

Part 2: Putting everything together

1. Attach the filter flask to the stand using a utility clamp.
2. Place the Buchner funnel on the filter flask. Add a filter paper inside the Buchner funnel.
3. Secure the tubing to the arm of the filter flask on the side and to the vacuum system. The tubing should be long enough to attach to the vacuum located in the fume hood.

Part 3: Carrying Out the filtration

1. For each filtration, use only one filter and one filter funnel. Make sure the system is set up.
2. Turn the vacuum on slowly and gradually pour in your solution to the Buchner funnel. The solution you are filtering should consist of both a solid and liquid, not just a solid.

SECTION 4: IMPORTANT THINGS TO REMEMBER

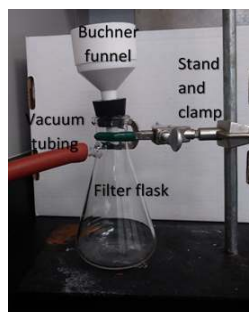
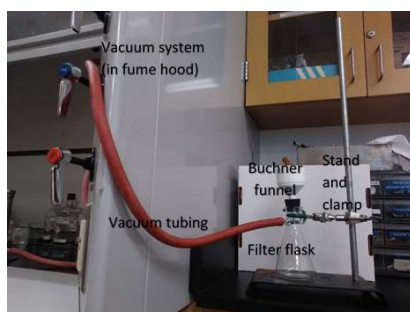
Requirements for Vacuum Filtration

Use filter paper that is the appropriate size for the funnel. It should lie flat on the inside of the Buchner funnel and cover all of the small holes of the funnel.

You may use deionized water to wet the filter paper enough so that it sticks to the Buchner funnel better and prevents solid substances from getting under the filter paper.

Use a utility clamp and stand to hold the filter flask down. Because you are using thick vacuum tubing for the filter flask, the tubing may cause the flask to be top-heavy. Not properly securing the filter flask can cause the filter flask to fall.

Some examples of how to set the vacuum filtration system are provided below:



When you are ready to filter, turn the vacuum on slowly and gradually pour in your solution.

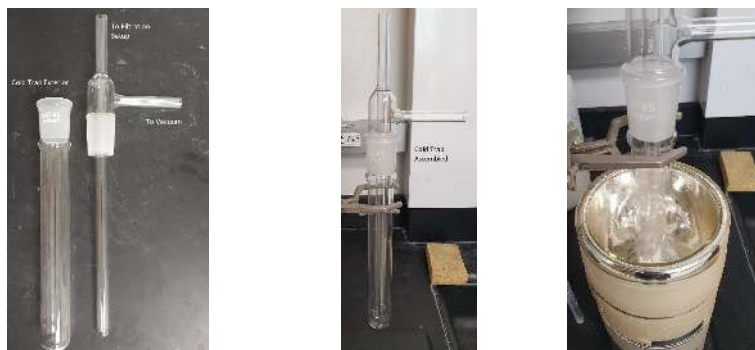
SECTION 5: SPECIAL CIRCUMSTANCES FOR LIQUID/VAPOR TRAPS

In special circumstances, a liquid trap may be connected between the system and the vacuum with vacuum tubing to prevent liquid from being suctioned into the vacuum system. Remember, **fluid is NOT to enter the vacuum system**. Using this technique serves as a precaution to prevent fluid from entering the vacuum system. Use of a secondary liquid trap is unlikely needed for this class.

An example of a vacuum trap assembly is shown below:



For certain high-vacuum systems, or when significant organic vapors may be present, additional protection for the vacuum system may be required. A cold trap is shown below, unassembled, assembled and clamped in place over an insulated Dewar flask. The Dewar flask is filled with an appropriate cold bath mixture such as ice, dry ice/acetone, or liquid nitrogen. It cools the exterior of the cold trap. Vapors and other contaminants will condense out on the inside of the cold trap, and the vacuum system is protected.



NOTE: Dewar flasks are evacuated double-walled glass vessels. They are fragile and need to be taped to avoid injury by glass in the event it implodes if broken. Use caution.

NOTE: If liquid nitrogen is used as the cryogen, some vacuum must be initially applied to the system to avoid accidental condensation of liquid oxygen and a resulting hazard. Consult with experienced personnel.

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E Volumetric Transfer Pipet Use

TECHNIQUE E: VOLUMETRIC TRANSFER OF REAGENT USING A PIPET

SECTION 1: PURPOSE OF TECHNIQUE

This technique describes how to use a volumetric pipet to collect, measure, and transfer a specific, precise volume of liquid. “Pipet” and “Pipette” are both correct spellings and “pipet” will be used here.

SECTION 2: OPERATIONS FOR A VOLUMETRIC PIPET TRANSFER

A common type of volumetric glassware is the volumetric pipet. It is an elongated glass tube with two, slim hollow glass tubes on either side of a glass bulb. The top end of the pipet is a fully-open tube, the bottom end of the pipet is tapered to a fine point. When held upright, above the glass bulb there is a calibration line that specifies the volume of liquid withdrawn at a specific temperature. The tapered tip that allows a precise stream of liquid to be dispensed from the bottom of the pipet. The volumes are fixed and marked on each pipet. The sizes often found in chemistry labs are 1.00-ml (milliliter), 2.00-ml, 3.00-ml, 5.00-ml, 10.00-ml, and 25.00-ml pipets. A rubber pipet bulb is used to generate suction and control the amount of reagent collected by the volumetric pipet.

1. Obtain two clean and dry beakers or other containers. One will be used for stock reagent. One will be used to receive the dispensed liquid.
2. Collect enough reagent for the procedure. Obtain a volumetric pipet of the appropriate volume for the experiment and a rubber pipet bulb.

Notes:

- Start with a dry pipet, or rinse the pipet with a little of the solution that you are measuring. If rinsing, try to coat all interior surfaces with the solution to be dispensed. Discard the rinse.
 - Several styles of pipet bulbs are available. The style shown does not affix to the top of the pipet, it is pressed briefly at the end of the pipet to generate a temporary seal to apply suction.
3. Hold the pipet vertically from the top. Hold the pipet bulb with the other hand and expel air from the bulb. Submerge the pipet tip below the surface of the reagent. Use the pipet bulb to provide suction and fill above the calibration line of the pipet.



4. Stop suctioning when the meniscus is a few centimeters above the calibration mark. Keep the pipet tip submerged in the solution. Be careful as the height of the liquid being sucked into the pipet gets faster as you get to the top.

5. When the liquid is above the calibration mark, remove the pipet bulb and quickly cover the opening of the pipet with a finger. Align your eye to the calibration mark so that it appears as a single paint line. Gently release the seal with your finger to allow the solution to drain from the pipet due to gravity. Let the meniscus move down until it is just touching the paint line of the calibration mark.



Correct

- Meniscus is right on the calibration line.



Incorrect

- Meniscus is not aligned with the calibration line.
6. Withdraw the pipet from the solution and quickly wipe the exterior of the pipet to make sure there are no liquid droplets on the exterior of the pipet. Place the pipet over the receiving container and allow the liquid to drain. Once the pipet has fully drained, wait an additional 15 seconds for any interior flow to complete.
 7. Tap the surface of the liquid three times with the tip of the pipet to reduce any surface tension built up in the beaker. Do not blow out the last drop from the pipet. The transfer pipet is calibrated to contain this last drop.

SECTION 3: TIPS FOR A PRECISE MEASUREMENT

A precise measurement of volume will not occur if the volumetric pipet is mishandled. When handling a volumetric pipet, there are certain factors that should be evaluated before throughout the procedure:

1. The tip of the pipet should be examined before the experiment begins to check if there is any damage that can affect the accuracy of the measurement. If the tip is chipped or shows any sign of damage that contacts the center hole of the pipet tip, insufficient surface tension will be present to maintain a constant size of the remaining drop in the transfer pipet. The pipet is likely not repairable and should be discarded.

2. It is important to hold the pipet from the top stem rather than in the middle. The middle of the pipet is made of very thin glass and can easily be broken when lateral force is applied.
3. Before placing the pipet bulb onto the opening of the pipet, squeeze the bulb to expel air before contacting it on top of the pipet. This will seal any air around the pipet that will make it hard to collect any liquid. It is not necessary to squeeze the bulb so vigorously to capture liquid. Slow presses on the bulb will allow enough liquid to come up and will give you control on the speed the liquid is collected at.
4. If the meniscus is not a little higher than the calibration mark while the pipet is submerged into the liquid, then the volume collected will not result in a precise mass. Once the pipet is taken out of the liquid, the change in volume allows the liquid to decrease which reduces the amount of mass the balance will read.
5. When transferring liquid using the volumetric pipet, it is important to use an index finger to stop the flow of liquid rather than the thumb. Thumbs tend to have larger areas of contact, which lead to less control of releasing pressure.
6. While the reagent is streaming down, do not let the tip of the pipet hit the liquid's surface or the sides of the beaker.

SECTION 4: FURTHER INFORMATION

Getting accurate results using the volumetric pipet takes practice. In order to achieve precise measurements of volume for an experiment, you must know how the pipet works and how to take care of it properly. For further information on using a volumetric pipet, check out the resources provided below.

https://chem.libretexts.org/Ancillary_Materials/Demos%2C_Techniques%2C_and_Experiments/General_Lab_Techniques/Use_of_a_Volumetric_Pipet

<http://faculty.sites.uci.edu/chem11/files/2013/11/RDGvolflaskpipet.pdf>

<http://dept.harpercollege.edu/chemistry/chm/100/dgodambe/thedisk/labtech/pipetech.htm>

<http://www.ehs.pitt.edu/workplace/pipetting.html>

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F Crucible use

Technique F: Use of Crucible

SECTION 1: PURPOSE OF TECHNIQUE

This technique describes how to use a laboratory crucible to safely heat a sample.

SECTION 2: OPERATIONS OF CRUCIBLE

There are several types of crucibles that can be used for heating up materials. Crucibles are often made from alumina, ceramic and porcelain. Most labs require the use of porcelain crucibles. Porcelain crucibles are glazed on the outside and inside, except there is usually no glaze on the bottom. Crucibles are meant to withstand high temperatures of 1020-1050°C. Crucibles can heat up a variety of materials like metals (copper, cobalt, etc.), hydrates and glass.

1. Use crucible tongs to pick-up crucibles and prevent fingerprints.
2. Put crucible on a clay triangle that has been placed over an iron ring attached to a stand and placed above a Bunsen burner.
3. Make sure that the crucible is directly in the center of clay triangle to prevent it from falling over.
4. After the crucible is in place, turn on the Bunsen burner flame (**CAUTION: DO NOT** use immediate high heat on the crucible) and begin heating.

SECTION 3: MAINTENANCE OF CRUCIBLE

The proper way to prepare the crucible and lid is to wash it with laboratory water **ONLY**. Use a gentle flame and heat the crucible for two minutes or until water is evaporated. This will allow the crucible time to be completely water free prior to more intense heating.

Place the clean crucible and lid on the wire gauze. **Do not** place the clean crucible or lid on the countertop. Clean it again if it is placed on the countertop.

SECTION 4: SAFETY PRECAUTIONS OF CRUCIBLE

- **DO NOT USE** broken or severely damaged crucibles. Thermal shock can break the crucibles if there are already defects in the crucible.
- **DO NOT** overheat the crucible. It may cause materials to spatter or to adhere the bottom
- **USE ONLY** crucible tongs to put on the lid when the crucible is over a flame.
- **DO NOT** touch a hot crucible with your hands. It can burn your hands.
- **NEVER** put down a hot crucible has been heated on the counter. **USE** a wire gauze.
- **NEVER** directly pick-up the crucible after it has been heated up. Allow the crucible to cool down first before using crucible tongs.
- **DO NOT** stand near the crucible, especially when material is heating up. Make sure to use the lid before turning up the flame on the material, as you **DO NOT** want the material to become ejected toward you.
- **DO NOT** allow cold water to interact with the hot water inside the crucible, it can cause thermal shock.
- **DO NOT** allow cold water to touch the crucible after it has been heated up. It can cause mechanical shock of the crucible.

Do's

Do use crucible tongs to pick up and put down the crucible and lid on the wire rack.



Do use crucible tongs to place the lid on the heated crucible.



Do place the lid and crucible after cleaning onto the wire gauze **and** when the crucible and lid are hot.



Do not's

Do not place the crucible and the lid on the counter at **ANYTIME** during the cleaning and experiment.



Do not place the crucible on the side. Make sure it is centered on the clay triangle.



<https://www.indianscientificinstrument.com/laboratory-porcelainware/porcelain-crucible.html>

<http://www.morganmms.com/media/1766/cruciblecareuseandsafetyinformation-mgam.pdf>

https://chem.libretexts.org/Ancillary_Materials/Demos%2C_Techniques%2C_and_Experiments/General_Lab_Techniques/Heating_a_Crucible_to_Constant_Weight

https://chem.libretexts.org/Ancillary_Materials/Demos%2C_Techniques%2C_and_Experiments/General_Lab_Techniques/HEATING_a_Crucible_to_Constant_Weight

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G Buret use

Technique G: Buret Use

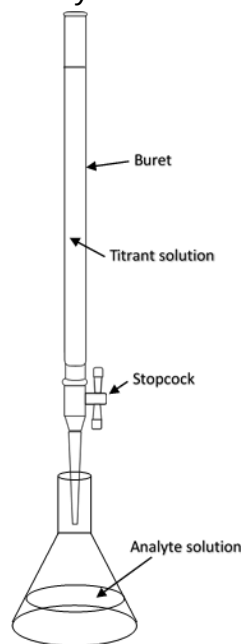
SECTION 1: PURPOSE OF TECHNIQUE

A buret is a glass apparatus for delivering precise, variable volumes of solution. It is primarily used for titration to determine the concentration of an unknown solution by adding a solution of known concentration. Titration is a type of quantitative analysis and the most common forms are acid-base, precipitation, complexometric, and redox titrations. Because the purpose of titration is to find the concentration of an unknown solution, it is very important that accurate measurements are taken to determine the volume of the known solution added to bring a reaction to an endpoint.

The *titrant* is the solution with known concentration and is placed in the buret. The *analyte* is the solution of unknown concentration that is being measured. The analyte is usually placed in an Erlenmeyer flask with an indicator. The equivalence point is the exact point where the moles of the titrant solution and the analyte solution are equivalent. The end point is when the indicator in the solution shows a color change. This usually occurs close to and just after the equivalence point is reached.

Using a buret as opposed to other measurable glassware, such as pipets and graduated cylinders, makes it easier to control the delivery of and measure the precise amount of solution being delivered. Slight turns of the stopcock located on the buret can control the flow of solution and there are methods to get as close as possible to the end point without going over. Because of its smaller meniscus, a buret is more accurate and precise than a graduated cylinder. A measurement made with a buret should be to the hundredths place (ex: 0.00). The burets generally used in the laboratory titration procedures hold up to 50 millimeters (ml). 10 ml, 25 ml and 1.0 ml burets are also commercially available. Refillable burets are available if

many routine titrations are needed. Automatic titrators are commercially available if needed.



SECTION 2: COMPONENTS FOR USING A BURET

Buret

Stand

Buret clamp

Deionized water bottle

Erlenmeyer flask

Waste beaker

SECTION 3: SETUP OF A BURET

Part 1: Inspect the equipment

1. Check the equipment for broken parts before use.
2. Inspect the glassware and the tip of the buret. As long as the bore of the buret tip is intact, the buret is functional.

Part 2: Preparing the Buret

1. Before titration, the buret must be rinsed with laboratory water and with the solution you are titrating with. Rinse the buret with small volumes laboratory water three times. Then, rinse the buret with the titrant solution three times, making sure to let the solution coat the inside of the buret completely and run through the tip.
2. Check for air bubbles in the tip of the buret. Air bubbles trapped in the tip can throw off your measurements. If you see an air bubble, you can get rid of it by filling the buret with a solution, firmly holding it over a sink, opening the stopcock fully for high flow and



vertically shaking the buret up and down in one quick motion.

Part 3: Setting up the Buret Apparatus

1. After the buret has been rinsed with the laboratory water and the solution, fill the buret with the solution, going above the 0.00 graduation mark.
2. Attach the buret to a stand using the buret clamp.
3. Place the waste beaker under the tip of the buret. Stand at eye level to the 0.00 graduation mark and turn the stopcock slightly to release some of the solution into the waste beaker. Try to get the meniscus to reach the 0.00 line as your initial measurement. It is okay if you go past the 0.00 line. Record the initial measurement.

Part 4: Performing the Titration

1. Fill the Erlenmeyer flask with the required amount of the analyte solution and make sure to **add indicator**. Adding indicator is the only way to visibly see whether you reached the end point or not. Place the Erlenmeyer flask under the tip of the buret.
2. Turn the stopcock of the buret, releasing the titrant solution into the Erlenmeyer flask. Control the flow of the titrant by adjusting the stopcock.
3. Observe the analyte solution for changes in color from the indicator. Swirl the solution in the flask to make sure the titrant is evenly distributed. The end point is close if the small bursts of color in the flask persist for more than a few seconds. Slow down the release of titrant into the flask by turning the stopcock as the endpoint is approached. Begin adding half drops instead of full drops if necessary. Titration is complete once the analyte solution reaches a uniform, persistent color. The end point is reached when the indicator color is faint. A more vibrant color may indicate that the end point was exceeded. If the endpoint was significantly surpassed, you will have to redo the titration.

The following photos show the general setup and a faint pink for the endpoint (phenolphthalein is the indicator used in the photos—other indicators may be used in the titration).



SECTION 4: IMPORTANT THINGS TO REMEMBER

Measure your buret to the nearest 0.01 ml

When taking measurements using the buret, make sure to view the meniscus at eye level. You should be looking straight at the line where the meniscus lies. The graduation line when looking directly at it should not form an oval.



In the left photo above, note that the inscribed circle at the 1.00 ml mark looks like an oval. In the right photo above, note the inscribed circle at the 20.00 ml mark looks like an oval, the 22.00 ml mark looks like a line, and the 24 ml mark again looks like an oval. If the mark closest to the meniscus is a line, it means that your eye is at the correct height to properly read the buret.

People are not generally good at estimating tenths, but are good at splitting by halves. Use this technique to estimate the final 0.01 ml reading between the tenths of a milliliter mark. Mentally draw a line $\frac{1}{2}$ way between the two marks. This will be the 0.05 ml between the two tenth milliliter marks. Repeat this process to split the distance between the 0.05 ml mark and the

inscribed mark. This will be a 0.025 ml increment. Evaluate where the meniscus is now relative to your estimates and round either up or down to the final reading.

For example, if your meniscus is between 22.30 ml and 22.40 ml, then your buret should be evaluated like this:

```

22.30  _____
22.325 - - - - -
22.35  - - - - -
22.375 - - - - -
22.40  _____

```

Now you should be able to accurately estimate where the meniscus falls on the buret to the nearest 0.01 ml.

Use your information

If you are doing multiple titrations, there is nothing wrong with doing calculations to estimate the volumes for your second, third, and fourth titrations. Recognize that your calculation is just an estimate and may not reflect the actual results of the titration procedure. The only way to know the actual concentration of the analyte is by doing the titration multiple times.

For example: You are performing an acid-base titration between hydrochloric acid (HCl) and sodium hydroxide (NaOH). The concentration of the sodium hydroxide is 2.1043 M and you are trying to determine the concentration of 15.00 ml of HCl. After your first titration, your final buret measurement is at 17.35 ml. Use this calculated estimate as a guide for the next titrations.



$$M_{\text{HCl}} = 2.1043 \text{ M NaOH} (17.35 \text{ mL NaOH}) / 15.00 \text{ mL HCl}$$

$$M_{\text{HCl}} = 2.434 \text{ M HCl}$$

For your second titration, you decide to use a 20.00 ml volume of HCl instead of a 15.00 ml volume. Using a little algebra, the amount of 2.1043 M NaOH that is expected to be added is given by:

$$V_{\text{NaOH}} = M_{\text{HCl}} V_{\text{HCl}} / M_{\text{NaOH}}$$

$$\text{For our example: } V_{\text{NaOH}} = 2.434 \text{ M HCl} \times 20.00 \text{ mL HCl solution} / 2.1043 \text{ M NaOH}$$

$$\text{For our example: } V_{\text{NaOH}} = 23.13 \text{ mL NaOH titrant (estimated)}$$

Therefore, for your second titration, dispense down about 20-22 ml of titrant before carefully adding the last drop to the endpoint.

Adding titrant by half-drops

If you think you are nearing the end point, instead of allowing the buret to release full drops one at a time, create half-drops. To do this, close the stopcock until the buret is no longer releasing

the titrant. Carefully open the stopcock slightly so that a drop forms at the tip of the buret, but not enough to fall. Turn the stopcock off. Use a deionized water bottle, and wash the tip of the buret so that the drop of titrant falls into the titration. This is about a half-drop. Remaining fluid on the tip of the buret at this point should be entirely laboratory water. The next full drop that the buret releases from the tip will actually be another half drop of titrant (combined with the remaining water on the tip). You can alternate back and forth to add half-drop quantities of titrant.



First titration quick method

Often, you have no idea where the first titration endpoint occurs. This makes it quite easy to accidentally exceed the endpoint and add too much titrant, with resulting accuracy error. One technique to work around this is to reserve a little bit of analyte solution. Pour a small volume of analyte into out into a clean, small beaker. It is considered reserve because if you accidentally over shoot the end point, you can use reserve to save your analysis. The titrant can be added relatively quickly until the indicator changes color.

The following photos show how this works. A small amount of analyte is temporarily transferred to a small beaker. The titration is rapidly done dropwise until the indicator changes. The reserve is returned to the Erlenmeyer beaker to change the indicator back, and you now know you are close to the endpoint.



Now, use the half-drop technique described earlier to finish the titration. Remember to return the reserve to the Erlenmeyer flask regardless if it were needed. After you have completed one titration, subsequent volumes can be estimated as was described above.

H Use of the Bunsen burner

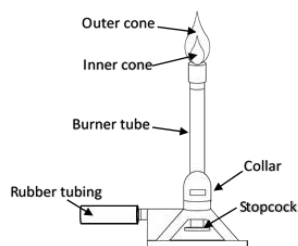
Technique H: Use of a Bunsen Burner

SECTION 1: PURPOSE OF TECHNIQUE

A Bunsen burner is a piece of equipment used for the general purpose of heating substances. When connected to a gas line and lit with either a striker or a match, Bunsen burners generate a controllable open flame that is hot enough for most classroom experimental procedures. The flame produced consists of a visible outer cone and inner cone, in which the topmost tip of the inner cone is the hottest part of the flame (See Figure 1).

In case of emergency, shutting off the gas line is the best way to stop the flame. Make sure the gas line is completely off at all times unless the Bunsen burner is in use.

The Bunsen burners used is a Humboldt model PN 6200.1. These are the standard Bunsen burners used for classroom laboratory experiments but other models and brands exist. Certain attachments are available for Bunsen burners such as fan adaptors (not shown) that can change the shape of the flame if desired.



SECTION 2: EQUIPMENT NEEDED FOR A BUNSEN BURNER

Bunsen burner

Gas source

Rubber tubing (gas tubing)

Striker or matches

SECTION 3: SETUP

Part 1: Inspect the equipment

1. Check the equipment for broken parts before use.
2. Inspect the rubber gas tubing for cracks by stretching it and looking for any major cracks throughout the tubing. If the tubing leaks, discard it and replace with new tubing.

Part 2: Assemble everything together

1. Attach one end of the rubber tubing to the arm of the Bunsen burner and the other end to the gas line located on your lab bench.
2. Get a striker or some matches.

Part 3: Lighting the Bunsen burner

1. Once the Bunsen burner and rubber tubing are secure, turn on the gas line.
2. If using a striker, squeeze the handle of the striker over the top of the Bunsen burner to generate a spark where gas is emitted. Do this until a flame appears. Adjust the size of the flame using the gas line knob or the stopcock as needed.
3. If using matches, strike the head of the match against the striking strip on the back of the pack. Hold the flame up to the top of the Bunsen burner where gas is emitted until a flame appears. Adjust the size of the flame using the gas line knob or the stopcock as needed.

SECTION 4: IMPORTANT THINGS TO REMEMBER

Precautions and tips for using a Bunsen burner

The gas lines located in the lab benches and the fume hood have a gas pressure of 4 pounds per square inch (psi). For reference, keep in mind that the pressure needed to blow up a balloon is 15 psi (atmospheric pressure). The pressure of the gas lines is not a dangerous amount, but always be aware of its use. When the gas line is on, you will normally hear a slight hissing sound. Natural gas has no smell, but a trace amount of a sulfur compound is added so that it does have odor. You might begin to smell it from unburned gas. To avoid leaks, you should not have the Bunsen burner or the gas line on when not in use.

Bunsen burners are generally safe to use as long as they are used properly and with common sense. For example, the Bunsen burner should be kept away from the edge of the table or anywhere it could easily be knocked over. Do **not** reach over the Bunsen burner, even if it is off. Be cautious when heating glass containers. If the container is too cold (from either a reaction, ice bath, or other freezing procedure), applying direct heat to the glass may cause it to crack or break.

There are several features on the Bunsen burner meant for controlling the size of the flame, but generally, the most effective way is by controlling gas valve located on your lab bench or in the fume hood. The gas valve controls the amount of gas entering the Bunsen burner. Allowing more gas to enter the Bunsen burner creates a larger flame. Twisting the collar clockwise will reduce the amount of air entering the Bunsen burner, cutting off oxygen supply to the flame. The flame can also be adjusted by using the stopcock located at the bottom of the Bunsen burner, which adjusts the flow of gas entering the Bunsen burner.

Here are some pictures of a Bunsen burner set up:



The hottest part of the flame is located at the tip of the inner blue cone. Bunsen burner flames are hot enough to soften soda-glass, and to fire-polish most other glass types. These flames are not hot enough to melt borosilicate glass, which is used for most other laboratory glassware.

Additional safety tips

Never leave a flame unattended.

Check the area around your Bunsen burner for flammable objects (paper towels, flammable solvents, etc.) prior to lighting the flame.

Hot objects look just like cold objects. Wait sufficient time for objects to cool before handling them.

Metals can conduct heat very well. Be aware that this may happen when heating objects with clamps.

Loose hair and clothing should be tied down to avoid contact with any flame.

Heating glass can create non-visible thermal stress that can weaken glass to mechanical shock, resulting in breakage. Be careful to avoid unnecessary force on areas of glass that have been heated.

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I Use of Spec 20/Spec 200

Technique I: Use of Spectrophotometer

SECTION 1: PURPOSE OF TECHNIQUE

This technique describes how to use a laboratory spectrophotometer.

SECTION 2: OPERATIONS OF SPECTRONIC 200

There are several types of spectrophotometers that are used to determine the amount of light absorbed within a given sample solution. The spectrophotometers used in the lab are the Spectronic 20, Spectronic 20+ and Spectronic 200. Before using the Spectronic 200, follow these steps to make sure that no damage or injury will occur due to faulty or careless behavior.

NOTE: Several terms are used for the container used to hold a sample in a spectrophotometer. This procedure will use the term 'optical cell' but other terms like 'cuvette' may be used in reference texts and instrument manuals.

1. Make sure the spectrophotometer is on a stable, flat surface free from vibrations.
2. Plug in the spectrophotometer and turn the power switch at the back of the machine on.

The following image will appear on the screen when it is on. (Figure 1)



Figure 1. On screen

3. Make sure that the sample compartment is empty. Then, press ENTER. The following image will appear on the screen. (Figure 2)



Figure 2

4. Look at the image below. Once at this point, wait for approximately 1 minute.
5. The screen will then display the Home Menu. Use the arrows to select the SPEC 200E MODERN INTERFACE option, then press ENTER. (Figure 3)



Figure 3. SPEC 200E MODERN INTERFACE

6. The screen will then show APPLICATION LIVE DISPLAY. Press the right arrow. (Figure 4)



Figure 4. APPLICATION LIVE DISPLAY

7. The screen will then show APPLICATION MULTI λ , press the right arrow again. (Figure 5)

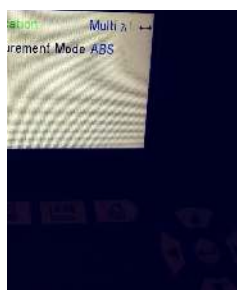


Figure 5. APPLICATION MULTI λ

8. Then the screen will show APPLICATION QUANT, press the right arrow again. Refer to figure 6.

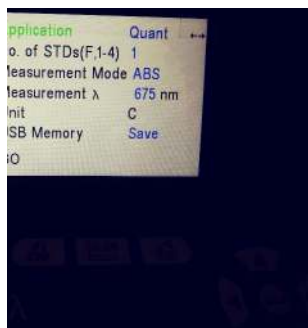


Figure 6. APPLICATION QUANT

9. Then the screen will show APPLICATION ANALYZER, press the right arrow again. Refer to figure 7.

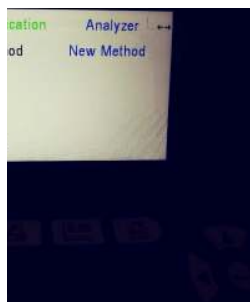


Figure 7. APPLICATION ANALYZER

10. Then the screen will show APPLICATION SCAN, press ENTER. Refer to figure 8.

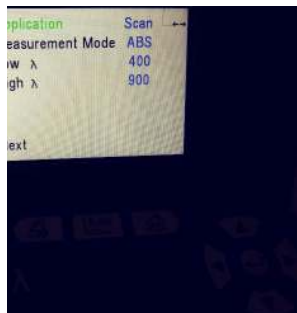


Figure 8. APPLICATION SCAN

11. Fill an optical cell half-way with laboratory water. Use Light-Duty Tissue Wipers, to wipe off any fingerprints on the outside of the cell.
12. Place the optical cell into the sample compartment, then press the AUTOZERO KEY.
13. After the spectrophotometer is zeroed, replace the optical cell filled with water with an optical cell filled with your chemical sample.
14. Wipe the optical cell with the Light-Duty Tissue Wipers and place it into the sample compartment. Press ENTER.
15. After the spectrophotometer is done analyzing the sample, a graph will appear to the right of the screen.
16. Use the right arrow to move cursor to the peak of the curve. This will show the peak absorption.
17. Press the HOME KEY when you are done. To turn the spectrophotometer off, flip the power switch on the back of the machine.
18. **DO NOT** move the spectrophotometer if it is plugged in, as it may cause damage to it. **DO NOT** yank the cord out of the wall, but gently pull from the base of the plug on the cord.
1. If there are any issues with the use of Spectronic 200, refer to the link below:

https://archive-resources.coleparmer.com/Manual_pdfs/02652-55.pdf



Figure 9. Spectronic 200

SECTION 3: HOW TO USE THE SPECTRONIC 20 AND 20+

To minimize any potential problems, follow these steps:

1. Turn on the spectrophotometer by turning the power switch located on the left side of the instrument. Allow for the instrument to warm up for 15 minutes.
2. After warming the machine up, use the Wavelength Control Knob to set to the wavelength that you want.
3. Adjust the filter lever to the needed position for the wavelength being used.
4. Set the mode to 100% T (Transmittance) or 0 A (Absorbance) by turning the knob under the wavelength knob. This may take several turns.
5. Fill a clean Spectronic 20 tube with laboratory water as a blank solution and wipe it with a KIMwipe™ to remove liquid droplets, dust, and fingerprint residue.
6. Place the Spectronic 20 tube in the sample compartment and align it with the guide mark within the compartment. Ensure that the optical cell is firmly placed in the compartment before closing the compartment cover.
7. Make sure that the absorbance of the water sample is at 0 by using the 100% T or 0A knob to adjust the absorbance accordingly.
8. Remove the tube from the sample compartment and discard the water.
9. Rinse the tube twice with whatever solution is being analyzed and then fill it half way with the solution.
10. Repeat steps 5 and 6, then read the screen to determine how much light is being absorbed by the solution (Absorbance).
11. For an accurate reading, make sure that the needle aligns with its reflection in the mirror.
12. Remove the Spectronic 20 tube from the sample compartment then close the lid of the compartment.
13. Turn off Spectronic 20 by turning the 100% T knob counterclockwise until it clicks.
14. If there are any issues with the use of Spectronic 20 and 20+, refer to the link below:
http://fg.cns.utexas.edu/fg/course_notebook_appendix_ii_files/Spectronic_20_D.pdf



Figure 10. Spectronic 20+



Figure 11. Spectronic 20

SECTION 4: TIPS FOR USING SPECTROMETERS

To minimize any potential problems, follow these steps:

1. Do not touch the bottom of the optical cell (or tube) when placing it in the spectrophotometer. To ensure that optical cell (or tube) is clean, use a paper towel or Light-Duty Tissue wipers to wipe away fingerprints.
2. Make sure to fill the optical cell (or tube) half way with the solution to ensure that all of the light path in the spectrophotometer is used.
3. When pouring the solution into the optical cell (or tube), make sure there are no bubbles present in the solution as this could causes potential errors.
4. For Spectronic 20+ make sure to align the solution to the mark in the front of the instrument.

SECTION 5: REFERENCES

[https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_\(Physical_and_Theoretical_Chemistry\)/Kinetics/Reaction_Rates/Experimental_Determination_of_Kinetcs/Spectrophotometry](https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Kinetics/Reaction_Rates/Experimental_Determination_of_Kinetcs/Spectrophotometry)

http://fg.cns.utexas.edu/fg/course_notebook_appendix_ii_files/Spectronic_20_D.pdf

https://archive-resources.coleparmer.com/Manual_pdfs/02652-55.pdf

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J: Use of Centrifuge

Technique J: Use of Centrifuge

SECTION 1: PURPOSE OF TECHNIQUE

This technique describes how to use a laboratory centrifuge.

SECTION 2: OPERATIONS OF CENTRIFUGES

There are many types of centrifuges available for laboratory use. Each centrifuge should have a specific manual for proper operation. If the manual is not available, an internet search for the manual under the specific model is advisable.

Centrifugation is used for the separation of solid-liquid mixtures that are stubborn to settle or difficult to filter. Centrifugal force is generated by rapidly spinning samples. Materials of higher density migrate solids to the bottom of the centrifuge tube.

A centrifuge can spin up to 10,000 rotations per minute. An unbalanced load will cause the centrifuge to knock and wobble. If severely unbalanced, the centrifuge can even wobble off the benchtop, potentially causing harm. To prevent wobbling, every sample in the centrifuge needs to be balanced by an equal mass of a vial in the opposite chamber in the centrifuge.

Special test tubes or centrifuge tubes must be used that exactly fit the width of the chambers in the centrifuge. Each tube should be filled to no greater than three-quarters full as the samples will be tilted in the centrifuge and could spill out.

SECTION 3: BALANCING A CENTRIFUGE

If only one sample is to be centrifuged, a tube of water containing an equal mass should be placed into an opposing position in the centrifuge to maintain balance. More than one sample can be centrifuged at a time, with the only requirement being that each opposing tube must have nearly the same mass.

If identical tubes are being used, filling the balance tube to the same level with water is usually close enough. If significantly different tubes are being used, or the centrifuge seems to be off-balance, weigh the individual tubes and add or subtract water from the balance tube until both are nearly the same mass.

SECTION 4: OPERATING A CENTRIFUGE

To operate the centrifuge, load the tubes, close the lid and turn on the centrifuge. Set the rotation speed if your instrument allows for it (a good general speed is 8,000 rotations per minute) and turn the dial to the recommended amount of time (set by your instructor or experimentation). Allow the system to spin for the designated time and turn it off.

Some centrifuges have an internal brake, but rapid breaking could stir up solids. It is better to simply let a centrifuge slow and come to a stop on its own. The liquid can then be decanted or pipetted off of the solid.

The time needed to separate solids from liquids in tubes varies on many factors, including but not limited to the density differences between the components and the size of particles of the solid. If the mixture is not separating after a reasonable amount of time, consider an alternative method.

SECTION 5: REFERENCES

[https://chem.libretexts.org/Bookshelves/Organic_Chemistry/Book%3A_Organic_Chemistry_Lab_Techniques_\(Nichols\)/01%3A_General_Techniques/1.05%3A_Filtering_Methods/1.5G%3A_Centrifugation](https://chem.libretexts.org/Bookshelves/Organic_Chemistry/Book%3A_Organic_Chemistry_Lab_Techniques_(Nichols)/01%3A_General_Techniques/1.05%3A_Filtering_Methods/1.5G%3A_Centrifugation)

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CHAPTER OVERVIEW

Part 3 Supplement

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[2505 Hydrates PreLab](#)

[2507 Stoichiometry of Reactions PreLab](#)

[2512 Periodic Properties PreLab](#)

[2516 Intermolecular Forces PreLab](#)

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PRELAB ASSIGNMENT

1. A student trying to determine if a white solid is a true hydrate heats the sample and finds that there is evolution of water, that the residue obtained is soluble in water and that the solution is colorless. Is this a true hydrate? Explain.
2. The chemical formula for Epsom salt is $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$. What is the IUPAC name for Epsom salt? Calculate the percent water of hydration.

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2507 Stoichiometry of Reactions PreLab

EXP_2507-STOICHIOMETRY OF REACTIONS

PRELAB ASSIGNMENT

1. Calculate the molar mass of calcium hydroxide and calcium chloride. (Make sure you use the correct chemical formula).
2. Write the balanced chemical reaction between HCl and NaOH.
3. If the reaction between water and calcium produces a basic solution, what products must be formed.

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2512 Periodic Properties PreLab

PRELAB ASSIGNMENT

Suppose that substances X, Y, and Z each potentially react with the anion (X^- , Y^- , or Z^-) of another species in the series. In solution, X is yellow, Y is blue, and Z is red. The anions are all colorless.

1. When a solution of Z is reacted with X^- the color changes from red to yellow, meaning that the reaction, $Z + X^- \rightarrow Z^- + X$ has occurred.

1. Which reactant was oxidized?
2. Which reactant was reduced?
3. What is the better oxidizing agent, Z or X?

2. When a solution of X is reacted with Y^- , the color changes from yellow to blue.

1. Write an equation for the reaction that occurred, assuming it is the same type of reaction as above.
2. Which reactant was oxidized?
3. Which reactant was reduced?
4. What is the better oxidizing agent, X or Y?

3. Rank X, Y, and Z in order of decreasing oxidizing ability. Best oxidizing agent first.

_____ > _____ > _____

4. Would you expect a reaction when Y is combined with Z^- ? Why or why not?

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2516 Intermolecular Forces PreLab

Exp_2516_Intermolecular Forces

Pre-Lab Questions

1. Provide the Lewis Dot structure and molar masses of each of the chemicals listed in the Chemicals and Solutions Table.

2. Which type(s) of IMFs occur within each chemical? Provide an explanation for your classification of intermolecular forces.

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 - J: Use of Centrifuge - *Undeclared*
 - Part 3 Supplement - *Undeclared*
 - 2505 Hydrates PreLab - *Undeclared*
 - 2507 Stoichiometry of Reactions PreLab - *Undeclared*
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